# MRI-DRIVEN EVALUATION OF PASSIVE MYOCARDIAL STIFFNESS

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# Abstract

Altered myocardial mechanical behavior is a component of maladaptive cardiac remodeling implicated in the etiology of heart failure. However, the specific underlying mechanisms and the extent to which the changes contribute to the progression of heart failure are poorly understood. Biophysical modeling of the heart has opened up significant possibilities for thoroughly investigating cardiac mechanics and could be a useful tool for further elucidating the mechanisms of cardiac dysfunction associated with heart failure. This thesis focuses on using a magnetic resonance imaging (MRI) approach combined with finite element modeling (FEM) to evaluate passive myocardial stiffness. This thesis contributes to the field in two specific ways: validating MRI-based passive myocardial stiffness estimation and using *in vivo* MRI to characterize healthy passive cardiac mechanics.

In order to validate an approach to passive myocardial stiffness estimation that combines MRI and FEM, we developed an experimental setup which incorporates synthetic heart phantoms with realistic cardiac geometry, as well as tunable heart-like stiffness and MRI-relaxation properties. Using the setup, we evaluated the degree of accuracy with which an MRI and FEM-based framework could identify the stiffness of heart phantoms of varying geometry and stiffness. Comparison of the simulated results to ground-truth stiffness values obtained independently through benchtop tensile testing, revealed errors that ranged from 1% to 6% across all models, thereby demonstrating the feasibility of highly accurate passive myocardial material stiffness estimation using MRI and FEM.

Regarding characterizing healthy passive cardiac mechanics using *in vivo* MRI, we established a tractable pipeline for developing personalized cardiac computational mechanics models. The pipeline combines MRI methods capable of estimating high-resolution cardiac geometry, *in vivo* microstructure, dynamic motion and local deformation kinematics over

the cardiac cycle. The *in vivo* passive myocardial stiffness estimation method was tested in healthy subjects in order to establish normal reference values for passive myocardial stiffness. Furthermore, we evaluated the sensitivity of passive mechanics to microstructural configuration. This kind of personalized biomechanical modeling, when extended to patients with heart failure, can provide new insights regarding the remodeling occurring in the disease. Such estimates of passive myocardial stiffness could help inform diagnosis and improve management of patients with a failing heart.

# Dedication

This thesis is dedicated to my dear brother:

## Fowokanmi Kolawole

$14^{th}$ December 1990 - $6^{th}$ October 2021

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Prior to starting my PhD, I heard from multiple sources that it would be difficult to simultenously optimize for 3 P's: Project, Principal Investigator (PI), and People in the lab. As it turns out, I am among the lucky few to have worked on an awesome project, under the supervision of an amazing PI, in a lab with incredible and collaborative people.

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# Chapter 1

# Introduction

## 1.1 Motivation

At the turn of the 21st century, when heart failure (HF) was initially designated as an emergent epidemic [11], in the USA, 4.7 million adults were estimated to be suffering from the condition [12]. Now, over 20 years later, almost 7 million US adults have HF [13], and the prevalence is projected to exceed 8 million by 2030 [14]. This increase in prevalence will further burden the healthcare system. Associated HF costs are projected to reach $70 billion within the next decade [14]. Moreover, HF survival rate remains low, with an estimated 5-year mortality rate of around 75% [15]. Given the increasing prevalence, substantial healthcare burden, and high mortality, it is critical that approaches aimed at HF diagnosis, prevention, and management are continually developed, improved, and translated to the clinic.

HF is a multifaceted clinical syndrome caused by inability of the heart to supply adequate blood to meet the metabolic demands of the body. This condition is commonly classified based on pumping ability, using the left ventricular (LV) ejection fraction (EF), a measure of the fraction of blood pumped during each heartbeat. Using this metric, patients can be clinically diagnosed as having HF with reduced ejection fraction (HFrEF) if their EF is <40%, or HF with preserved ejection fraction (HFpEF) if their EF is >50%. Recently, another sub-type, HF with mildly reduced ejection fraction (HFmrEF), has been defined as an EF between 40 and 50%. It is important to note that there is still some debate over the

current classification of HF according to EF, with some clinicians and researchers arguing that the EF-based phenotypes do not represent distinct forms of HF but exist as part of a single HF spectrum [16, 17].

Across all HF phenotypes, the symptoms are consistent - dyspnea, fatigue, exercise intolerance, and edema in the distal extremities. However, the nature of ventricular dysfunction underpinning the different HF phenotypes are distinct. HFrEF is characterized by systolic (pumping) dysfunction or reduced heart contractility, and is typically accompanied by progressive LV dilation and wall thinning. In HFpEf, ventricular filling during diastole is compromised, leading to inadequate cardiac output despite maintenance of systolic function as measured by EF. In patients with HFpEF, there is also typically concentric LV remodeling and ventricular wall hypertrophy. HFmrEF shares characteristics of HFpEF and HFrEF and may be associated with mild systolic and diastolic dysfunction [18].

There exist several prognostically beneficial pharmacologic interventions for HFrEF such as angiotensin-converting-enzyme (ACE) inhibitors, beta-blockers, and mineralocorticoid receptor antagonists [19]. These therapies, which can reduce mortality by up to 35% [20, 21], have been successful because they target the neurohormonal responses to reduced cardiac output in HFrEF, thus slowing, and in some cases, reversing the associated pathophysiological cardiac remodeling [22, 23]. Moreover, device therapy and cardiac rehabilitation have also been shown to improve HFrEF prognosis [24, 25, 26]. On the other hand, evidence-based therapeutic options to improve outcomes for patients with HFpEF are limited. Currently, there are very few successful clinical trials, and only in recent times [27, 28]. This failure to identify effective pharmacologic intervention is largely due to an incomplete understanding of the multi-scale remodeling underlying each patient's HFpEF progression, and significant pathophysiologic heterogeneity in the HFpEF patient population. Although ventricular dysfunction and adverse cardiac remodeling are at the heart of HFpEF, management of the disease is mostly directed at related conditions and symptoms (e.g. hypertension) [29, 30]. This approach to therapy has largely been unsuccessful at improving clinical outcomes in HFpEF. To achieve similar success in HFpEF management, as has been achieved in HFrEF, there is need for a new clinical approach that targets the various mechanisms of ventricular dysfunction and cardiac remodeling unique to HFpEF. Identification of these mechanisms requires a better understanding of HFpEF

pathophysiology.

HFpEF is the culmination of a complex interplay of risk factors and comorbidities, cellular and organ level pathology, ventricular structural remodeling, and hemodynamic abnormalities that lead primarily to LV diastolic dysfunction. This dysfunction could present as slowed active ventricular relaxation, when the cardiomyocytes, in an energy intensive process, relax from their contracted states at end-systole [31]. The diastolic dysfunction could also be in the passive relaxation phase, when the ventricles passively fill under the action of atrial systole [32].

Increased LV filling pressure, a marker of reduced LV chamber compliance, is a consistent finding in HFpEF patients. Implicated in HFpEF etiology, are two significant cardiac remodeling mechanisms that can lead to this reduction in LV chamber compliance - ventricular hypertrophy and elevated passive stiffness of the LV myocardium (heart muscle tissue). Patients can exhibit varying levels of increased passive myocardial stiffness and ventricular hypertrophy, while exhibiting similar degrees of HFpEF. While ventricular hypertrophy can readily be assessed via many medical imaging modalities, there currently exists no accepted clinical method to measure passive myocardial stiffness. It has been hypothesized that there are sub-types within the HFpEF spectrum characterized by changes in passive myocardial material stiffness [33]. Identification of patients with this sub-type can improve characterization of HFpEF, enabling more targeted clinical diagnosis and treatment of the disease. Thus, to improve HFpEF management, there is a pressing clinical need for a method to adequately measure subject-specific passive myocardial stiffness.

One emerging approach to measure *in vivo* passive myocardial stiffness is through finite element modeling based on medical imaging data [34]. For these medical imaging-based passive myocardial stiffness estimation methods to effectively transition from lab to clinic, there is a need to: (1) establish a tractable pipeline to develop personalized cardiac computational mechanics models from clinical data, (2) extensively validate these passive myocardial stiffness estimation frameworks to understand the accuracy and repeatability of the passive myocardial stiffness estimates, (3) test the methods in healthy subjects and patients with heart failure in order to establish normal reference values for passive myocardial stiffness and characterize changes in disease progression.

## 1.2 Specific Aims

The overall objective of my thesis is to develop and validate a clinically translatable *in vivo* passive myocardial stiffness evaluation framework by integrating advanced Magnetic Resonance Imaging (MRI) and Finite Element Modeling (FEM). To this end, the specific aims of this research were to:

1. develop a tunable experimental system for validating MRI-based passive myocardial stiffness estimation approaches by using synthetic heart phantoms with realistic cardiac morphometry, and heart-like stiffness and MRI-relaxation properties;

2. evaluate the accuracy of an MRI and FEM-based framework for estimating passive myocardial stiffness using synthetic heart phantoms of varying stiffness and geometry;

3. establish a tractable pipeline for developing personalized cardiac mechanics models for evaluating passive myocardial stiffness using *in vivo* MRI methods from which cardiac geometry, deformation kinematics, and microstructure can be evaluated;

4. test the *in vivo* passive myocardial stiffness estimation approach in healthy human subjects in order to establish reference values for passive myocardial stiffness, and evaluate the sensitivity of passive mechanics to microstructural configuration.

The passive myocardial stiffness estimation framework developed in this thesis can be extended to HFpEF patients to enable the establishment of quantitative links between patient-specific passive myocardial stiffness and the progression of the disease. This could revolutionize HFpEF management and diagnosis and provide a useful metric for evaluating the sub-type and severity of disease, the effectiveness of therapy, and provide an end point for clinical trials aiming to improve clinical outcomes in HFpEF.

## 1.3 Thesis Overview and Contributions

**Chapter 2** provides background information on healthy cardiac anatomy, the cellular components of myocardium, myocardial mesostructure, the cardiac cycle, and cardiac function

in the healthy state. Some of the major cardiac derangements in Heart Failure are detailed, with a particular focus on the dysfunctions in HFpEF.

**Chapter 3** reviews MRI phsyics, discusses some pulse sequences, and highlights specific imaging methods used for assessing cardiac structure, function and kinematics.

**Chapter 4** presents some of the key elements of continuum mechanics necessary for biomechanical modeling of myocardium including mathematical descriptions of the kinematic quantities, existing constitutive models of passive myocardium, and mechanical laws of conservation. The finite element method is described and key points in the literature on *ex vivo* and *in vivo* passive myocardial mechanical behavior are summarized.

**Chapter 5** describes the development of a custom-built highly controlled experimental setup for validating MRI-based evaluation of passive myocardial stiffness, using synthetic heart phantoms of realistic cardiac morphometry and known myocardium-like stiffness. The synthetic heart phantom manufacturing procedure is detailed, as well as the approach used for simulating diastolic LV filling in the heart phantoms using the MRI-compatible *in vitro* system.

**Chapter 6** extends the work in Chapter 5 by using the experimental framework developed therein to validate an MRI-driven computational constitutive modeling approach for evaluating passive myocardial stiffness. The accuracy of the stiffness estimation approach was evaluated in heart phantoms of varying geometries and stiffness by comparing the phantom material stiffness estimates with a ground-truth obtained independently from benchtop tensile testing.

**Chapter 7** details the establishment of a method to integrate multi-phase *in vivo* cardiac geometric data (cine bSSFP) and kinematic data (tagged MRI) with single phase *in vivo* microstructural data (cDTI) for the development of a personalized LV mechanics model for evaluating subject-specific passive myocardial mechanical behavior.

**Chapter 8** extends Chapter 7 by testing the passive myocardial stiffness estimation framework in multiple healthy human subjects in order to establish reference human *in vivo* passive myocardial stiffness values. The sensitivity of passive mechanics to microstructural configuration is also assessed.

**Chapter 9** concludes by summarizing the findings from this thesis, outlining limitations of the work, and stating longer term goals of this research. The chapter is ended with a note on the outlook of the field.

## 1.4 Publication List

Publications arising directly from this work are highlighted.

### 1.4.1 Articles and Book Chapters

**Kolawole, F.O.**, Wang V.Y., Freytag B., Loecher M., Cork T.E., Nash M.P., Kuhl E., Ennis D.B., 2023. "Integrating in vivo cDTI, cine, and tagged MRI to Characterize Healthy Passive Cardiac Mechanics." (To be submitted in 2023)

**Kolawole, F.O.**, Wang V.Y., Freytag B., Loecher M., Cork T.E., Nash M.P., Kuhl E., Ennis D.B., 2023. "Evaluating Passive Myocardial Stiffness using in vivo cine, cDTI, and tagged MRI." in *Functional Imaging and Modeling of the Heart*, Lecture Notes in Computer Science. Springer Nature, Switzerland, pp. 527-536

**Kolawole, F.O.**, Peirlinck, M., Cork, T.E., Levenston, M., Kuhl, E., Ennis, D.B., 2023. "Validating MRI-Derived Myocardial Stiffness Estimates Using in vitro Synthetic Heart Models." *Annals of Biomedical Engineering*. Springer International Publishing, pp. 1–14.

**Kolawole, F.O.**, Peirlinck, M., Cork, T.E., Wang, V.Y., Dual, S.A., Levenston, M.E., Kuhl, E., Ennis, D.B., 2021. "A Framework for Evaluating Myocardial Stiffness Using 3D-Printed Heart Phantoms." in *Functional Imaging and Modeling of the Heart*, Lecture Notes in Computer Science. Springer International Publishing, Cham, pp. 305–314.

### 1.4.2 Conference Abstracts

**Kolawole F.O.**, Cork T.E., Loecher M., Wang V.Y., Kuhl E., Ennis D.B., 2023. "Framework for Characterizing *in vivo* Passive Myocardial Stiffness using *in vivo* MRI." *32nd International Society for Magnetic Resonance in Medicine Annual Meeting and Exhibition*, Ontario Canada (digital poster)

**Kolawole F.O.**, Pierlinck M., Cork T.E., Levenston M.E., Kuhl E., Ennis D.B., 2022. "Accuracy of MRI-driven myocardial stiffness estimation in heart phantoms of varying stiffness." *9th World Congress of Biomechanics*, Taipei Taiwan (oral presentation)

**Kolawole F.O.**, Pierlinck M., Cork T.E., Levenston M.E., Kuhl E., Ennis D.B., 2022. "Impact of cardiac geometry segmentation on MRI-driven estimates of myocardial stiffness in an *in vitro* synthetic heart model." *31st International Society for Magnetic Resonance in Medicine Annual Meeting and Exhibition*, London England (digital poster)

**Kolawole F.O.**, Cork T.E., Loecher M., Zimmermann J., Dual S.A., Levenston M.E., Ennis D.B., 2021. "System for Validating MRI-based Myocardial Stiffness Estimation Techniques Using 3D-Printed Heart Phantoms." *30th International Society for Magnetic Resonance in Medicine Annual Meeting and Exhibition*, Virtual meeting (oral presentation, **Magna Cum Laude Merit Award**)

**Kolawole F.O.**, Cork T.E., Loecher M., Dual S.A., Levenston M.E., Ennis D.B., 2021. "Experimental Framework for MRI-Based Measurement of Myocardial Stiffness in 3D-Printed Heart Phantoms." *7th Summer Biomechanics, Bioengineering, and Biotransport Conference*, Virtual meeting (oral presentation)

# Chapter 2

# Cardiac Structure and Function

## 2.1 Whole Organ Anatomy

The heart (figure 2.1) is the primary organ of the circulatory system. The heart wall consists of three layers: the epicardium, myocardium, and endocardium. The epicardium is the outermost layer and the endocardium lines the inner surfaces of the heart chambers. The myocardium, the muscular tissue of the heart, is the main constituent of heart wall and is its thickest layer. The whole heart and the roots of the major cardiac vessels are enclosed by the pericardial sac, a thin membranous tissue that restricts motion of the heart while allowing it to also smoothly slide.

Heart size is affected by race, age, gender, and physical activity levels, but is generally described as being slightly bigger than the size of a clenched fist in healthy adult humans. The heart's main tasks are to distribute oxygenated blood throughout the body (systemic circulation) and to deliver deoxygenated blood from the body to the lungs for reoxygenation (pulmonary circulation). Maintaining both pulmonary and systemic circulation necessitates the separation of the heart into four distinct chambers: two ventricles and two atria. The left side of the heart is connected to the systemic circulation and the right side is connected to the pulmonary circulation. The left ventricle (LV) is the largest of the four chambers and its geometry can be approximated as a truncated prolate spheroid. The right ventricle (RV) is crescent-like in shape [35] and connects to the left ventricle at the interventricular septum. Due to physiological adaptation to the greater systemic resistance compared with

the pulmonary resistance, the LV myocardium (∼10-12mm), in adults, is thicker than the RV myocardium (∼4-6mm) [36]. The left and right atria are situated at the base of the ventricles and are separated by the interatrial septum. Atrial myocardium is typically substantially thinner than the ventricular myocardium. Atrioventricular valves enable undirectional blood flow from the atria to the ventricles. During ventricular systole, the pulmonary valve maintains blood flow from the right ventricle to the pulmonary artery and lungs, while the aortic valve allows for blood flow from the left ventricle to the aorta and body.

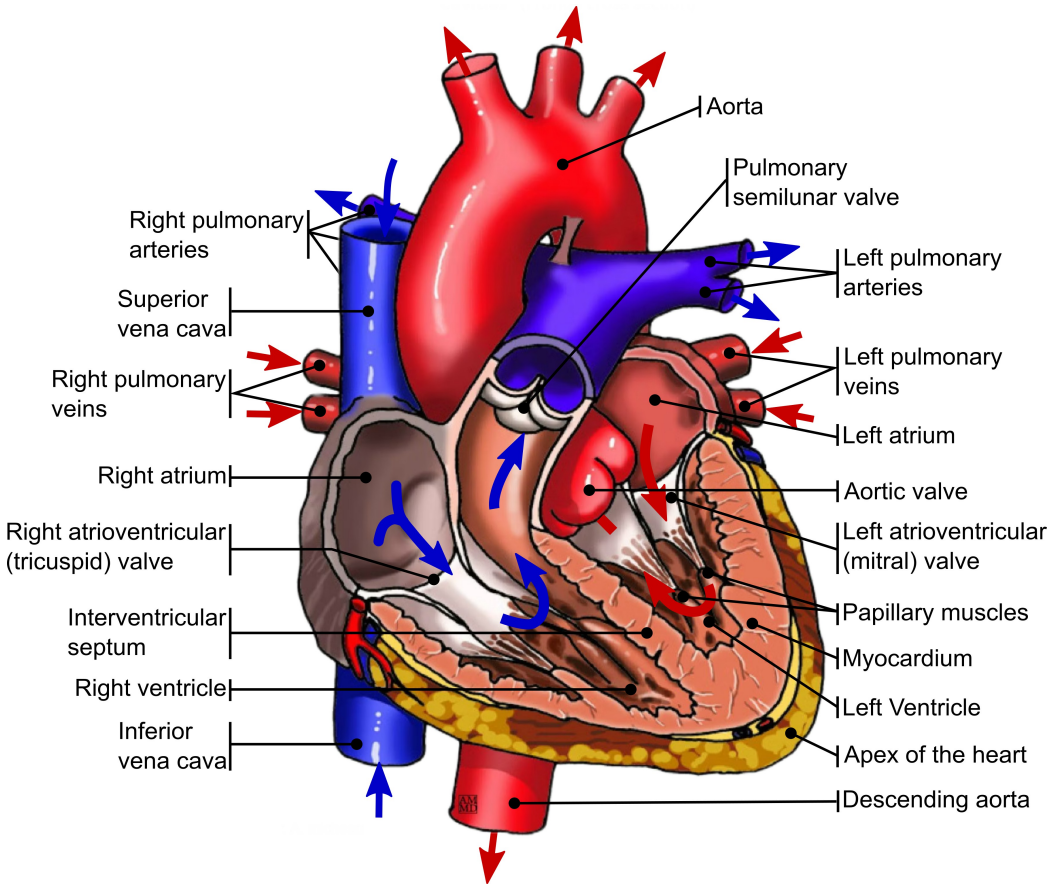



Figure 2.1: **Cardiac Anatomy.** Direction of blood flow is indicated by the arrows (colored blue and red to indicate deoxygenated and oxygenated blood respectively). Courtesy of IMAIOS © "Micheau A, Hoa D, e-Anatomy, www.imaios.com, DOI: 10.37019/e-Anatomy"

## 2.2 Cellular Components

Cardiomyocytes and fibroblasts are the two major cell types in the myocardium. Fibroblasts are primarily responsible for synthesising collagen within the extracellular matrix, which helps connect cardiomyoctyes to one another. Cardiomyocytes are muscle cells which occupy over 70 percent of myocardial volume [37] and are responsible for cardiac contraction (Figure 2.2). These cardiac muscle cells are tubular and are composed of bundles of myofibrils connected in series. Myofibrils are rod-like and consist of sarcomeres arranged in series and parallel. Sarcomeres, which are the fundamental contractile units of cardiomyocytes, contain many interlacing thick and thin filaments. Each thick filament is composed of bundles of myosin, and the myosin heads project towards the thin filaments at regular intervals. The thin filaments are comprised of actin, tropomyosin, and troponin. Actin provides binding sites for the myosin heads, and tropomyosin and troponin regulate force production by exposing the binding sites only during electrical activation in the presence of calcium. Titin is another protein in the sarcomere which connects the thick filaments to the ends of the sarcomere at the z-lines. Myocardial hierarchy from sarcomeres to tissue level is illustrated in figure 2.3.

The sliding filament theory initially proposed in the 1950s by two independent research teams [38, 39] is the prevailing explanation of force generation in sarcomeres. According to the theory, during muscle contraction, the thick and thin filaments slide past each other as the sarcomere changes length, thus modulating the extent of overlap between the actin and myosin i.e. the number of cross-bridges between the myosin heads and the actin binding sites. The force generated by the sarcomere is thus related to the number of cross bridges. Notably, the force-length relationship (figure 2.4) up to the optimal contraction length ($L_{max}$) is much steeper in cardiac muscle than in skeletal muscle because cardiac myofibrils are increasingly more activated as muscle length increases [40].

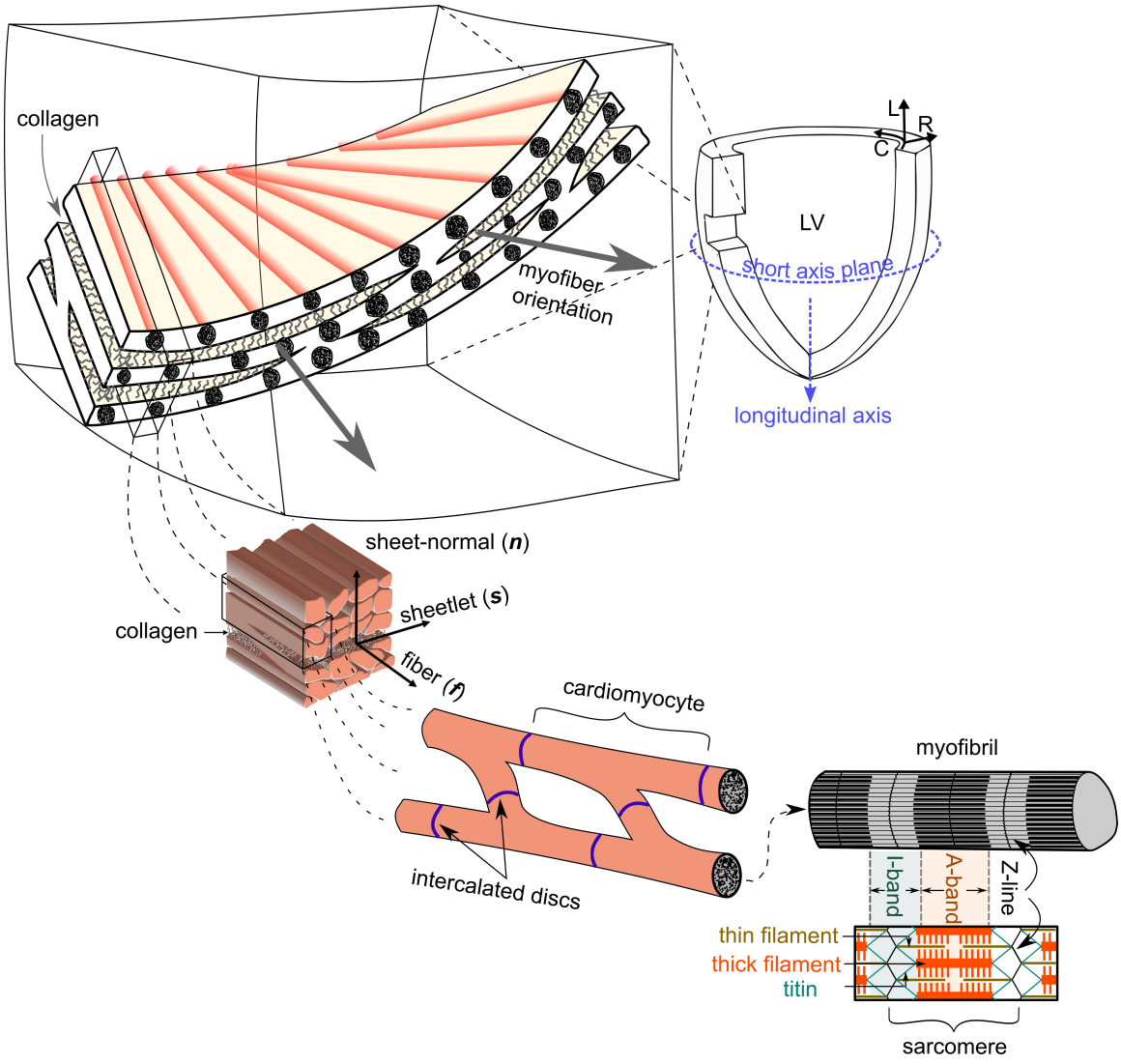



Figure 2.2: **Cardiomyocyte organization.** Cardiomyocytes are composed of bundles of myofibrils, which contain sarcomeres in series. Sarcomeres comprise interlacing thick and thin filaments enabling contraction following the sliding filament theory. Abbreviations: L - longitudinal, C - circumferential, R - radial.

## 2.3 Myocardial Mesostructure

Myocardial mesostructure concerns the structural organization of cardiomyocytes in the myocardium on the scale of 100s of micrometers [5]. The complex arrangement of cardiomyocytes is critical to cardiac studies as it indicates the direction of the synchronized

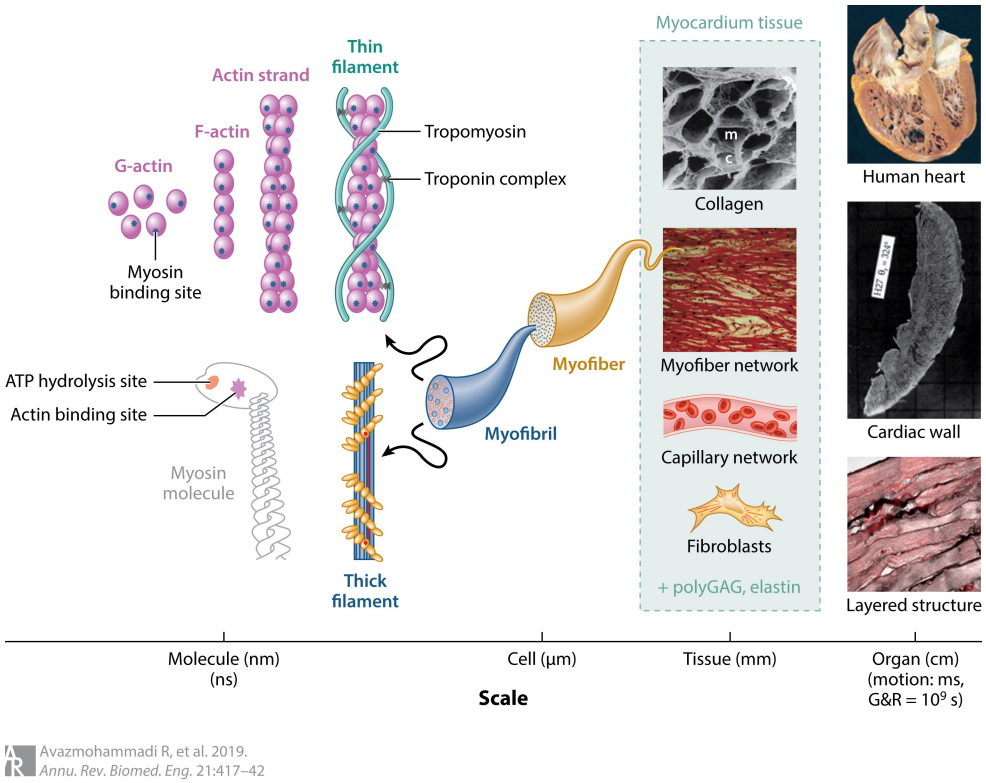



Figure 2.3: **Myocardial structural hierarchy** from molecular to organ scale. Reproduced with permission from Avazmohammadi *et. al.* [4]. Abbreviations: ATP - adenosine triphosphase, c - capillary lacunae, G&R - growth and remodeling, m - myofiber lacunae, polyGAG - polyglycosaminoglycan.

electrical impulse conduction and underlies the distinct ventricular systolic and diastolic kinematics. Cardiomyocytes are embedded within an extracellular matrix, a complex meshwork of structural and nonstructural proteins. The most prominent structural protiens are collagen and elastin, which influence the mechanical microenvironment and can affect myocardial cell activity through mechanotransduction pathways [41]. In addition, the passive mechanical response of myocardium is influenced by these structural proteins. It is important to note that proteins within cardiomyocytes (e.g. titin) also contribute to passive myocardial mechanical behavior [33].

Ventricular mesostructure is described with reference to the cardiac coordinate system which is defined relative to the LV geometry (figure 2.2). By one definition, the longitudinal axis passes through the LV apex and the commissure between the left and right coronary

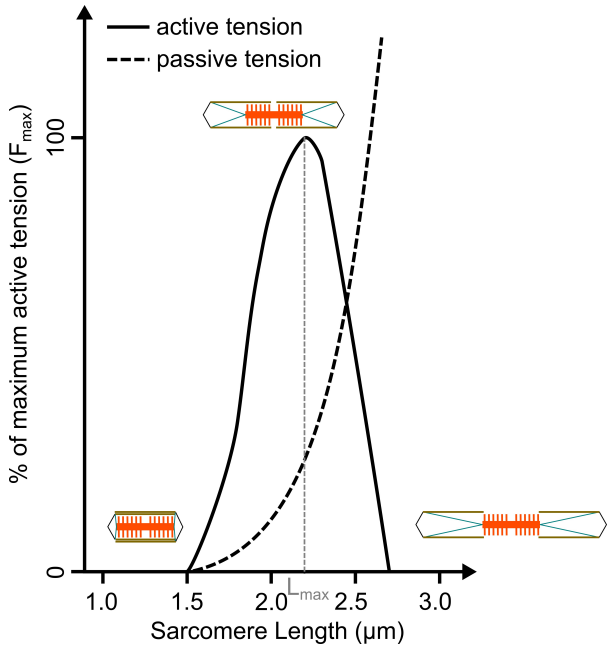



Figure 2.4: **Sarcomere length-tension relationship.** As thick and thin filaments slide past each other, the sarcomere length changes. The active force generated by a sarcomere is a function of its length and the force peaks at length ($L_{max}$) when the maximum number of cross-bridges are formed. As the sarcomere length changes, passive tension is also generated in large part due to to stretching out of titin.

cusps of the aortic valve [42]. Within the short-axis plane (orthogonal to the longitudinal axis) are the radial axis, which extends from the LV cavity center to the epicardial surface, and the circumferential axis, which is mutually orthogonal to the radial and longitudinal directions and follows the curvature of the epicardial surface (Figure 2.2).

LV myocardium can be described as a composite of sheetlets of 3 to 5 cardiomyocytes [43] connected in parallel by endomysial fibers. Cardiomyocyte sheetlets are joined by perimysial fibers, and the sheet orientations vary transmurally from the sub-endocardial to sub-epicardial boundary in a "helical" fashion. If a small cubic block of tissue (~8 $mm^3$) is excised from the LV myocardium, a "local" myocardial material coordinate system can be defined: the fiber axis ($\boldsymbol{f}$) is the mean cardiomyocyte direction, the sheetlet axis ($\boldsymbol{s}$) corresponds to the direction the cardiomyocytes connect to form sheetlets, and the sheet-normal axis ($\boldsymbol{n}$) is normal to the sheetlet plane as the name suggests (Figure 2.5). While in computational modeling, for simplicity, these axes are assumed to be mutually orthogonal, there

is no anatomical basis for this assumption. The fiber axis or the mean cardiomyocyte orientation is of particular interest because it indicates the principal direction of force generation during cardiomyocyte contraction and also indicates the principal direction along which the electric potential spreads throughout the heart.

Given the manner in which cardiomyocytes are connected to form myocardium, the orientation of cardiomyocytes in the LV changes transmurally (i.e. with radial position). As a consequence of this transmural variation in cardiomyocyte orientation, limited cardiomyocyte contraction (∼15% [44, 45]) can produce up to 70% ejection [46] in normal cases. The 3D orientation of a cardiomyocyte (or small group of cardiomyocytes) in space is commonly described using three angles: the helix angle is between the fiber direction projected onto the longitudinal-circumferential plane and the short-axis plane, the transverse angle is between the fiber direction projected unto the short-axis plane and the local circumferential direction, the secondary eigenvector angle (E2A) is between the sheet direction projected onto the cross-fiber plane (defined by sheet and sheet-normal axes of myocardial material coordinate system) and the longitudinal-circumferential plane (Figure 2.5). In the literature, of these three descriptors of cardiomyocyte orientation, the helix angle has been the most commonly reported. In the mammalian heart, the helix angle varies from a negative value on the epicardial surface to positive on the endocardial surface. Most studies report angles of 60° to 80° at the subendocardium and -40° to -60° at the subepicardium [47]. Different transmural helix angle trends are seen across different animal species however, from approximately linear in porcine hearts [42] and rats [48, 49] to roughly tangential in dogs [50].

## 2.4 Cardiac Cycle

The cardiac cycle encompasses the series of events (phases) that comprise one heart beat. The normal cardiac cycle starts with atrial contraction, as a pacemaker impulse originating in the sinoatrial (SA) node spreads rapidly across the atria, causing depolarization of atrial cardiomyocytes (P wave on electrocardiogram in Figure 2.6). The right and left atria contract almost simultenously, which causes blood to be propelled across the atrioventricular (mitral and tricuspid) valves into the ventricles. During atrial contraction, the pulmonary

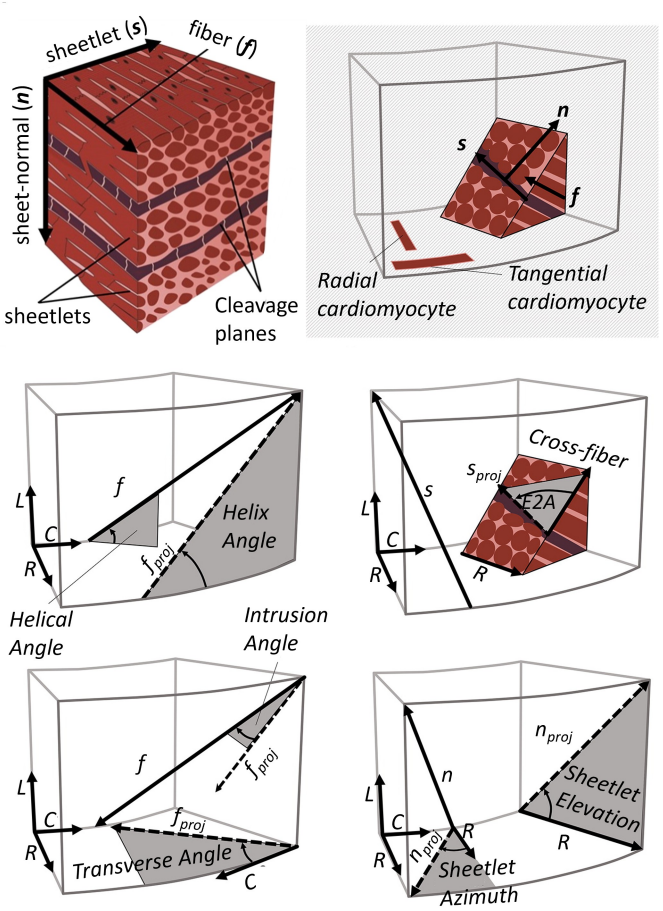



Figure 2.5: **Mesostructural coordinates and angles.** Adapted from Wilson *et. al.* [5] with permission. From a reference tissue block, a myocardial coordinate system, with the mutually orthogonal fiber (***f***), sheetlet (***s***), and sheet-normal (***n***) directions, can be defined. Helix angle is the projection of the ***f*** onto the longitudinal-circumferential plane. Secondary eigenvector angle (E2A) is the projection of the sheetlet direction ***s*** onto the cross-myocyte plane. The transverse angle is the projection of ***f*** onto the short axis plane (radial-circumferential plane).

and aortic valves are closed. The excitation impulse is delayed at the atrioventricular (AV) node before it is conducted to the bundle of His down the interventricular septum, ensuring that the atria have enough time to fully eject blood into the ventricles. The impulse then passes from the bundle of His to the Purkinje fibers, spreading the excitation wave to the

ventricles (QRS complex on electrocardiogram in Figure 2.6), which results in nearly synchronized right and left ventricular contraction. As the intraventricular pressure exceeds the atrial pressure, the atrioventricular valves close. Papillary muscles, which also contract during this phase, apply tension to the atrioventricular valve leaflets, preventing then from swinging into the atria. All heart valves are closed during this phase, which is termed isovolumetric contraction. When the intraventricular pressure exceeds the pressure in the great arteries, the aortic and pulmonary valves open, enabling rapid ejection of blood into the great arteries that deliver blood to the lungs (right heart) and the body (left heart). Coronary arteries, originating at the base of the aorta perfuse the myocardium itself with blood. When the ventricles are maximally contracted, the heart is said to be in end-systole. As the cardiomyocytes repolarize (T wave on Figure 2.6), ventricular active tension and intraventricular pressures reduce, leading to reduced ejection rate. Though the ventricular pressure starts to fall below the outlow tract pressure, some outward blood flow still occurs (at a slowed rate) because of the residual kinetic energy of blood from the rapid ejection phase. When intraventricular pressures drop sufficiently below the pressures in the great arteries, the aortic and pulmonary valves close, and the ventricles isovolumically relax. Driven by active relaxation of ventricular cardiomyocytes, the intraventricular pressures continue to drop, and eventually fall below atrial pressures. This causes the atrioventricular valves to open and consequentially the ventricles rapidly fill with blood. Eventually, the pressure gradient across the atrioventricular valves reduces, and the ventricles fill at a slower rate (diastasis). This relatively quiescent state is then interrupted by atrial contraction that further increases the blood volume in the ventricles and restarts the cardiac cycle. The cardiac configuration at the end of atrial contraction corresponds with the maximum ventricular volume and is termed end-diastole. The Wiggers diagram in Figure 2.6 illustrates the pressure trace in the left ventricle, atrium, and aorta, and the left ventricular volume evolution over the cardiac cycle. The phonocardiogram is a graphical representation of the double heartbeat (“lup-dup”) sound which results from closure of the heart valves. The normal cardiac cycle is also illustrated on the LV pressure-volume loop (Figure 2.7).

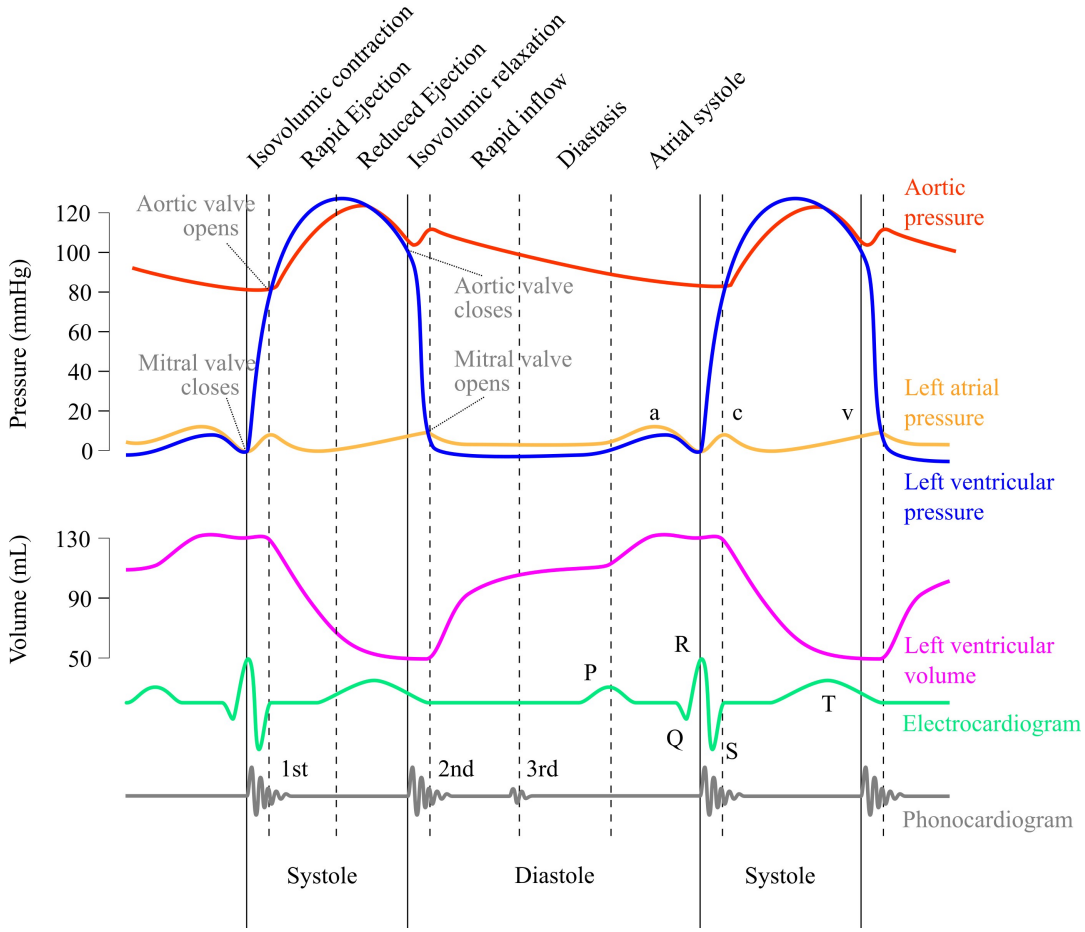



Figure 2.6: **Wiggers Diagram.** Illustrates the events of the cardiac cycle and the synchronized variation of left heart and great artery pressure, left ventricle volume, electrical activity, and heart sounds. Adapted from Yerachmiel C / Wikimedia Commons / CC BY-SA 2.5

## 2.5 Cardiac Function

Quantitative assessment of cardiac performance is important for clinical categorization and diagnosis of a patient's heart health. LV ejection fraction (EF) is an important and widely used clinical index of cardiac systolic performance. LV EF is the percentage of blood ejected by the LV during each cardiac cycle calculated as the ratio of stroke volume (LVSV) to end-diastolic volume (LVEDV). Stroke volume is the volume [mL] of blood pumped during each cardiac cycle, and is the difference between end-diastolic volume and end-systolic

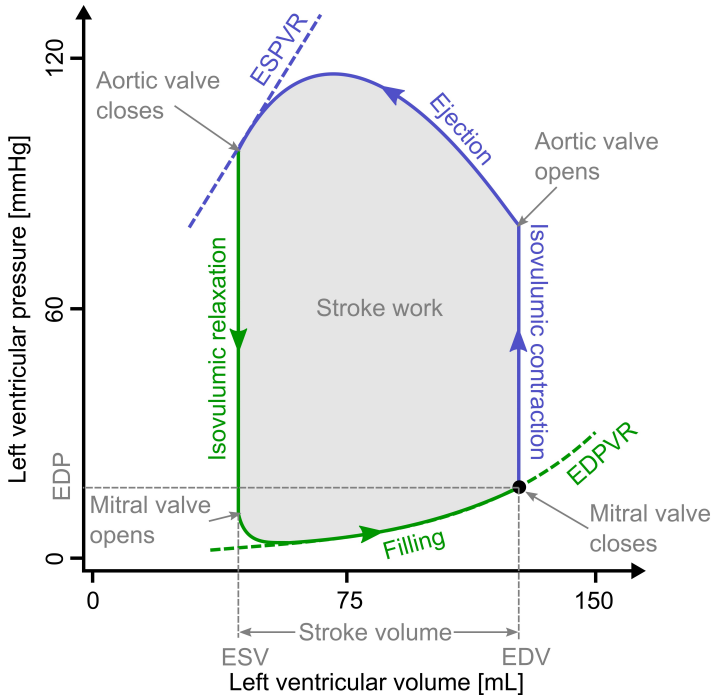



Figure 2.7: **Left ventricular pressure-volume loop** during the cardiac cycle showing systolic (blue) and diastolic (green) phases. The stroke work is the area contained within the loop and the stroke volume is the difference between the end-diastolic (EDV) and end-systolic volume (ESV). The end-diastolic pressure (EDP) is an important index of diastolic function. The end-systolic pressure volume relationship (ESPVR) and the end-diastolic pressure volume relationship (EDPVR) determine the maximum and minimum possible ventricular pressures.

volume. The stroke work is the area inside the pressure-volume loop. Cardiac output, which is the amount of blood pumped by the LV per minute, is also used to quantify ventricular function. Cardiac output is the product of stroke volume and heart rate. Another important index of cardiac function is the filling pressure (end-diastolic pressure), which when elevated ($\geq$16 mmHg [51]) is indicative of diastolic dysfunction.

## 2.6 Cardiac pathophysiology in HFpEF

HFpEF is a complex clinical syndrome with heterogenous pathophysiology that includes cardiac dysfunction as well as extra-cardiac multiorgan defects. While numerous mechanisms underly HFpEF, diastolic LV dysfunction leading to inadequate filling is the most distinguishing feature of HFpEF, which motivated the historical name - diastolic heart failure.

### 2.6.1 Increased Passive Myocardial Stiffness

Diastolic dysfunction could result from elevated passive myocardial stiffness, which reduces the extent of passive ventricular filling during atrial systole. Determinants of passive myocardial stiffness are contained within the extracellular matrix as well as cardiomyocytes. Collagen is the most significant contributor to extracellular matrix stiffness. Type I collagen fibers, which primarily determine tensile strength of the extracellular matrix, are the most abundant collagen in healthy hearts [52]. Most of the remaining collagen in the extracellular matrix is type III, which is thinner and more flexible that type I [52]. In HFpEF, excessive collagen deposition by fibroblasts and an increase in the proportion of type I collagen compared with the more flexible collagen III [53, 54] contributes to elevated passive myocardial stiffness. HFpEF patients also show increased collagen cross-linking [53], which reduces LV compliance (i.e. increases stiffness). Titin, the giant elastic protein contained within cardiomyocytes, also contributes to passive myocardial stiffness. In HFpEF, there is a shift towards the stiffer isoform of titin, N2B, compared with the more compliant isoform, N2BA [55]. Hypophosphorylation of N2B titin has also been found in HFpEF [56, 55]. This is further evidence of the contributions of titin to increased passive myocardial stiffness in HFpEF since titin dephosphorylation is linked with increased cardiomyocyte passive tension.

### 2.6.2 Slowed Active Relaxation

Another characteristic feature of HFpEF is slowed active relaxation, which limits cardiac filling, thus contributing to reduced stroke volume especially during exercise [57]. Different from passive filling, active relaxation is the energy dependent recoil of cardiomyocytes after contraction. The rate of relaxation is determined mainly by an ATP-dependent $Ca^{2+}$ pump that actively removes calcium from the cardiomyocytes. Increased myocardial $Ca^{2+}$ is observed in HFpEF patients, indicating slowed active relaxation [58].

### 2.6.3 Ventricular Hypertrophy

LV hypertrophy is a common myocardial structural anomaly in HFpEF [59]. Hypertrophic remodeling could be maladaptive or compensatory depending on the biomechanical driving stimulus. For example, cardiac hypertrophy is observed in athletes [60] and during pregnancy [61] but in these cases, there is no associated fibrosis and the adaptations do not reduce cardiac function. In HFpEF however, maladaptive hypertrophic remodeling is promoted by comorbidities, (e.g. hypertension and diabetes), neurohormonal activation, and pressure overload. Myocardial hypertrophy in HFpEF is driven, on the cellular scale, by increase in the size of cardiomyocytes by parallel addition of sarcomeres. This is exemplified by the increased thicknesss of cardiomyocytes in myocardial biopsy samples of HFpEF patients [62].

Maldaptive cardiac hypertrophy may be initiated as a compensatory mechanism, but sustained remodeling eventually leads to ventricular compromise. In the case of pressure overload due to hypertension, concentric myocardial hypertrophy (increased thickness) initially helps to stabilize wall stress and maintain normal ejection fraction even against the increased afterload. This adaptation however, when sustained, leads to diastolic dysfunction due to increased chamber stiffness and reduced filling.

## 2.7 Summary

This chapter covered cardiac anatomy, the cellular components of myocardium, myocardial mesostructure, the cardiac cycle, and cardiac function in the healthy state. We conclude

with a discussion on the cardiac derangements that occur in HFpEF. Medical imaging can be used to non-destructively evaluate this pathophysiological remodeling. Magnetic Resonance Imaging (MRI) in particular, has proven to be a powerful tool for the assessment of cardiac structure and function in health and disease, and is being used to aid management of heart disease, including HFpEF. The next chapter will discuss cardiac MRI, including the underlying theory and some of the applications.

# Chapter 3

# Cardiac MRI (CMR)

Magnetic Resonance Imaging (MRI) is a technology that non-invasively produces detailed anatomical images of the body. Modern MRI operates on the principle of Nuclear Magnetic Resonance (NMR), which underpins the behavior of certain atomic nuclei (e.g. hydrogen) in the presence of a strong external magnetic field, when exposed to electromagnetic waves of specific resonance frequency. Ever since it was first described in 1938 by Isidor Rabi [63], who won the Nobel prize in Physics for his work, NMR has revolutionized numerous fields, including medicine through MRI, which is now one of the cornerstone medical imaging modalities.

MRI possesses unique advantages over other medical imaging modalities including the abscence of ioniozing radiation, exceptional soft tissue contrast, contrast modulation capacity, multiplanar imaging of arbitrary orientation, 3D imaging, and high resolution capabilities. The major weakness of MRI is that it typically takes appreciably longer to acquire images of similar body parts than other imaging modalities like ultrasound, X-ray, and CT. However, technical developments on the hardware and software are addressing this issue. Decades of cardiac MRI (CMR) research have now enabled comprehensive assessment of cardiac anatomy, function, and microstructure, making it a powerful tool for developing biophysical computational models of the heart. In this section, the physics underlying MRI will be briefly reviewed, basic pulse sequences will be described, and some common cardiac MR sequences will be highlighted.

## 3.1 MRI Physics Basics

MRI takes advantage of the high prevalence of hydrogen in the body, since the body is largely composed of water. The nuclei of hydrogen atoms, which only contain one proton, are the primary atomic nuclei of concern in MRI. These hydrogen protons are slightly magnetic because they possess a quantum mechanical quantity known as spin in addition to being positively charged.

In the presence of a strong external magnetic field ($\vec{B}_0$), the magnetic moments of a group of hydrogen protons aligns with the external field (Figure 3.1). The protons will exist in one of two spin states, such that their moments are either aligned with the magnetic field ("spin up") or directly opposed to it ("spin down"). The two different spin states possess a small energy difference which is proportional to the $\vec{B}_0$ strength. At thermal equilibrium, there is a slight preference for the "spin-up" lower energy state. This leads to a net magnetization ($\vec{M}$) in the group of protons which is aligned with the $\vec{B}_0$ field.

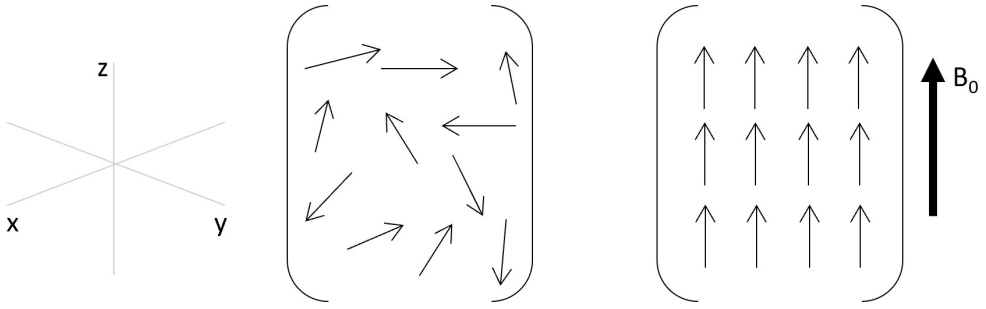



Figure 3.1: **Spins align in the presence of a strong magnetic field**

The magnetic moment possessed by a spin is proportional to its Larmor frequency ($\omega_L$, frequency of precession) defined as:

$$\omega_L = \gamma |\vec{B}_0| \tag{3.1}$$

where $\gamma$ is the gyromagnetic ratio (units of MHz/Tesla) which is unique to the atomic nuclei of interest. A weak magnetic field ($\vec{B}_1$) oscillating at the Larmor frequency can perturb $\vec{M}$ ("tip away" from the $\vec{B}_0$ axis), and excite protons from the lower energy, "spin-up" to the

higher energy "spin-down" state. This causes $\vec{M}$ to increasingly possess a transverse component orthogonal to $\vec{B}_0$ ($M_{xy}$), as it precesses around $\vec{B}_0$, and the component of $\vec{M}$ parallel to $\vec{B}_0$ ($M_z$) gradually decreases (Figure 3.2). The $M_{xy}$ component of the magnetization can be detected by nearby receiver coils because the precession of $\vec{M}$ leads to an oscillating magnetic field. The extent to which $\vec{M}$ tips away from $\vec{B}_0$ relates to the duration and strength of the radiofrequency (RF) pulse which produces $\vec{B}_1$ and is referred to as the flip angle. After the RF pulse is turned off, the net magnetization continues to precess about the $\vec{B}_0$ at $\omega_L$ as the magnetization gradually returns exponentially to the thermal equilibrium state $\vec{M}_0$ which is aligned with $\vec{B}_0$. This process is termed longitudinal relaxation and is characterized by an exponential growth function with a time constant $T_1$:

$$M_z(t) = M_0 + (M_z(0) - M_0)e^{-t/T_1}. \tag{3.2}$$

Here, $M_z(0)$ is the initial longitudinal magnetization after excitation and depends on the flip angle. $M_0$ is the magnetization at thermal equilibrium which is entirely longitudinal.

After excitation, as longitudinal relaxation occurs, $M_{xy}$ simultaneously returns gradually to its equilibrium state, which is $M_{xy} = 0$. This transverse relaxation can be represented by an exponential decay curve with time constant, $T_2$:

$$M_{xy}(t) = M_{xy}(0)e^{-t/T_2} \tag{3.3}$$

with $M_{xy}(0)$ representing the initial transverse component of the magnetization after excitation. Due to inhomegeneity of the main magnetic field, there is accelerated spin dephasing leading to an effective decay time denoted as $T_2^*$ ($\leq T_2$). The different characteristic $T_1$, $T_2$ and $T_2^*$ time constants of biological tissues as well as the proton densities in these tissues contributes to the image contrast in MRI. By varying the sequence of RF pulses applied and the data collection time, different image contrasts can be created between the same tissues.

Repetition time (TR) is the amount of time between successive RF pulse sequences applied to the same slice and echo time (TE) is the time between the RF pulse and the acquisition of the resulting echo signal from the slice of interest. Different combinations of TR and TE can modulate image contrast. For example, generally, a short TR and short TE leads to a $T_1$-weighted image, where the contrast is dependent on the characteristic $T_1$

relaxation time constants of the tissues of interest. The contrast of a $T_1$-weighted image can be enhanced by gadolinium, a non-toxic paramagnetic agent. Similarly, a $T_2$-weighted image can be produced with a long TR and TE. Another possible contrast is proton density-weighting, which is produced using a long TR and short TE to minimize $T_1$ and $T_2$ contrast respectively.

## 3.2 Basic Pulse Sequences

### 3.2.1 Free Induction Decay

A single RF pulse induces a free induction decay (FID) signal in the receiver coil [64, 65] (Figure 3.2). Any excitation pulse ($\neq$180°) will generate some level of transverse magnetization and by extension, an FID signal in the receiver coils. The FID signal is a damped sinusoidal wave which oscillates at $\omega_L$, with an amplitude that decays exponentially with time at the $T_2^*$ time constant.

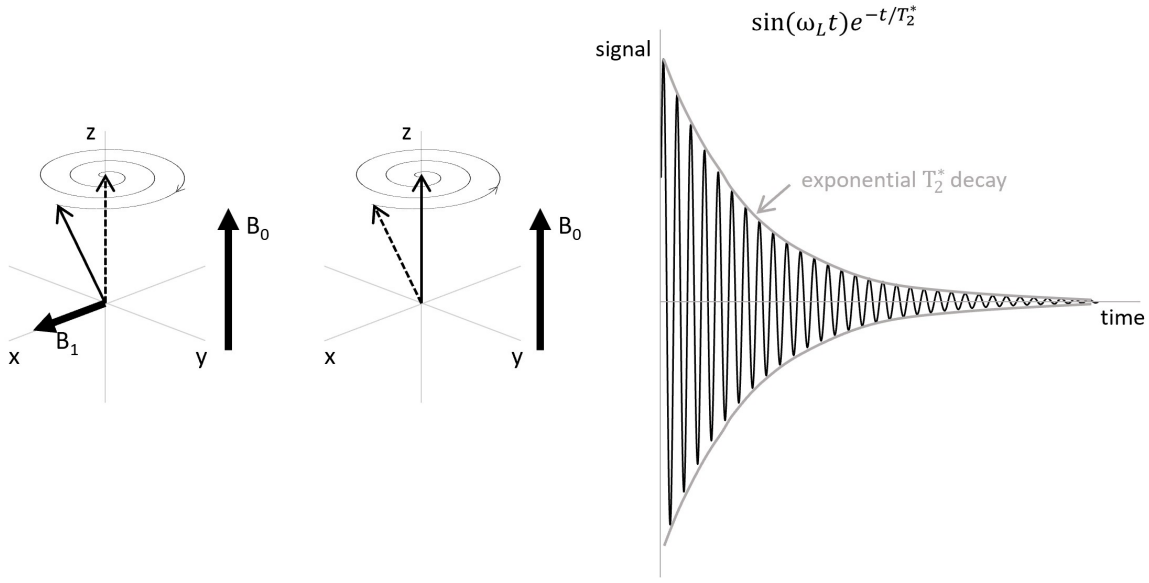



Figure 3.2: **Free Induction Decay.** An excitation RF pulse produces an oscillating signal which decays exponentially according to $T_2^*$

### 3.2.2 Gradient Echo (GRE)

The gradient echo pulse sequence is also known as a gradient recalled echo (GRE). In this method, a slice-selective RF pulse with a flip angle between 0 and 90° is first applied. Then an external dephasing gradient is applied to dephase the spins. This dephasing is followed by a rephasing gradient which reverses the phase scrambling, and causes the initially dephased spins to become in-phase at the center of the signal acquisition period. The reversal of dephasing leads to the formation of a gradient recalled echo (GRE). In the spoiled gradient recalled echo (SPGR) pulse sequence, the remaining transverse magnetization is destroyed at the end of each TR by strong gradients. As only one RF pulse is required and a small flip angle can be used, GRE sequences can have relatively short TRs and by extension can be quite fast, given the dependence of imaging duration on TR.

### 3.2.3 Spin Echo (SE)

Spin echo sequences are used to introduce $T_2$ dependence to the signal, instead of $T_2^*$ dependence. In a spin echo sequence, typically, a slice-selective 90° RF pulse is applied, leading to a FID signal which decays according to $T_2^*$. The decay of the FID is partly due to loss of phase coherence of the individual spins comprising the FID. Some of this lost phase coherence can be reversed by a 180° RF pulse applied at a period of TE/2 after the 90° pulse [66]. This causes the decayed FID signal to "regrow" (sources of off-resonance are rephased) and a spin echo occurs at time TE.

### 3.2.4 Stimulated Echo (STE)

Stimulated echos differ from spin echoes, in that they are composed of three 90° RF pulses. The first RF pulse generates an FID signal, which starts to decay according to $T_2^*$, as the spins dephase. After some delay, a second 90° pulse tips the dephased spins into the vertical plane, after which a third RF pulse flips the spins back into the transverse plane. Dephased spins then rephase in the transverse plane, leading to formation of a stimulated echo. This stimulated echo appears at a time delay after the third pulse that equals the interval between the first two pulses. While classically, flip angles of 90° are used, any triplet of RF pulses

that generate flip angles between 0 and 90° will lead to a stimulated echo.

## 3.3 Common Cardiac MRI Sequences

### 3.3.1 Steady State Free Precession (SSFP)

In this MRI sequence, a steady, residual transverse magnetization is maintained between successive cycles. For this to occur, special conditions must be satisfied, including the maintenance of a TR that is significantly shorter than $T_2$. In SSFP, a long train of evenly spaced RF-pulses is applied to the sample. Note that each RF pulse leads to an FID signal, each pair of RF pulses generates a spin echo, while each set of three or more RF pulses produces a stimulated echo coinciding with the spin echoes. If the RF pulses are close enough together (TR $<<$ $T_2$), the MR signal never completely decays i.e. spins in the transverse plane never completely dephase. After several TR periods, a steady state of magnetization is reached.

The most commonly used technique of SSFP in cardiac imaging is referred to as balanced SSFP (bSSFP), and it is used in nearly every CMR clinical examination. This is because bSSFP offers exceptional signal to noise ratio and $T_2/T_1$ weighting, enabling high contrast between myocardium and blood [67, 68]. The specific use of bSSFP to observe myocardial motion by acquiring multiple images of the heart throughout the cardiac cycle is termed cine bSSFP (cine from cinematography). Due to the nature with which the image signal is acquired, cine bSSFP requires consistent heart cycles to reproduce high quality images. Moreover subjects typically need to hold their breath during image acquisition so that the images are not affected by respiratory motion. Thus, cine bSSFP imaging is not robust in the cases of arrhythmias and dyspnea.

Cine bSSFP is primarily used to extract anatomic and functional information about subject's hearts. Due to the high myocardial-blood pool contrast and SNR, the endocardial and epicardial borders are easily identified from bSSFP images (Figure 3.3). Recent developments have focused on deep-learning bSSFP segmentation approaches [69] to eliminate the need for manual contouring. By combining endocardial and epicardial contours from a series of short and long axis cine images, 3D geometric surface representations of the heart

can be developed, enabling rigorous quantitative assessment of anatomy. For example, morphological indices, such as wall thickness [70], LV sphericity index [71, 72], LV mass [73], ventricular and atrial volume indices [74, 75] deviate from normal values in disease, and can be effectively assessed using cine bSSFP.

Qualitative assessment of ventricular contraction and relaxation can also be based on cine bSSFP. Furthermore, indices of ventricular function such as LV stroke volume and ejection fraction are effectively assessed using cine bSSFP. Feature tracking has also been used to assess global strain indices from cine bSSFP images [76]. The method works by propagating myocardial surface contours and guide points over the cardiac cycle and interpolating myocardial displacements based on the contours. The method however, cannot capture local displacements and rotations as well as tissue tagging or cine DENSE as exemplified by poor reproducibility of segmental strains [77, 78, 79] and moderate accuracy when compared with gold standard strain assessment methods [77, 80]. However, newer approaches to cine bSSFP-based myocardial strain analysis using deep learning trained with DENSE data have been developed [81], and are significantly more accurate than feature tracking alone.

While cine bSSFP is an essential CMR sequence, the imaging method is still undergoing technical developments. Recent research is focused on limiting the number and duration of breath holds or eliminating them altogether [82, 83], tackling image quality degradation due to respiratory and cardiac motion [84], adding robustness to cardiac cycle inconsistency (e.g. arrhythmia) [85], rapid imaging and reconstruction [86], and real time cine imaging [87].

### 3.3.2 Cardiac Tagged Magnetic Resonance Imaging

Spatial Modulation of Magnetization (SPAMM) or tagged MRI is a technique where RF pulses are used to place stripes or grids on the heart to follow its motion during the cardiac cycle [88]. An imaging sequence (typically SPGR) follows after the RF-pulse has made an imprint on (“tagged”) the tissue. The tag lines, which run throughout the entire myocardial wall, deform in conjunction with movement of the underlying tissue, thus tagged MRI enables analysis of regional myocardial deformation. The tagged images are usually presented

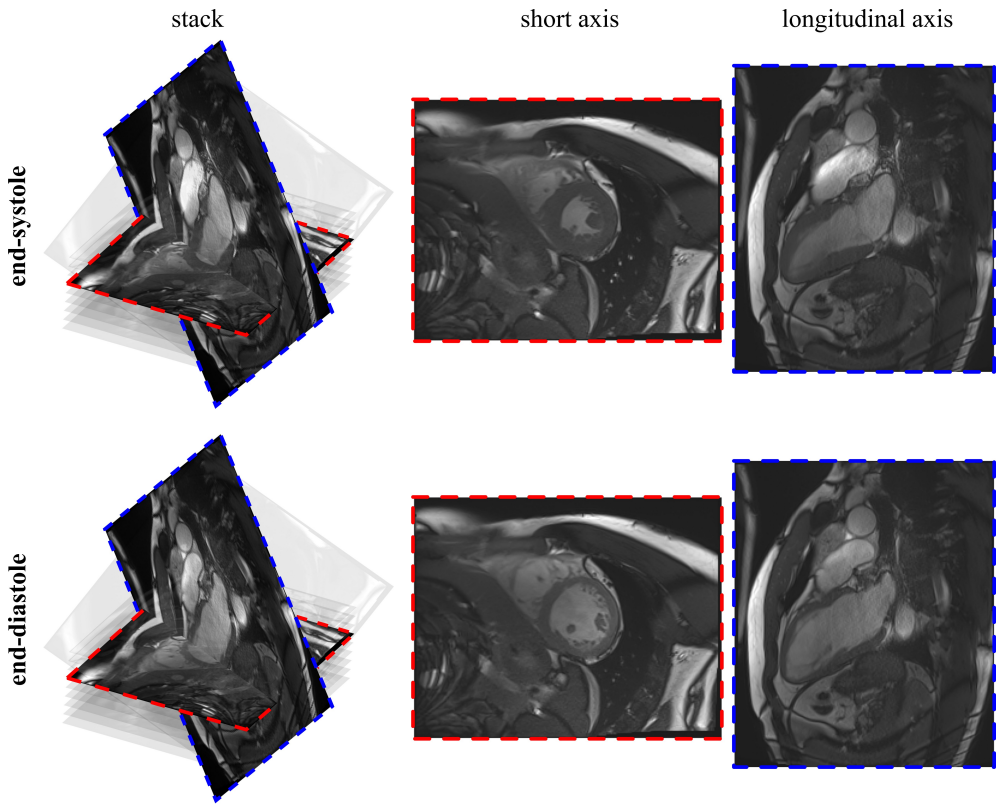



Figure 3.3: **Example of cine bSSFP MR images**

in cine form. As such, like cine bSSFP, this CMR sequence enables the qualitative inspection of gross heart motion over the cardiac cycle. However, unlike cine bSSFP, tagged MRI also encodes quantitative information about local (i.e. regional) myocardial displacements.

To produce a series of parallel stripes, at least two RF pulses are required. The first RF pulse rotates the magnetization such that it is 45° from the longitudinal direction ($\vec{B}_0$). A tagging gradient is then played that disperses the spins such that the phases of the spins are now a function of position. A second RF pulse then tips the now dephased spins by another 45°. The spins now have a spatially varying longitudinal and transverse component. Finally, a large spoiler gradient is applied to all the spins, eradicating the transverse component of all the spins. Now the spins only contain a spatially varying longitudinal component whose magnitudes vary in a sinusoidal pattern. The spatially modulated magnetizations are reflected as parallel bands of varying signal intensity in the tissue. To create grid tags,

a second set of RF pulses and gradients are made to induce tag lines in the perpendicular direction. Typically, grid tags are used in the short axis, and line tags in the long axis because of the challenge of acquiring acceptably high SNR grid tagged images in the long axis (Figure 3.4). Additional RF pulses and gradients can also be applied to refine and sharpen the tag-line profile.

While tagged MRI is an extensively validated tool for quantitative assessment of regional myocardial kinematics, a drawback of the method is the extensive post-processing that can be required to develop kinematic models of the heart from the images. Additionally, the tag lines fade due to $T_1$ relaxation which can sometimes lead to imperceptible tag lines towards the end of the cardiac cycle. The contrast between blood and myocardium can also be relatively poor in tagged MRI, making it challenging to dileneate the endocardial borders.

### 3.3.3 Diffusion-Weighted Magnetic Resonance Imaging

Diffusion, or Brownian motion refers to the random movement of molecules due to thermal energy. Diffusion weighted MRI (DWI) 3.5 provides a visual representation and quantitative measure of the diffusion of water in a region. In the body, extracellular water experiences relatively free diffusion while movement of intracellular water is relatively constrained (restricted diffusion) due to higher chances of collisions with the cellular components and cell membrane. Different tissues have varying relative proportions of intracellular and extracellular compartments and are composed of varying cellular architecture. As a consequence, the tissues of the body have different characteristic diffusion properties and as such, different apparent diffusion weightings as evident in images.

In Diffusion Weighted Imaging (DWI), after a 90° slice-selective RF pulse, diffusion sensitizing gradients are applied on either side of a 180° refocusing pulse. The signal equation for DWI can be described by the following equation:

$$S = S_0 e^{-bD} \tag{3.4}$$

where $S$ is the diffusion weighted signal, $S_0$ is the non-diffusion weighted signal, the scalar $b$ represents the b-value, and $D$ is the scalar-valued apparent diffusion coefficient (also termed

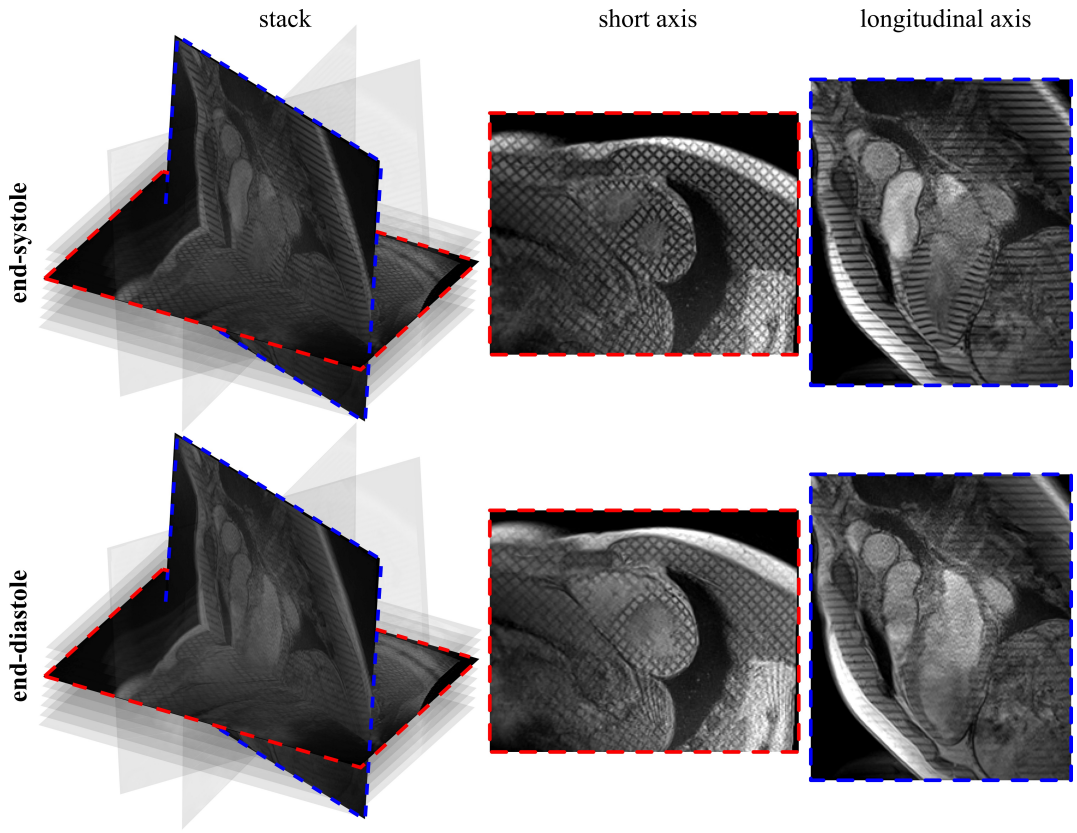



Figure 3.4: **Example of tagged MR images.** Grid tags are used in the short axis, and line tags are used in the longitudinal axis

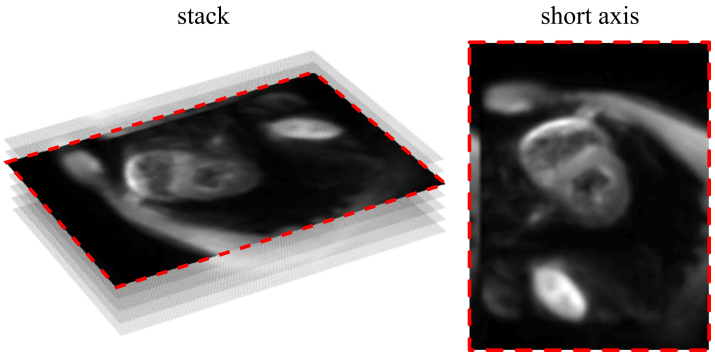



Figure 3.5: **Example of diffusion weighted MR images (b=0)**

ADC). The b-value (units of s/mm$^2$) is an important parameter that determines the sensitivity of the resulting images to the diffusion of water molecules in the tissue. The b-value

is proportional to the square of the amplitude ($G$) and duration ($\delta$) of the diffusion gradient applied, and also relates with the time duration between application of the two diffusion gradients ($\Delta$):

$$b = \gamma^2 G^2 \delta^2 (\Delta - \delta/3). \tag{3.5}$$

From equation 3.4, it is clear that for any given pixel in a diffusion weighted image, the signal is attenuated relative to the non-diffusion weighted signal by an amount which is proportional to the extent of diffusion ($D$) and the b-value. This signal attenuation in the presence of diffusion gradients is the basic principle of DWI. The magnitude of diffusion can be quantified from DWI by calculating, on a pixel-by-pixel basis, the apparent diffusion coefficient (ADC), which has units of $mm^2/s$. Lower ADC values correspond to more restricted diffusion. Since the b-value is predetermined, at least two images are required to reconstruct a pixel-wise ADC map - one diffusion weighted image ($S$) and a non-diffusion weighted image ($S_0$).

In highly structured tissues with well defined anisotropy (e.g. cardiomyocytes that compose the myocardium or nerves in the brain), there exists diffusion anisotropy where water preferentially diffuses in certain directions more than others. For example, faster diffusion is apparent along the longer direction of cells, and slower diffusion perpendicular to them. This anisotropic diffusion is exploited by diffusion tensor imaging (DTI) which uses DWI acquired with six or more directions of diffusion encoding to construct a diffusion tensor on a pixel by pixel basis. Using the diffusion tensors, DTI, is able to more completely describe anisotropic diffusion in tissues of interest, unlike DWI which provides a single number that only quantifies the magnitude of diffusion. The matrix representation of the diffusion tensor in the MRI scanner (x-y-z) coordinates could be represented as:

$$\mathbf{D} = \begin{bmatrix} D_{xx} & D_{xy} & D_{xz} \\ D_{xy} & D_{yy} & D_{yz} \\ D_{xz} & D_{yz} & D_{zz} \end{bmatrix} \tag{3.6}$$

Here, the diagonal components represent diffusion along the principal axes, and the off-diagonal terms represent the coupled diffusion between each pair of principal directions.

When diffusion is perfectly isotropic, the off-diagonal terms are all zero and the diagonal components are all equal.

Since the diffusion tensor is symmetric, it can be constructed if its six independent components can be estimated. To do this, we can reformulate equation 3.4 as:

$$S_i = S_0 e^{-b\hat{g}_i^T \mathbf{D} \hat{g}_i}. \tag{3.7}$$

As described previously, $S_0$ refers to the non-diffusion weighted signal intensity at the voxel and $S_i$ refers to the signal intensity measured after application of a diffusion-sensitive gradient in the direction $\hat{g}_i$. $\mathbf{D}$ is the diffusion tensor (which at this point is unknown), and the scalar valued $\hat{g}_i^T \mathbf{D} \hat{g}_i$ is the projection of the diffusion tensor in the direction $\hat{g}_i$ i.e. the diffusivity in the direction $\hat{g}_i$. Given the fact that equation 3.7 has 6 unknowns (6 independent components of the diffusion tensor, $\mathbf{D}$), to construct the diffusion tensor, there is a need to repeat a diffusion-weighted sequence to probe diffusion along at least 6 non-collinear directions. There is also a need to acquire a non-diffusion weighted image (b $\approx$ 0 s/mm$^2$) which is used to remove the non-diffusion based scaling from the diffusion weighted images. While seven images are the minimum required to construct a diffusion tensor, diffusion-weighted images along more directions can be used to construct the diffusion tensors using least squares linear regression.

The diffusion tensor is a symmetric, positive-definite matrix. Symmetry of the matrix means that all the eigenvalues are real, and the positive-definite property means the eigenvalues are all positive. Furthermore, the tensor is invertible, and so it can be decomposed and represented in terms of three linearly independent (mutually orthogonal) eigenvectors and three corresponding eigenvalues. A (nonzero) vector $\mathbf{v}$ is an eigenvector of the diffusion tensor $\mathbf{D}$, if it satisfies a linear equation of the form:

$$\mathbf{D}\mathbf{v} = \lambda \mathbf{v} \tag{3.8}$$

for a scalar $\lambda$ which is the corresponding eigenvalue. The primary eigenvector (eigenvector corresponding with the greatest eigenvalue) of the diffusion tensor corresponds to the direction of maximal diffusion. In cardiac DTI (cDTI), the primary eigenvector has been confirmed histologically to align with the mean cardiomyocyte orientation [89, 90, 91, 92].

Some studies have confirmed that the secondary eigenvector corresponds to the sheetlet orientation [93, 94]. The eigenvalues quantify the amount of diffusion in the direction of their associated eigenvectors. The mean diffusivity (MD) is the average of the eigenvalues and is a scalar representation of the overall diffusion in a pixel. The fractional anisotropy (FA) is another common cDTI metric used to asses the degree of isotropy of the diffusion and ranges from 0 (completely isotropic) to 1 (completely anisotropic).

## 3.4 Summary

In this chapter, we reviewed MRI physics and discussed some basic pulse sequences with a particular focus on applications to cardiac imaging. Cardiac MRI provides a wealth of information which can be leveraged to develop biophysical models of the heart for rigorous assessment of mechanics-based biomarkers of heart health (eg heart tissue stiffness). The mechanical behavior of the heart tissue is conveniently described using the mathematical framework of continuum mechanics discussed in the next chapter.

# Chapter 4

# Biomechanical Modeling of Passive Myocardium

## 4.1 Essential Elements of Continuum Mechanics

Continuum mechanics describes the deformations and related stresses in materials that can be approximated as continuous for certain length and time scales. Myocardial mechanical behavior is usually described within the mathematical framework of continuum mechanics, given the large deformations myocardium undergoes *in vivo*. Central to myocardium mechanics modeling is the development of rational constitutive equations that relate kinematic quantities (deformations) to kinetics (forces). This section reviews elements of continuum mechanics necessary for describing myocardium, including kinematic quantities, stress tensors, and finally existing constitutive models.

### 4.1.1 Kinematic Quantities

The deformation gradient tensor ($\mathbf{F}$) is the fundamental kinematic quantity in continuum mechanics. $\mathbf{F}$ relates points in the reference (undeformed) configuration ($\mathbf{X}$) to points in the deformed configuration ($\mathbf{x}$). The deformation gradient tensor is defined as:

$$\mathbf{F} = \nabla \mathbf{x}(\mathbf{X}, t) = \frac{\partial \mathbf{x}}{\partial \mathbf{X}}. \tag{4.1}$$

Thus, $\mathbf{F}$ carries an infinitesimal, undeformed line segment, $\partial\mathbf{X}$, to a new length and orientation, $\partial\mathbf{x}$ in the deformed configuration (Figure 4.1). The deformation gradient tensor, $\mathbf{F}$, completely characterizes the deformations in the vicinity of a particle $\mathbf{X}$. This means that $\mathbf{F}$, also describes the deformation of reference lengths, orientations, angles, areas, and volumes. The volumes of a differential volume element in the reference ($\partial V$) and deformed ($\partial v$) configurations are related by the following:

$$\partial v = \det \mathbf{F}\, \partial V. \tag{4.2}$$

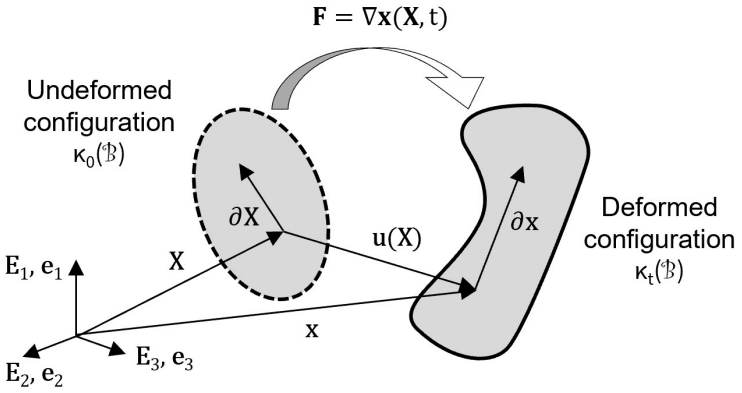



Figure 4.1: Given an **arbitrary deformation of a continuum body** $\mathscr{B}$, the reference and deformed configurations in Euclidean space can be represented by $\kappa_0(\mathscr{B})$ and $\kappa_t(\mathscr{B})$ respectively. A particle $\mathbf{X}$ in the undeformed body, with components $X_i, i \in (1,2,3)$ with respect to the basis $\{\mathbf{E}_1, \mathbf{E}_2, \mathbf{E}_3\}$ is displaced by $\mathbf{u}(\mathbf{X})$ to position $\mathbf{x}$, with components $x_i, i \in (1,2,3)$ with respect to the basis $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$. Here the material and spatial bases ($\mathbf{E}_i$ and $\mathbf{e}_i$, $i \in (1,2,3)$) are shown to coincide, as is often done for mathematical simplification. The deformation gradient tensor ($\mathbf{F}$) transforms a referential line segment $\partial\mathbf{X}$ to its deformed configuration $\partial\mathbf{x}$.

In general, we assume the myocardium is incompressible, like most biological tissues, due to the high water content. Incompressibility is enforced by the constraint:

$$\det \mathbf{F} = J \equiv 1. \tag{4.3}$$

Using polar decomposition, $\mathbf{F}$ can be represented as the product of a rotation and stretch tensor according to:

$$\mathbf{F} = \mathbf{RU} = \mathbf{VR} \tag{4.4}$$

where $\mathbf{V}$ and $\mathbf{U}$ are the left and right stretch tensors from alternate polar decompositions, and $\mathbf{R}$ is the rotation tensor. Note that the stretch tensors are positive definite, and so can be decomposed into three orthonormal eigenvectors that have three corresponding real eigenvalues. Thus, $\mathbf{V}$ and $\mathbf{U}$ can be interpreted as tensors which stretch an infinitesimal line segment in principal directions defined by the eigenvectors, by amounts given by the corresponding eigenvalues. For equivalence with the deformation gradient tensor, the right stretch tensor must be applied before rotation, and the left stretch tensor must be applied after rotation.

Given, $\mathbf{F}$, we can define the right and left Cauchy-Green tensors respectively:

$$\mathbf{C} = \mathbf{F}^T\mathbf{F} \tag{4.5a}$$

$$\mathbf{b} = \mathbf{F}\mathbf{F}^T. \tag{4.5b}$$

Physically, the right Cauchy-Green tensor relates the magnitude of a deformed differential line segment to the undeformed differential line, i.e. $|\partial\mathbf{x}|^2 = \partial\mathbf{X}\cdot\mathbf{C}\partial\mathbf{X}$, and the left Cauchy-Green tensor relates a referential line segment's magnitude to its deformed counterpart as $|\partial\mathbf{X}|^2 = \partial\mathbf{x}\cdot\mathbf{b}^{-1}\partial\mathbf{x}$.

Furthermore, we can use the Cauchy-Green deformation tensors to define the Green-Lagrange strain tensor ($\mathbf{E}$) and the Eulerian-Almansi strain tensor ($\mathbf{e}$):

$$\mathbf{E} = \frac{1}{2}(\mathbf{C} - \mathbf{1}) \tag{4.6a}$$

$$\mathbf{e} = \frac{1}{2}(\mathbf{1} - \mathbf{b}^{-1}) \tag{4.6b}$$

where $\mathbf{1}$ is the identity tensor which has entries equal to the Kronecker delta i.e. $\delta_{ij}$ (equals 1 if i = j and equals 0 otherwise). Note that $\mathbf{E}$ and $\mathbf{e}$ are strain tensors referenced to the undeformed (reference) and deformed (spatial) configurations respectively. Both of these strain tensors operate on a line element by returning half the change in the squares of the undeformed and deformed lengths:

$$\frac{|\partial \mathbf{x}|^2 - |\partial \mathbf{X}|^2}{2} = \partial \mathbf{X} \cdot \mathbf{E} \partial \mathbf{X} = \partial \mathbf{x} \cdot \mathbf{e} \partial \mathbf{x}. \tag{4.7}$$

The principal scalar invariants of $\mathbf{C}$ and $\mathbf{b}$ coincide, and are defined as:

$$I_1 = \operatorname{tr} \mathbf{C} = \lambda_1^2 + \lambda_2^2 + \lambda_3^2 \,, \tag{4.8a}$$

$$I_2 = \frac{1}{2}[I_1^2 - \operatorname{tr}(\mathbf{C}^2)] = \lambda_1^2\lambda_2^2 + \lambda_2^2\lambda_3^2 + \lambda_3^2\lambda_1^2 \,, \text{ and} \tag{4.8b}$$

$$I_3 = \det \mathbf{C} = \lambda_1^2\lambda_2^2\lambda_3^2 \tag{4.8c}$$

where $\lambda_i, i \in (1, 2, 3)$ represents the principal stretches defined as $\lambda_i = \frac{\partial \mathbf{x}_i}{\partial \mathbf{X}_i}$. For a material undergoing pure stretch (e.g. uniaxial, biaxial or triaxial tension), the deformation gradient tensor only has diagonal elements and has components $F_{ij} = \lambda_i \delta_{ij}$ (no sum). Note that for an incompressible material, $I_3 = J^2 = 1$.

Myocardium exhibits anisotropic behavior arising from the structured orientation of cardiomyocytes and the collagen network. Given a local mean fiber direction in the reference configuration ($\mathbf{f}_0$), we can define two transversely isotropic invariants:

$$I_4 = \mathbf{f}_0 \cdot (\mathbf{C}\mathbf{f}_0) \text{ and} \tag{4.9a}$$

$$I_5 = \mathbf{f}_0 \cdot (\mathbf{C}^2\mathbf{f}_0). \tag{4.9b}$$

If we describe myocardium as orthotropic with the local sheet direction represented as $\mathbf{s}_0$, we can define two additional orthotropic invariants:

$$I_6 = \mathbf{s}_0 \cdot (\mathbf{C}\mathbf{s}_0) \text{ and} \tag{4.10a}$$

$$I_7 = \mathbf{s}_0 \cdot (\mathbf{C}^2\mathbf{s}_0) \tag{4.10b}$$

This implies that there is now a second preferred direction, denoted by $\mathbf{s}_0$. We can introduce a useful invariant $I_8$ that is dependent on the other invariants and is defined as:

$$I_8^2 = [\mathbf{f}_0 \cdot (\mathbf{C}\mathbf{s}_0)]^2 = I_2 + I_4I_6 + I_5 + I_7 - I_1(I_4 + I_6) \tag{4.11}$$

These invariants are so called because they do not change under coordinate transformations. The use of invariants is an essential part of constitutive modeling described in the following section 4.1.2. This is because a material's behavior should be independent of the coordinate system, and expressing material models in terms of invariants ensures this property.

### 4.1.2 Strain Energy Laws (Constitutive Models)

Assuming the material behavior is elastic, it can be described by a strain-energy function, $\psi$, which depends on $\mathbf{F}$. We can represent the 2nd Piola Kirchoff Stress Tensor as

$$\mathbf{S} = \frac{\partial \psi}{\partial \mathbf{E}}. \tag{4.12}$$

The cauchy stress tensor is given by $J\boldsymbol{\sigma} = \mathbf{F}\mathbf{S}\mathbf{F}^T$, and in the case of incompressibility, the hydrostatic pressure constant is introduced and $J = 1$ is enforced giving:

$$\boldsymbol{\sigma} = \mathbf{F}\frac{\partial \psi}{\partial \mathbf{E}}\mathbf{F}^T - p\mathbf{1}. \tag{4.13}$$

Provided we can represent a material adequately with a strain energy law, we can fully describe the combination of stresses and strains that the material can undergo. Any rational strain energy law needs to capture essential characteristics of myocardial mechanical behavior, including nonlinearity and anisotropy. A number of constitutive equations have been developed to describe myocardial tissue [4, 95]. Some of the proposed constitutive models are transversely isotropic with strain energy laws defined in terms of invariants [96, 97] or in terms of components of the Green-Lagrange strain tensor [98, 99]. Evidence for orthotropy in myocardium [43, 100, 101] prompted the development of orthotropic models [102, 103]. There are also structurally based models [104, 105, 106, 107] that account for contributions of the various constituents of myocardium, and their distinct structural and material properties. For the case of these structural models, the strain energy law is typically written as a sum of the mechanical contributions of the extracellular matrix (ECM) and the embedded constituents, including cardiomyocytes and connective tissue:

$$\psi = \psi^g + \psi^f = \psi^g + \psi^m + \psi^c + \psi^i, \tag{4.14}$$

where the contributions of the ground matrix ($g$) is isotropic and the fiber contributions ($f$) can be split into contributions from myocytes ($m$), collagen ($c$), and interactions between them ($i$).

Of note, though myocardium exhibits viscoelastic behavior as illustrated by *ex vivo* experiments showing creep, stress relaxation, hysteresis and frequency dependence [108, 109, 110, 111], most of the currently used models treat the heart muscle as hyperelastic. The viscous effects may be negligible in healthy tissue functioning at regular physiological rates [112], but viscoelasticity could be an important consideration in pathology such as in atrial fibrillation [113] when the strain rates in myocardium are elevated or when there are highly disspiative structures in the myocardium such as in excess fatty infiltration [114]. A computational study performed recently by Tikenoğulları *et. al.* demonstrated that while hyperelasticity is an adequate assumption for single-beat simulations of cardiac mechanics, viscocity may be an important consideration for simulations involving multiple beats [115]. Recently Nordsletten *et. al.* developed a viscoelastic model of myocardium [116], which could be important for modeling passive myocardial mechanical behavior in cases where viscous effects are appreciable.

While a number of constitutive equations have been developed to describe passive myocardial tissue, two of the most widely used are described here - the orthotropic invariant-based model introduced by Holzapfel and Ogden [95] and the transversely isotropic exponential model as introduced by Guccione *et. al.* [98].

### 4.1.3 Holzapfel Ogden (HO) Model

This is a structurally-motivated invariant-based orthotropic constitutive model [95] and the strain energy law is given by:

$$\psi = \frac{a}{2b}\exp[b(I_1 - 3)] + \sum_{i=f,s} \frac{a_i}{2b_i}\{\exp[b_i(I_{4i} - 1)^2] - 1\} + \frac{a_{fs}}{2b_{fs}}[\exp(b_{fs}I_{8fs}^2) - 1]. \tag{4.15}$$

The HO constitutive model includes an isotropic exponential term based on the first

principal invariant, $I_1$. This term is included to model the underlying non-collagenous and non-muscular extracellular matrix. The exponential relationship in terms of $I_{4f}$ reflects the stiffening behavior in the muscle fiber direction. The stiffening in the sheet direction, which is partly due to endomysial collagen fibers connecting parallel cardiomyocytes, is modeled by the exponential function of the $I_{4s}$ invariant. These terms are only included in tension because they do not support compression, and so are not expected to significantly contribute to the stored energy in compression. It also includes an exponential function of the invariant $I_{8fs}$ to account for shear effects in the sheet plane partly due to coupling of the cardiomyocytes and collagen fibers.

There are eight material parameters - $a$, $b$, $a_f$, $a_s$, $b_f$, $b_s$, $a_{fs}$ and $b_{fs}$ which are all positive. The $a$ parameters have dimensions of stress and the b parameters are dimensionless.

### 4.1.4 Transversely Isotropic Exponential Model (Guccione)

For a generalized Fung-type exponential relation with orthotropic symmetry [117], the strain energy function is given as:

$$\psi = \frac{1}{2}c(e^Q - 1) \tag{4.16}$$

where

$$Q = b_1E_{ff}^2 + b_2E_{ss}^2 + b_3E_{nn}^2 + 2b_4E_{ff}E_{ss} + 2b_5E_{ss}E_{nn} + 2b_6E_{ff}E_{nn} + b_7(E_{fs}^2 + E_{sf}^2) + b_8(E_{sn}^2 + E_{ns}^2) + b_9(E_{fn}^2 + E_{nf}^2)$$

and $E_{ij}$ are the components of the Green-Lagrange strain tensor represented in terms of the local myocardial coordinate system where the fiber, sheet, and sheet-normal directions are represented with subscripts f, s, and n respectively. Guccione *et. al.* proposed a special case of transverse isotropy and used it to model cardiac tissue [98]:

$$Q = b_fE_{ff}^2 + b_t(E_{ss}^2 + E_{nn}^2 + 2E_{sn}) + 2b_{ft}(E_{fs}^2 + E_{fn}^2). \tag{4.17}$$

Here, the strain tensor is symmetric, so $E_{ij} = E_{ji}$. The terms defining the mechanical

response in the sheet (s) and sheet-normal (n) directions are now equivalent, as well as the term characterizing the shear response between these two directions (sn) so $b_2 = b_3 = b_8 = b_t$. Additionally, the fiber-sheet (fs) shear response is now equivalent to the fiber-sheet normal shear response (fn), so $b_7 = b_9 = b_{ft}$. By setting $b_4 = b_5 = b_6 = 0$, the interaction terms $E_{ff}E_{ss}$, $E_{ss}E_{nn}$ and $+ E_{ff}E_{nn}$ are not accounted for. The four positive material parameters are $c$, $b_f$, $b_t$, $b_{ft}$. The $c$ parameter scales the overall material response (bulk stiffness) and has dimensions of stress, while the b parameters are all dimensionless and control the exponentiality of the response.

### 4.1.5 Governing Principles

There are fundamental principles governing continuum mechanics and these are described here.

### 4.1.6 Conservation of Mass

This principle states that the mass of the object remains constant over time. The continuum body is considered as a closed system, and mass is neither added nor removed from it, so the mass in the body in the referential configuration must equal its mass in the deformed configuration. In the Eulerian (spatial) point of view, this principle is mathematically represented as:

$$\dot{\rho} + \rho(\nabla \cdot \mathbf{v}) = 0 \tag{4.18}$$

where $\rho(\mathbf{x}, t)$ is the spatial mass density, $\dot{\rho}$ is the material time derivative of $\rho$, and $\mathbf{v}$ is the particle velocity. The above equation is the local balance of mass for any volume within the continuum body. In cardiovascular solid mechanics, we consider the domain of the body as fixed, so the conservation of mass reduces to $\dot{\rho} = 0$, because $\nabla \cdot \mathbf{v}$, which is interpreted as the volumetric strain rate equals zero. Consequentially, conservation of mass is typically fulfilled automatically.

### 4.1.7 Conservation of Linear Momentum

The linear momentum of a body is a vector quantity measured as a product of its mass and velocity. According to Newton's second law of motion, the rate of change of a body's momentum equals the net force acting on it. This principle ensures that the vector sum of all external forces acting on a material equals the rate of change of the total linear momentum and is expressed mathematically as:

$$\rho\dot{\mathbf{v}} - \nabla \cdot \boldsymbol{\sigma} - \rho\mathbf{b} = 0 \tag{4.19}$$

where $\boldsymbol{\sigma}$ is the Cauchy stress tensor and $\mathbf{b}(\mathbf{x}, t)$ is the spatial body force density. In solid mechanics, oftentimes, accelerations ($\dot{\mathbf{v}}$) and body forces ($\rho\mathbf{b}$) are assumed to be negligible, so the linear momentum balance reduces to $\nabla \cdot \boldsymbol{\sigma} = 0$.

### 4.1.8 Conservation of Angular Momentum

The angular momentum is the rotational motion equivalent to linear momentum. It is a vector quantity, which is the product of a body's rotational inertia and angular velocity about a particular axis. This principle ensures that the angular momentum of a rotating object does not change unless some external torque is applied to it. Conservation of angular momentum is satisfied by symmetry of the Cauchy stress tensor, which reduces the independent stress tensor components from 9 to 6:

$$\boldsymbol{\sigma} = \boldsymbol{\sigma}^T \tag{4.20}$$

### 4.1.9 Conservation of Energy

This principle states that the work done by external loads per unit time $P_e(t)$ (external mechanical power) must equal the sum of the rate of change of kinetic energy ($\mathscr{K}$) with time and the internal stress power ($P_\sigma$). The global energy balance is expressed mathematically as:

$$P_e(t) = \frac{\partial}{\partial t}\mathscr{K} + P_\sigma. \tag{4.21}$$

The local energy balance is expressed mathematically as:

$$\rho\dot{e} - \boldsymbol{\sigma} : (\nabla\mathbf{v}) + \nabla \cdot \mathbf{q} - \rho s = 0 \tag{4.22}$$

where $e(\mathbf{x}, t)$ is the internal energy per unit mass, and $\dot{e}$ is the spatial derivative of $e$, $\mathbf{q}$ is the heat flux vector, $s$ is the rate of energy generation by sources in the volume per unit mass, and other variables are represented as previously described. For any volume in the continuum body, the sum of the internal power ($\rho\dot{e}$) and the heat power ($\nabla \cdot \mathbf{q}$) must equal the sum of the stress power ($\boldsymbol{\sigma} : (\nabla\mathbf{v})$) and the rate of energy generated by sources inside the volume ($\rho s$).

## 4.2 Finite Element Method

Finding direct, analytical solutions to complicated differential equations over complex domains can be infeasbile. The finite element method (FEM) seeks an approximate solution to such problems and has applications in many engineering disciplines, including solid mechanics. Broadly, in FEM, the problem is discretized into smaller parts (finite elements) which simplifies the behavior over each element. The FEM is based on the weak form of the boundary value problem where the goal is to determine a function that satisfies given differential equations within a domain, while also satisfying prescribed boundary conditions. Additionally, the FEM is formulated such that the full problem can be represented as a system of algebraic equations by assembling the contributions of all elements.

This thesis makes use of Lagrange basis functions (shape or interpolation functions) to approximate quantities that vary over the domain (e.g. stress or strain). The linear Lagrange interpolation is the simplest form. In 1-D, this means we represent a function as piecewise linear polynomials over smaller domains. In 3-D, this means we represent the spatial variation in a quantity using elements within which the quantity varies linearly. Higher order Lagrange interpolation represents a function using piecewise higher order polynomials (e.g. quadratic, cubic, etc).

The FEM requires the prescription of essential or Dirichlet boundary conditions that

specify the value of the unknown function along some boundary of the domain (e.g. prescribed surface displacement). Higher order boundary conditions (e.g. Neumann, Robin, etc) can be prescribed along other boundaries. Loading conditions can also be prescribed as surface tractions, pressures, forces, etc. The classical FEM problem involves solving for the stress (or strain) state of a body of defined reference geometry and a known strain energy law (constitutive model), given certain loading and boundary conditions. In inverse FEM, however, through a series of iterative forward simulations, the goal is to find an optimal constitutive model that is able to reproduce a given stress (or strain) distribution in a body under certain loading and boundary conditions. Challenges with inverse FEM problems include the possible nonlinearity of the inverse mapping function, the multiplicity and identifiability of solutions, the effects of sparsity and noise of the input data, and the significant computational expense [118, 119]

Since FEM provides an approximate solution, it is important to perform an application-dependent convergence analysis to understand the solution accuracy. In FEM, the greater the number of nodal parameters, the higher the level of accuracy. However, the computational cost grows non-linearly with increasing number of nodes.

The FE method as implemented in FEBio was used [120]. Further details on FEM theory can be found in [121].

## 4.3 *Ex vivo* Passive Myocardial Mechanical Behavior

Passive Myocardium is an orthotropic, heterogenous, viscoelastic, non-linear material, that exhibits non-recoverable strain softening. These key properties have been revealed by *ex vivo* testing of left ventricular myocardium.

Dokos *et. al.* performed simple shear tests on passive ventricular myocardium [7] from six pig hearts using *in vivo* physiological maximum shear displacements. Three specimen from each heart were tested in different orientations to cover all six modes of simple shear deformation (Figure 4.2).

A key result from their paper is shown in Figure 4.3 which clearly demonstrates the anisotropy in the material response. As expected intuitively, myocardium is most resistant to shear deformations that extend the cardiomyocyte (FS and FN shear modes). The resistance

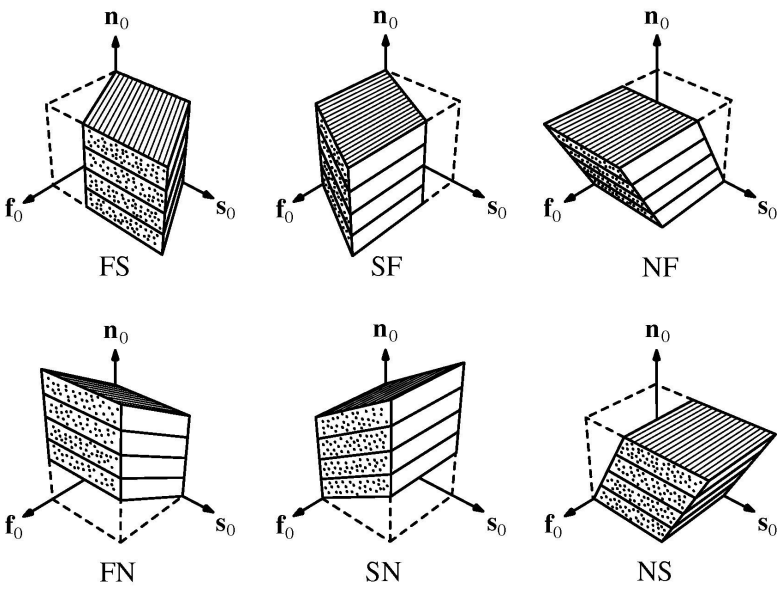



Figure 4.2: **Simple shear modes of myocardium.** Reproduced with permission from Sommer *et. al.* [6]. Shear mode ij represents application of shear on the face orthogonal to the $i^{th}$ axis in the $j^{th}$ direction.

to shear on the face normal to the fibers is the least and is considered equal between shear in the fiber direction (NF) and in the sheetlet direction (NS). The SN and SF shear modes have intermediate properties, and the difference in shear stiffness between these modes may be down to the effects of endomysial collagen. The results show that the axial stiffness is greatest in the fiber direction and least in the sheet-normal direction. The hysteresis in cyclic testing indicates the viscous energy dissipation in myocardium, a phenomenon that is commonly observed in soft biological tissues [122]. Their results also showed that myocardium exhibits non-recoverable strain softening which may be due to disruption of the extracellular meshwork surrounding cardiomyocytes [123] or stress-dependent damage to the endomysial collagen [124], or disruption of myofilaments (e.g. titin, actin and myosin) and cytoskeletal proteins [125, 126]. Lastly, myocardium is quite soft as demonstrated by the peak shear stresses. Notably, the *ex vivo* behavior is unlikely to transfer directly to *in vivo* conditions because of the effects of rigor mortis, mechanical damage to the tissue from cutting, and strain-softening. Nevertheless, *ex vivo* testing has helped to provide key insight into the material response of passive ventricular myocardium.

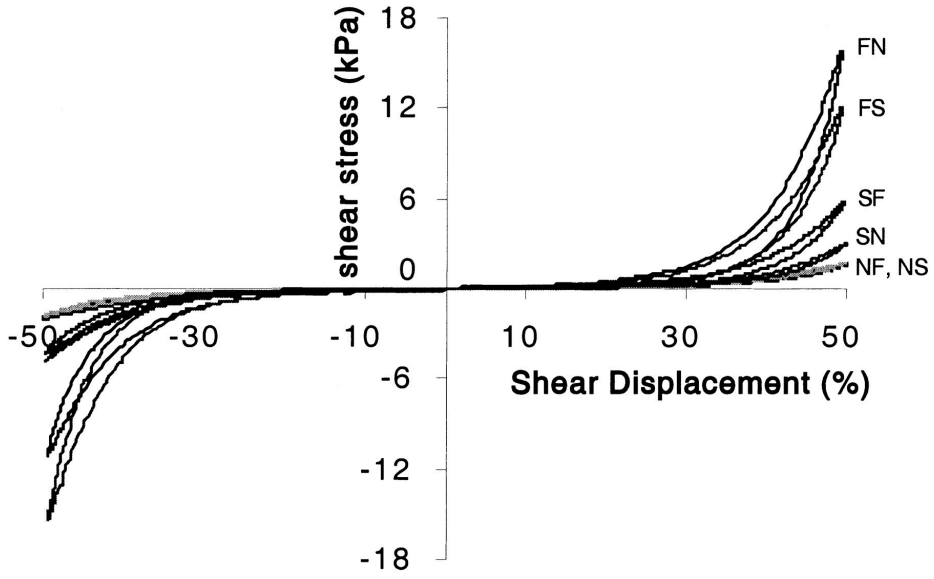



Figure 4.3: **Shear Behavior of Passive Myocardium.** Reproduced with permission from Dokos *et. al.* [7]. Shear modes are described in Figure 4.2

The most recent comprehensive *ex vivo* mechanical tests on passive myocardium were performed by Sommer *et. al.* on excised specimen from human ventricles [6] (Figure 4.4). Sommer *et. al.* performed 'true' biaxial extension testing by using different stretch ratios between the different directions, thus covering a large range of possible deformation modes and obtaining information about the cross-coupling of the mechanical response. Their work further demonstrated key properties of myocardium that had been shown by earlier papers including orthotropic material response, heterogeneity, viscoelasticity, non-linearity, and strain softening. Results from biaxial testing showed that the stiffness in the mean fiber direction was about 50% greater than in the cross fiber direction. Additionally, they found significant energy dissipation between loading and unloading of up to 38%. While they found similar qualitative behavior in myocardium excised from different portions of the heart (LV free wall, RV free wall, septum) in terms of nonlinearity and hysteresis, there were appreciable differences in the peak stresses at different stretch levels between the LV and RV. Specifically, myocardium excised from the RV free wall was significantly stiffer than LV myocardium. On the other hand. the peak Cauchy stresses were similar for the LV free wall and septum, and throughout the LV wall from base to apex. Additionally, comparison of

their results to results from *ex vivo* testing of porcine [127] and canine myocardium [97, 128, 129] revealed that human myocardium may be significantly stiffer than that of commonly used large animal models of heart disease.

Avazmohammadi *et. al.* developed a new approach to drive optimal experimental design for quantifying the mechanical behavior of cardiac tissue [130]. This is important because there exist an optimal set of 3D displacement paths that are most robust to noise in the experimental data and variations in the boundary conditions used. By identifying and subjecting the test tissue to these displacement paths, ability to determine material parameters, is increased. Currently, there is a shortage of suitable experimental data on the passive mechanical behavior of myocardium, including shear and true biaxial testing. The work of Avazmohammadi *et. al.* could inform future *ex vivo* testing of cardiac tissue.

## 4.4 *In vivo* Passive Myocardial Mechanical Behavior

Medical image-driven biophysical modeling of the heart enables the assessment of passive myocardial mechanical behavior. This is essentially an inverse FEM problem where the objective is to identify an optimal material parameter set of a representative constitutive model that reproduces certain kinematics under prescribed boundary and loading conditions (e.g LV end-diastolic pressure). To achieve this, the ventricular reference geometry and microstructural configuration identified from medical imaging (e.g. MRI, ultrasound, CT) are used to develop an FE geometric mesh. The myocardium is represented by a constitutive model (see section 4.1.2), and the boundary and loading conditions are prescribed. After initialization of the material parameters, optimization is used to identify a constitutive model that reproduces some measured kinematic-based criteria (e.g. displacements from tagged MRI). There is significant interest in biophysical modeling of the heart by combining medical imaging with ventricular pressure measurements. Through these patient-specific simulations, certain metrics of cardiac function that are otherwise difficult to measure directly, such as wall stresses and strains, can be estimated. This personalized understanding of cardiac function can reveal disease mechanisms that can be therapeutic targets [131].

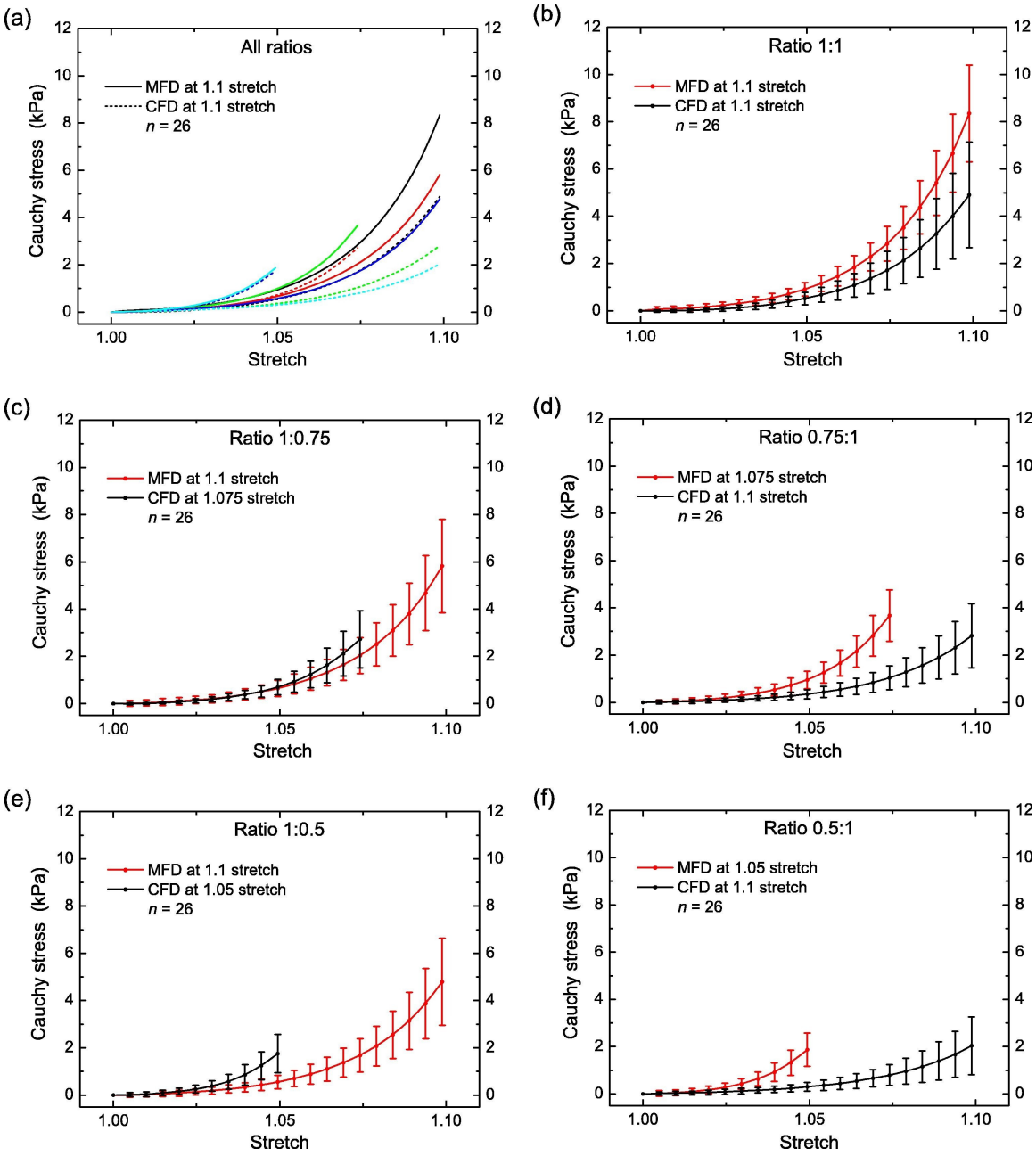



Figure 4.4: **Biaxial Behavior of Passive Myocardium.** Reproduced with permission from Sommer *et. al.* [6]. Abbreviations: CFD - cross fiber direction, MFD - mean fiber direction

### 4.4.1 Reference Geometry evaluation

Simplified geometries have been used to represent the LV, including prolate spheroid models [132, 133, 134], cylindrical models [98, 135], and spherical models [136]. These approaches offer the benefit of rapid computation of mechanics-based parameters of ventricular performance since the LV shape is represented by few parameters. However, these models may be unable to acceptably represent ventricular anatomy, so accuracy can suffer.

Currently, most studies obtain the cardiac reference geometry for FEM from segmented cardiac medical imaging data. FEM requires the prescription of an unstressed reference configuration, which presents a challenge since *in vivo* myocardium is known to exhibit prestresses [137]. Due to these prestresses, the geometry obtained from medical images don't represent an actual unstressed reference configuration for finite element analysis even if the images represent an externally unloaded state. Inclusion of the influence of these myocardial prestresses, which are thought to facilitate diastolic filling [138] and homogenize the ventricular wall stresses [137], can significantly impact passive myocardial stiffness estimation [138, 139, 140].

Peirlinck *et. al.* proposed an inverse elastostatics approach to resolving the unstressed ventricular configuration using *in vivo* medical images [141]. An iterative method was implemented by Krishnamurthy *et. al.* [142] to determine the unstressed configuration from the end-diastolic biventricular geometry. Of note, these inverse prestress estimation procedures require assumptions about the underlying material properties and full validation is still lacking. Another approach to including the effects of residual stress on passive ventricular mechanics was proposed by Wang *et. al.* through a modification to the Holzapfel-Ogden material model [143]. Most studies assume that the effects of prestresses are negligible, and directly extract the reference geometry from medical-images, but the choice of cardiac phase ranges from the end of isovolumic relaxation [144, 145] to early diastole [3], and diastasis [9].

Using an MRI-based framework, the ventricular surface geometry is typically extracted from cine bSSFP images, since they offer good identification of endocardial and epicardial borders (see section 3.3.1). Some studies have extracted the reference geometry from images typically used to extract kinematics (e.g. tagged MRI [144]), eliminating the need to register

kinematic and geometric data. However, surface geometries extracted from tagged MRI may overestimate wall thickness due to the difficulty associated with dileneating endocardial borders from papillary muscles.

Most modeling has been focused on the left ventricle, with comparatively few biventricular models in the literature [146, 147, 148, 149, 142]. There is evidence that the LV has nearly homogenous material properties and the RV is significantly stiffer than the LV [6]. Thus, in the absence of localized tissue damage (e.g. ischemia), the assumption of homogeneity appears to be appropriate in LV models. On the other hand, biventricular models may require allowance for variation in material properties between the ventricles.

### 4.4.2 Cardiac microstructural configuration

The FE model also needs to incorporate the cardiac microstructural configuration. However, subject-specific microstructure, such as that which can be obtained through diffusion tensor MRI (see section 3.3.3), is not always readily available in patients or research subjects. Thus, mathematical models of myofiber orientation have been proposed, including the widely used Laplace-Dirichlet rule-based algorithm, from Bayer *et. al.* [10], which was extended to biventricular geometries with outflow tracts by Doste *et. al.* [150], the minimum-distance rule-based algorithm proposed by Potse *et. al.* [151], the mathematical models proposed by Rijcken *et. al.* to optimize cardiac ejection [152], and by Nielsen *et. al.* based on the structure of canine hearts [153]. While these rule-based and mathematical approaches are convenient, their use affects the fidelity of the estimated passive myocardial behavior to the subject, given the sensitivity of ventricular mechanics to fiber orientations.

### 4.4.3 Kinematics

Material parameter optimizations are typically based on a geometry-based cost function. Comprehensive geometry-based cost functions incorporating local deformation kinematics can be constructed. For example, computational models have been calibrated using local displacements extracted from tagged MRI [144, 154] (see section 3.3.2) or cine DENSE MRI [155]. The target function can also be defined based on regional averaged strain which could dampen the effect of noise in the kinematic data, as was done by Finsberg *et. al.* who

used regional strains from speckle tracking [156]. In the absence of local kinematic data, some authors have matched end-daistolic surfaces or volumes [157, 145], but this method may not adequately constrain the material parameter solution.

### 4.4.4 Boundary and Loading Conditions

Kinematic boundary conditions are typically applied at the base, with some studies choosing to fully constrain basal displacement [158] and others applying image-derived displacements at the basal plane [144] and yet others constraining rotations and displacement of the base in specified directions [142]. Some authors prescribe apical motion as well [3]. These boundary conditions impact ventricular mechanics [159], thus care should be taken to apply physiological boundary conditions. Pfaller *et. al.* also highlighted the effect of pericardial boundary conditions on simulated ventricular mechanics [160].

LV end-diastolic pressure(s) is the most widely used loading condition in cardiac passive mechanics modeling. Direct measurement using a catheter is the gold standard ventricular pressure estimation technique, but these invasive measurements are rarely available. Thus comparatively few studies have used catheter-obtained pressure traces for material parameter optimization [144, 9]. Non-invasively estimated LV end-diastolic pressure based on mitral flow velocity obtained from 4D flow MRI was used by Asner *et. al.* as the ventricular loading condition [161]. Such estimates present a viable alternative in the absence of invasively acquired ventricular pressures. The Klotz curve has also been used to determine end-diastolic pressures, but there is evidence that the Klotz relationship may not generalize adequately [162, 163].

### 4.4.5 Uncertainty Quantification

Passive myocardial mechanics depends on multiple measurements that may be uncertain, noisy, or sparse. Though the uncertainty in the input data unavoidably leads to errors in material parameter estimates, most studies do not present confidence in the results. For example, ventricular mechanics is affected by myofiber orientation [164], wall thickness [165], kinematic boundary conditions [159], geometry-based cost function [1], and LV end-diastolic

pressures. Some authors have quantified the sensitivity of LV passive mechanics models and uncertainty of inverse-FEM obtained material parameters [166, 167]. Given the subject-specific nature of passive ventricular mechanics modeling, wherever possible, authors should quantify the uncertainty associated with their models, with respect to the input data.

## 4.5 Summary

In this chapter, we reviewed some essential elements of continuum mechanics which are important for biomechanical modeling of passive myocardium. This overview included descriptions of kinematical quantities, constitutive modeling, and conservation principles. We also described the finite element method and presented some key points in the literature on *ex vivo* and *in vivo* passive myocardial mechanical behavior. The following Chapters 5-8, will be focused on some of the major contributions made in this work to the field of passive myocardial mechanics modeling.

# Chapter 5

# A Framework for Evaluating Myocardial Stiffness Using 3D-Printed Hearts

## 5.1 Abstract

MRI-driven computational modeling is increasingly used to simulate *in vivo* cardiac mechanical behavior and estimate subject-specific myocardial stiffness. However, *in vivo* validation of these estimates is exceedingly difficult due to the lack of a known ground-truth *in vivo* myocardial stiffness. We have developed 3D-printed heart phantoms of known myocardium-mimicking stiffness and MRI relaxation properties and incorporated the heart phantoms within a highly controlled MRI-compatible setup to simulate *in vivo* diastolic filling. The setup enables the acquisition of experimental data needed to evaluate myocardial stiffness using computational constitutive modeling: phantom geometry, loading pressures, boundary conditions, and filling strains. The pressure-volume relationship obtained from the phantom setup was used to calibrate an *in silico* model of the heart phantom undergoing simulated diastolic filling. The model estimated stiffness was compared to a ground-truth stiffness obtained from uniaxial tensile testing. Ultimately, the setup is designed to enable extensive validation of MRI and FEM-based myocardial stiffness estimation frameworks.

## 5.2 Introduction

Heart failure (HF) is a condition in which the heart is unable to meet the metabolic demands of the body. The US public health burden of heart failure is expected to grow significantly in the next decade with the prevalence projected to reach 8 million by 2030 [14]. As HF often occurs because of deteriorating cardiac function due to persistent remodeling, pathophysiological cardiac remodeling has been identified as a therapeutic target in HF [131]. Thus, understanding the various mechanisms and manifestations of remodeling is fundamental for formulating appropriate clinical intervention [168]. A significant consequence of cardiac remodeling is changes to the passive (diastolic) stiffness of ventricular myocardium. Measuring *in vivo* passive myocardial stiffness requires a comprehensive evaluation of the mechanical behavior (stress-strain) in diastole.

Passive stiffness of the left ventricle (LV) is commonly inferred from the LV end diastolic pressure-volume relationship (EDPVR) [169]. However, EDPVR only provides a global estimation of the apparent LV chamber stiffness. It is therefore inappropriate to infer intrinsic myocardial mechanical behavior from EDPVR. Cardiac Magnetic Resonance Elastography (MRE) has been used for direct measurement of myocardial shear stiffness [170], but its implementation requires assumptions that make solutions challenging for *in vivo* cardiac physiology.

Computational modeling using MRI data and Finite Element Modeling (FEM) enables the estimation of *in vivo* myocardial passive stiffness [34]. Provided ventricular geometry, microstructural organization, boundary conditions, kinematics from MRI, and endocardial filling pressures from catheterization, continuum balance laws and optimization techniques can be leveraged in FEM to inversely obtain the parameters of the constitutive model governing the material's mechanical behavior. Currently, however, validation of MRI and FEM-based myocardial stiffness estimation frameworks is exceedingly difficult *in vivo* as ground-truth *in vivo* myocardial stiffness remains elusive.

We have developed a 3D-printed heart phantom with a myocardium-mimicking material of known stiffness and MRI properties. The phantom was incorporated within an MRI compatible *in vitro* diastolic filling setup. We estimated stiffness of the heart phantom using an MRI-derived computational model and compared with ground-truth measures obtained

from uniaxial tensile testing.

## 5.3 Methods

### 5.3.1 Phantom Material Selection

The phantom material was selected to have similar stiffness and MR relaxation properties to myocardium. The suitability of different Sylgard blends was assessed, and the tissue-mimicking material was chosen based on mechanical and MR relaxation properties. Blends with different mass compositions of Sylgard 184 and 527 were mechanically characterized according to the ASTM D412 standard using an Instron 5848 Microtester (100N load cell).

Uniaxial tensile testing samples of the Sylgard blend were produced in parallel with the phantom development. Samples were punched out (ASTM cutting die A) from cured sheets (thickness $3.13 \pm 0.11$mm) and three samples were mechanically tested according to the ASTM D412 standard using an Instron 5848 Microtester (100N load cell). Samples were mounted by first clamping the specimen to the upper grip, zeroing the load cell, then clamping the specimen to the lower grip while taking care to avoid load application. The extensometer was then clamped to the samples with the gauge length set at 50.8mm. The test was performed at ambient conditions with a strain rate of 500 mm/min. The Elastic Modulus was evaluated from a linear regression of the stress-stretch relation at stretches from 1.0 to 1.2.

MRI relaxation properties were measured using T1-mapping (MOLLI 5-3-3, spatial resolution $1.00\times1.00\times5.0\text{mm}^3$) and T2-mapping (T2-prep FLASH; flip angle 12º; spatial resolution, $1.00\times1.00\times5.0\text{mm}^3$).

### 5.3.2 Phantom Development

High-resolution T1-weighted images from a healthy *ex vivo* porcine heart (restored to *in vivo* mid-diastasis geometry) [171] were used to generate a 3D geometric heart model. The heart phantoms were subsequently manufactured using a lost-wax casting technique which has been used to manufacture anatomically detailed heart phantoms [172]. The epicardial surface of the 3D geometric heart model was used to create a negative epicardial mold. The

ventricular blood pool segmentations were used to create LV and right ventricle (RV) blood pool casts. The epicardial mold and blood pool casts were converted into stereolithography files and 3D printed (Ultimaker 3 Extended) using tough polylactic acid and water-soluble polyvinyl alcohol, respectively. The heart phantom was cast using a silicone elastomer blend containing mass ratio of Sylgard 184:527 (Dow Corning) of 1:4 (Figure 5.1). Heart phantoms were cast by curing the Sylgard blend in the 3D printed mold for 48 hours at room temperature. Ventricular basal ports and an apical anchor were added to the heart phantom to facilitate loading and motion stabilization.

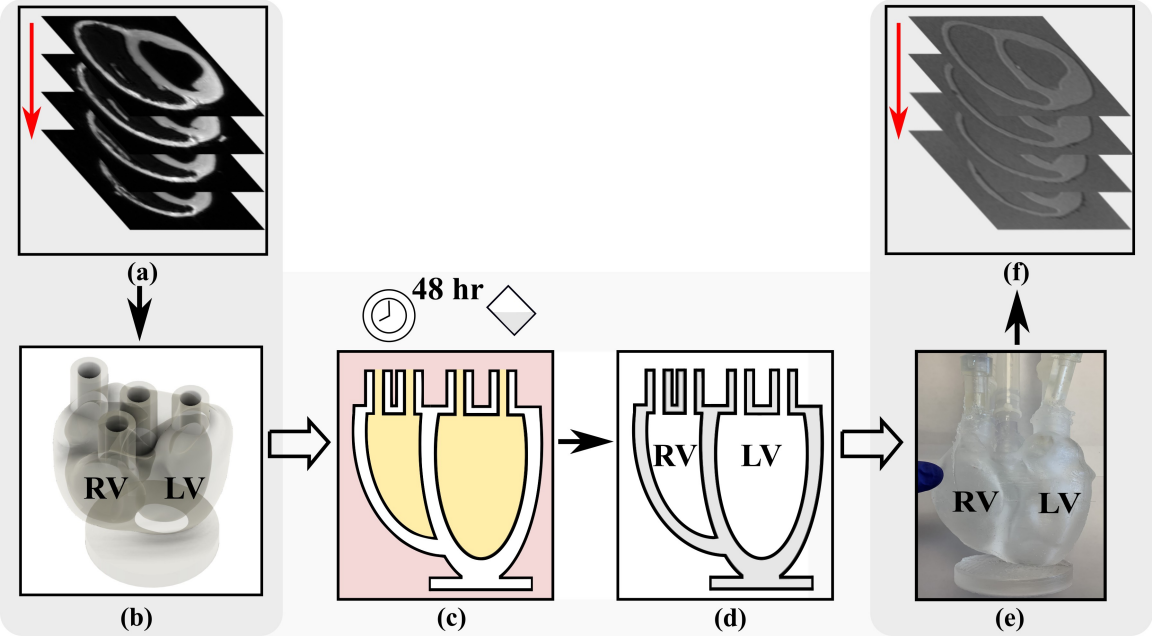



Figure 5.1: **Heart Phantom Development.** (a) *ex vivo* porcine T1-weighted images segmented to develop a geometric model. (b) model fitted with ports and apical anchor. (c) Sylgard blend cured in mold for 48 hours. (d) epicardial mold (pink) removed and ventricular blood-pool casts (yellow) dissolved in water. (e) subject-specific heart phantom with ports. (f) phantom 3D-SPGR images

### 5.3.3 *In vitro* Diastolic Filling Setup

The final heart phantom was embedded in a flow loop (Figure 5.2) controlled by an MRI-compatible linear motion stage (MR-1A-XRV2, Vital Biomedical Technologies). To simulate *in vivo* diastolic LV filling, the loop was designed to deliver a cardiac-like late-diastolic filling cycle in the LV (sinusoidal flow: 13 mL/cycle mean, 1s period).

Each ventricle was connected in a closed loop and the fluid volume in each loop was fixed. The motion stage was used to deliver the sinusoidal flow into the LV through a fluid filled syringe within the LV loop. The RV was kept at a constant volume throughout LV filling. LV pressure was acquired continuously in PowerLab (ADInstruments) using MRI-compatible pressure transducers (Micro-Tip SPR 350S, Millar). LV filling volume was recorded from the motion stage and synchronized with pressure measurements. The phantom was fixed at the apex and basal ports.

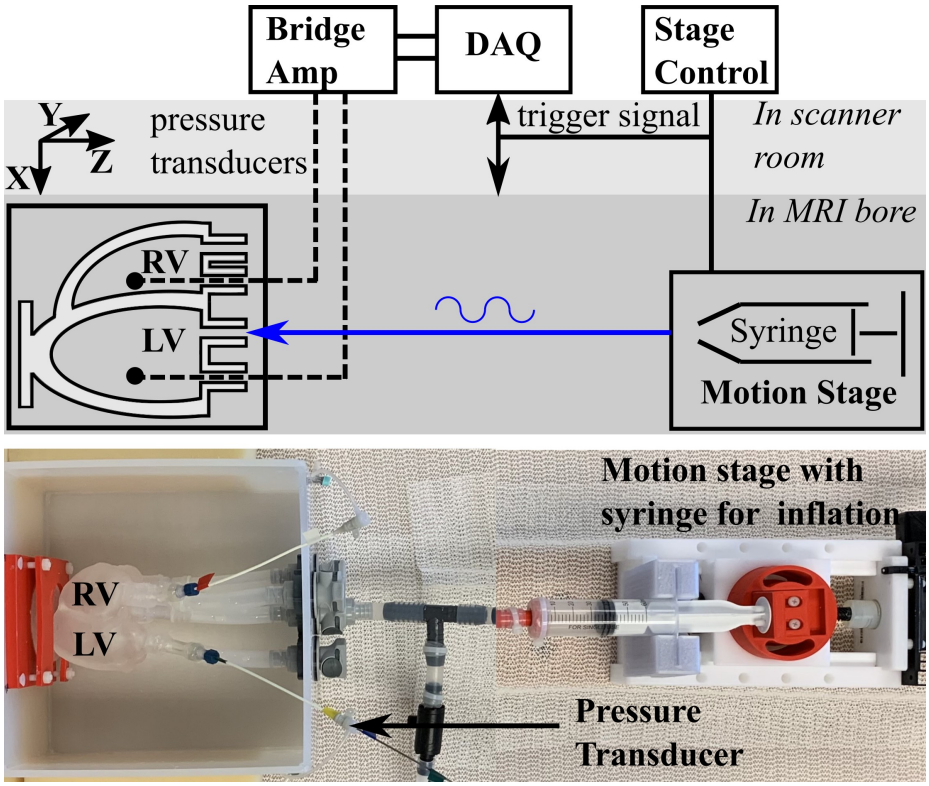



Figure 5.2: **Experimental Setup.** Schematic (top) and picture (bottom)

### 5.3.4 Image Acquisition and Processing

All *in vitro* images were collected on a 3T MRI (Skyra, Siemens) using a 32-channel chest and spine coil. In a static equilibrium phase, a 3D spoiled gradient recalled echo (3D SPGR; TE/TR=2.17/5.5ms; FA=20º; isotropic 1.00mm$^3$; Static equilibrium phase) was performed over the entire volume of the phantom.

The 3D phantom geometric fidelity was evaluated using *in vitro* and *ex vivo* images semi-automatically segmented to extract binary masks of the myocardium, LV blood pool, and RV blood pool using Otsu's method and manual clean up (MITK Workbench). *In vitro* myocardium masks were registered to *ex vivo* myocardium using an automated 3D rigid regular-step gradient descent algorithm (Matlab, Mathworks). This transformation was applied to all *in vitro* binary masks. Volumetric accuracy and dice similarity coefficient (DSC) of binary masks were assessed by comparing the volume of the myocardium, LV blood pool, and RV blood pool from the *in vitro* images to *ex vivo* images.

### 5.3.5 *In silico* modeling and stiffness quantification

Starting from the 3D geometrical model (Section 5.3.2), a volumetric quadratic tetrahedral mesh with an average edge size of 1.5mm was constructed using the 3-Matic meshing software suite (Materialise). For computational efficiency, the constrained apical anchor was removed from the mesh. The resulting mesh consisted of 118,337 nodes and 61,272 tet10 elements, summing up to 355,011 degrees of freedom to be solved in the finite element analysis software suite Abaqus (Dassault Systemes, Simulia Corp). Previous studies on mechanical characterization of Sylgard 184 and 527 indicate that the materials are nearly incompressible, or weakly compressible ($\nu$=0.495) [173]. Thus, we assumed near incompressibility in the Sylgard blend. Given that we are working in a lower strain regime (strains under 40%), the Sylgard silicone elastomer blend was assumed to be linearly elastic [174] (consistent with our benchtop data). In accordance with the experimental setup, the computational heart phantom was kinematically constrained by constraining the apical bottom surface and the top surfaces of the ventricular basal ports. The port openings were virtually closed off to create enclosed fluid cavities, allowing an efficient computation of the temporal pressure and volume evolutions in the left and right ventricular 'blood' pools. Using a

volume-driven boundary condition on the left ventricular fluid cavity, the deformation of the phantom following the sinusoidal inflow and outflow was virtually simulated. Similarly, for the RV fluid cavity a volume-driven boundary condition enforced the RV volume to remain constant, in accordance with experimental conditions. To include large displacement effects, our simulation was set up to account for geometric nonlinearity.

The resulting pressure field was used to compute the elastic stiffness of the 3D casted Sylgard blend. More specifically, we found the Young's Modulus (E) using Abaqus as the forward solver wrapped inside Python's Nelder-Mead optimization algorithm [175, 176]. Starting from an initial value (E=150 kPa), we computed the mismatch between the experimentally measured and simulated LV pressure evolution, and iteratively updated the Young's Modulus until this error was minimized.

## 5.4 Results

### 5.4.1 Material Selection

Mechanical behavior in uniaxial extension is shown for all evaluated Sylgard blends in figure 5.3. MR relaxation properties of the Sylgard blends are shown in figure 5.4. The comparison between the blends and human myocardium in terms of stiffness, $T_1$ and $T_2$ relaxation times are reported in table 5.1. Though the blend containing 10% Sylgard184 exhibits the most comparable mechanical [177] and MR properties [178] to myocardium, the blend containing 20% Sylgard184 was the more feasible phantom material choice due to easier workability.

Table 5.1: Comparison of Sylgard Blends to Myocardium

| Material | | Stiffness [kPa] | $T_1$ [ms] | $T_2$ [ms] |
|---|---|---|---|---|
| Myocardium | | 8-70 [177] | 1158.7 [178] | 45.1 [178] % |
| % Sylg.184 | 10 | 088.3 ± 14.1 | 979.5 ± 07.7 | 434.8 ± 18.0 |
| | 20 | 202.6 ± 10.8 | 959.9 ± 05.0 | 313.8 ± 12.3 |
| | 30 | 383.3 ± 22.4 | 866.2 ± 20.3 | 232.7 ± 10.7 |
| | 40 | 523.1 ± 32.8 | 816.5 ± 11.5 | 214.7 ± 14.1 |

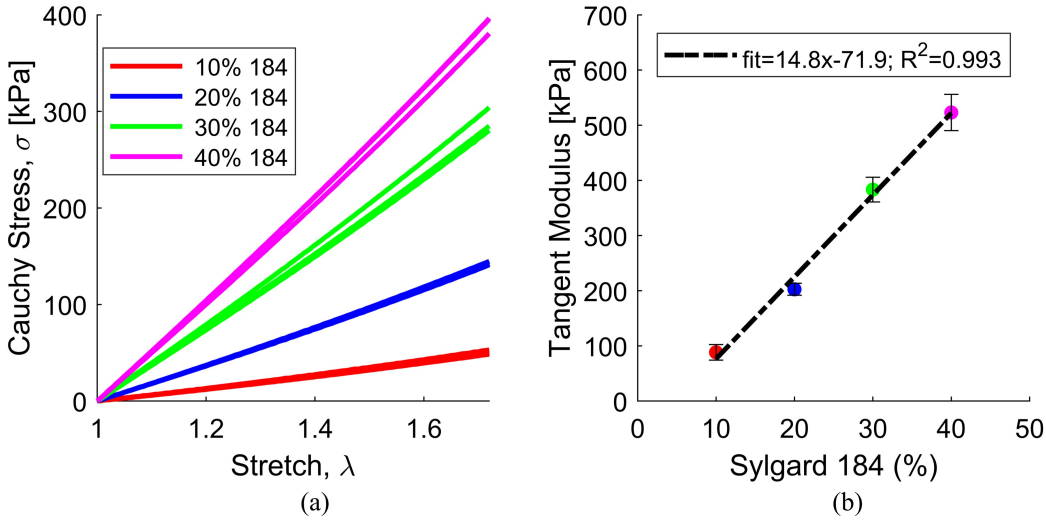



Figure 5.3: **Mechanical Properties of Sylgard Blends.** The stiffness of a blend is linearly proportional to the quantity of Sylgard184 in the mixture

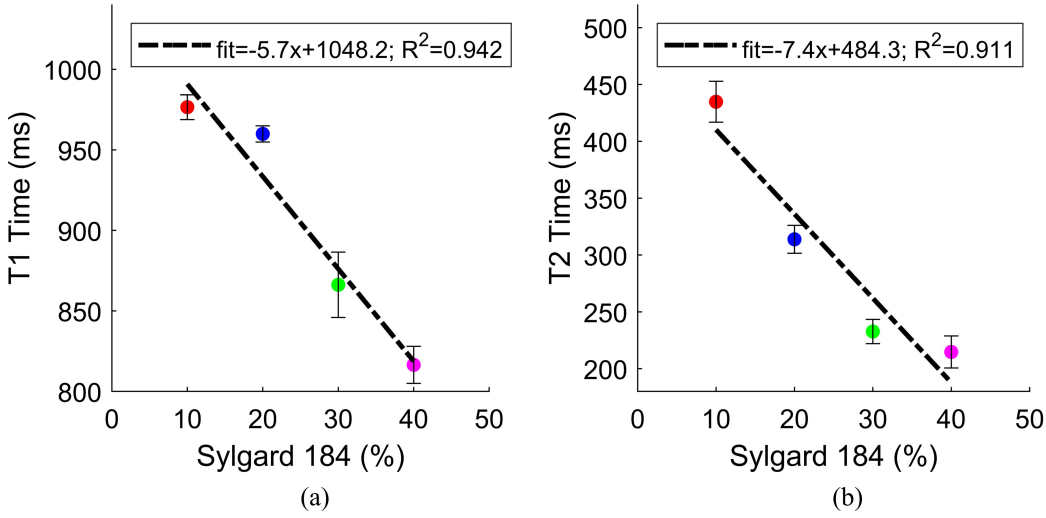



Figure 5.4: **MR relaxation properties of Sylgard blends.** The T1 and T2 relaxation times of a blend are related to the quantity of Sylgard184 in the mixture

### 5.4.2 Phantom Geometric Accuracy

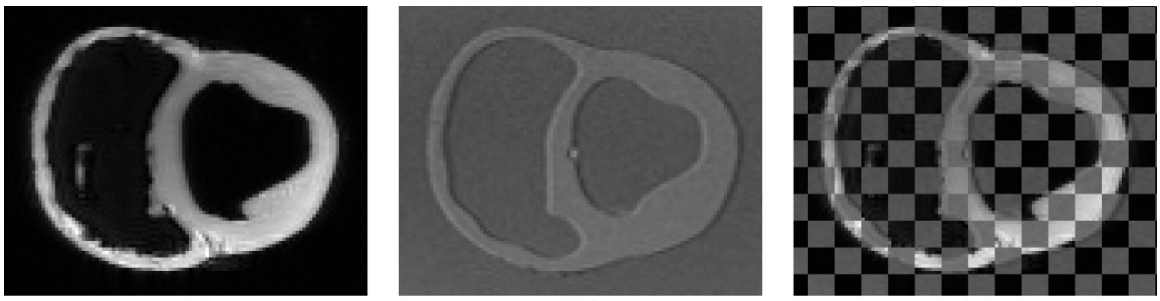
Analyses of the 3D *in vitro* heart phantom and *ex vivo* porcine subject MR images showed that the phantom development procedure adequately reproduces the porcine subject heart

geometry. The *in vitro* and *ex vivo* images were all well registered.

In general, qualitatively, images of slices of the heart phantom and porcine subject heart were well matched. Qualitative comparison of the porcine subject and heart phantom images at a midventricular slice is shown (Figure 5.5). Table 5.2 reports quantitative measures of the geometric fidelity of the heart phantom compared to the porcine subject heart. DSC is reported for the LV blood pool, RV blood pool, and myocardium. Geometric fidelity was also assessed by comparing the LV blood pool, RV blood pool, and myocardium volumes between the phantom and the swine subject.

Table 5.2: Geometry comparison between in vitro (phantom) and ex vivo (porcine subject)

| Tissue | DSC | *In vitro* volume | *Ex vivo* volume | Volume Error |
|---|---|---|---|---|
| LV Blood Pool | 0.92 | 45.5 mL | 43.9 mL | 3.6 % |
| RV Blood Pool | 0.95 | 67.5 mL | 68.9 mL | -2.0 % |
| Myocardium | 0.89 | 90.8 $mm^3$ | 93.0 $mm^3$ | -2.4 % |



(a) (b) (c)

Figure 5.5: **Phantom Geometric Accuracy.** Short-axis slices showing (a) T1-weighted *ex vivo* porcine subject (b) 3D SPGR *in vitro* heart phantom (c) qualitative checkerboard of *ex vivo* and *in vitro*

### 5.4.3 Uniaxial Tensile Testing

The Cauchy stress versus the stretch ratio for a representative sample of the Sylgard blend used to cast the heart phantom is shown in Figure 5.6a. The elastic modulus was calculated for the three samples by fitting a linear model to the stress-stretch data for stretches between

1.0 and 1.2 and estimated to be 235 ± 6kPa (mean ± SD). T1 and T2 times were 959.5 ± 5.0ms and 313.8 ± 12.3ms, respectively.

### 5.4.4 Pressure-Volume

Recorded pressures were sinusoidal (mean 34 mmHg, min. 0 mmHg, max. 68 mmHg) as were the volumes (peak 13mL; mean 6.5mL) (Figure 5.6). The LV pressure-volume (P-V) relation was obtained after the loading and unloading cycles were steady and repeatable. The loading P-V curve was used for the *in silico* stiffness calibration.

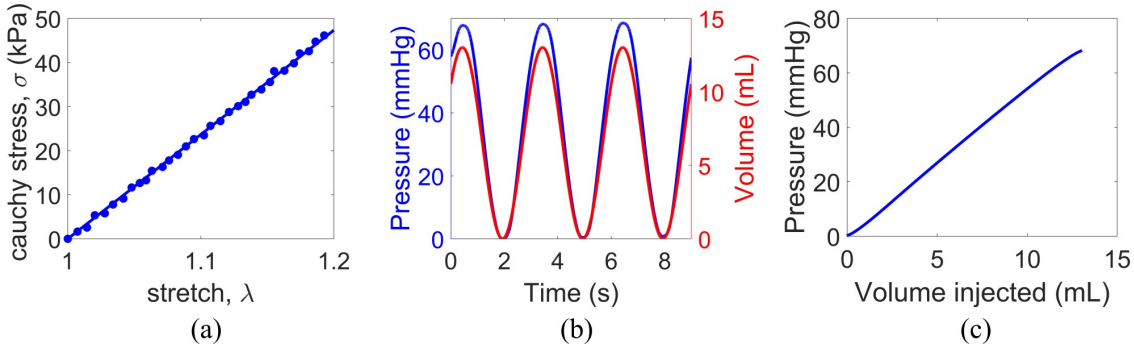



Figure 5.6: (a) Cauchy stress vs Stretch. (b) Pressure & Volume vs Time (c) P-V Experimental

### 5.4.5 *in silico* stiffness calibration

Figure 5.7a shows the computed deformation and the spatial variation in maximum principal stretch state at the maximal loaded volume during the sinusoidal loading protocol. The maximal stretch in the phantom is located at the endocardial LV surface, reaching maximum principal stretches up to 1.2. The *in silico* stiffness calibration is shown in Figure 5.7b. Starting from an initial elastic modulus of 150 kPa, the elastic modulus was iteratively updated, minimizing the pressure differences between the computational (line plot) and the experimental (x markers) results. The line plot represents the converged forward simulation from which an elastic modulus of 328 kPa was obtained.

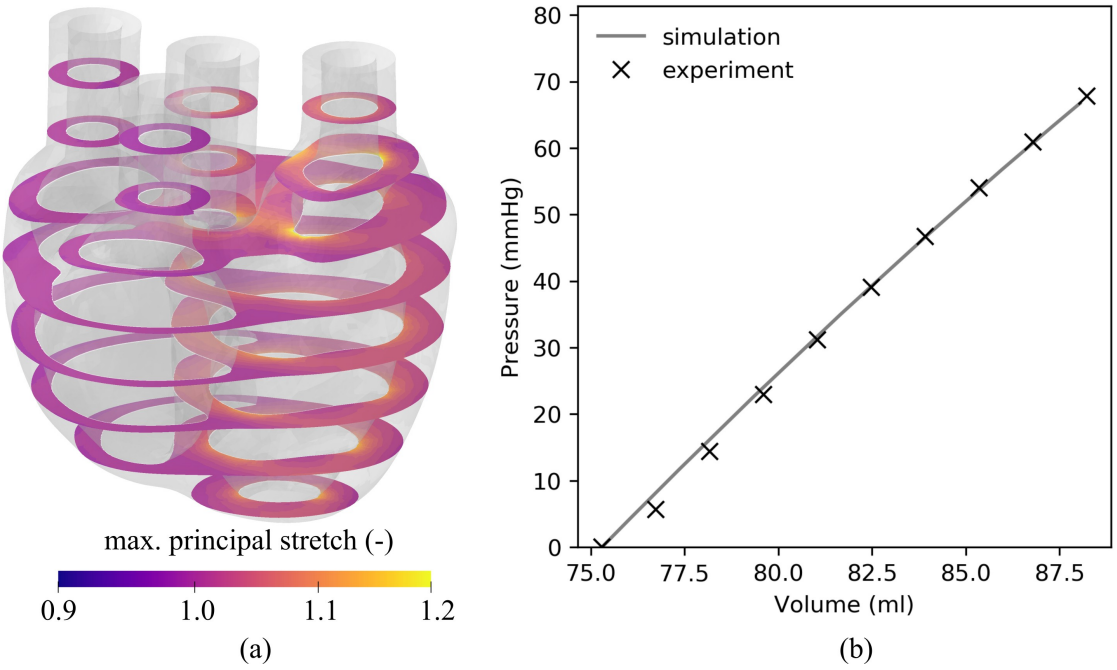



Figure 5.7: (a) Computed phantom deformation at maximal inflation. Short-axis slices depict the spatially varying stretch of the Sylgard blend. (b) Calibrated elastic modulus of 328 kPa.

## 5.5 Discussion

To validate our MRI and FEM-based stiffness estimation framework, we have developed subject-specific heart phantoms with passive ventricular myocardium-mimicking mechanical and MRI relaxation properties. The heart phantom adequately replicates the porcine subject geometry as shown by the reported high DSC between registered 3D SPGR *in vitro* (heart phantom) and the *ex vivo* (porcine subject, geometrically restored to *in vivo* mid-diastasis) images.

The properties of the material used to construct the phantom were comparable to myocardium in some respects, but lacked some key features. Most notably, unlike myocardium, which is anisotropic and nonlinear, the phantom material is isotropic and linear in the strain range considered. These fundamental differences make it difficult to directly compare mechanical properties of the phantom material to *in vivo* myocardium. By comparison, Sommer et al. in their work on *ex vivo* mechanical characterization of myocardium obtained

maximum Cauchy stresses in the fiber direction of about 8 kPa at stretch of 1.1 [6]. Our material, on the other hand, exhibited Cauchy stress of about 22 kPa at the same stretch. Although Sylgard blends with more Sylgard527 are softer, this blend was chosen for its far superior workability. Additionally, the T1 relaxation time of the Sylgard blend was identified as 959.9ms at 3T compared with reported myocardium T1 times of 1158.7ms at 3T [178]. The T2 relaxation time, which is less important for our study was however, significantly higher than that of myocardium at 313.8ms compared with 45.1ms [14].

Although we have yet to replicate *in vivo* myocardial anisotropy and nonlinearity in the heart phantom, our main objective, which is to validate MRI and FEM-based methods for estimating myocardial passive stiffness, does not require the phantom material properties to match perfectly with the properties of passive myocardium. This project aims to understand the ability of an MRI and FEM-based framework to reproduce accurate and precise estimates of the stiffness of soft heart phantoms under loading conditions similar to *in vivo* hearts in diastole. Necessary for the project are accurate quantification of ground-truth phantom material properties, and appropriate reproduction of experimental loading and boundary conditions in the computational model of simulated inflation. Validating the stiffness estimation framework in a controlled system, with isotropic, linear, heart shaped phantoms will build confidence in stiffness estimates of more complex materials like *in vivo* myocardium.

We estimated the phantom stiffness in an *in silico* model by matching the simulated compliance-pressure curve to the *in vitro* experimental P-V relation obtained from the heart phantom diastolic filling setup. The simulated solution converged to an elastic modulus of 328kPa compared with the phantom material stiffness ground-truth of $235 \pm 6$kPa obtained through tensile testing. A good match between experimental and simulated pressure-volume points was recorded, with some minor mismatch in the low-pressure regime which might suggest that the experimental pressure volume points do not form a perfect linear relationship. At this point, we saw no reason to assign additional weight to the pressure-volume points in the low-pressure regime to improve the fit. Palchesko et al. characterized blends of Sylgard 184 and 527 [8]. For our blend, their experimentally derived relationship between mass composition of a blend and its elastic modulus gives a stiffness of 215 kPa, comparing favorably with results from our tensile testing (235 kPa). This, and our measurement data,

suggests that our stiffness simulation is overestimating the material stiffness.

The discrepancy between simulated and benchtop estimate of phantom mechanical properties could be due to inaccuracies in the P-V used for stiffness calibration. The connection between the syringe and the phantom has inherent compliance that is not currently modelled by our simulation. Hence, it is possible the experimental pressure-volume relationship overestimates the volume injected into the LV. Future work aims to eliminate this effect by directly obtaining the ventricular kinematics and volumes from MRI images, as opposed to obtaining the filling volumes from the inflation unit. Future studies will also use MRI tagging or displacement encoding to reconstruct local phantom displacements during the filling cycle. The filling pressures and kinematics will be used for *in silico* stiffness calibration, providing more constraints for stiffness estimation, thus improving the accuracy of simulated phantom stiffness.

It is also possible that the discrepancy is due to inaccurate replication of the effect of the RV in the simulation. In the simulation, the RV is modeled as a constant volume. The RV however is not closed off directly at the basal ports. Hence, it is possible that due to the compliance of the tubing connected to the phantom, some fluid volume escapes into the tubing from the RV during LV inflation. To more accurately model RV effects, future work will acquire RV and LV pressures simultaneously during inflation and use the RV pressure data to model the effect of the RV at the septal wall.

The discrepancy between simulated and benchtop phantom material stiffness may also be indicative of variations in mechanical properties between the heart phantom and tensile testing specimen. Although the heart phantom and tensile testing specimen were developed in parallel from the same batch, their mechanical properties may differ due to different curing conditions. The mechanical properties of Sylgard are sensitive to curing temperature and aging time between manufacture and testing [179]. This effect may be accelerated in thinner samples, which may explain variability in mechanical properties between the tensile testing samples and the heart phantom. To verify ground-truth phantom stiffness, in the future, testing strips could be cut out directly from the heart phantom after MRI and pressure data have been obtained.

In conclusion, we have developed an experimental setup for validating MRI and FEM-based myocardial passive stiffness in 3D-printed heart phantoms. The setup enables acquisition of MRI and pressure data necessary to estimate phantom material properties using an MRI and FEM-based constitutive modeling framework. The phantom development procedure effectively reproduces the subject geometry. The experimental setup will be refined to minimize discrepancy between the phantom mechanical properties and those of the tensile testing strips used to quantify ground-truth stiffness. The phantom stiffness can be tuned, and different subject-specific phantoms can be developed, enabling extensive quantification of the accuracy and repeatability of stiffness estimates obtained through our MRI and FEM-based stiffness estimation framework.

## 5.6 Acknowledgements

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# Chapter 6

# Validating MRI-derived myocardial stiffness estimates using *in vitro* synthetic heart models

## 6.1 Abstract

Impaired cardiac filling in response to increased passive myocardial stiffness contributes to the pathophysiology of heart failure. By leveraging cardiac MRI data and ventricular pressure measurements, we can estimate *in vivo* passive myocardial stiffness using personalized inverse finite element models. While it is well-known that this approach is subject to uncertainties, only few studies quantify the accuracy of these stiffness estimates. This lack of validation is, at least in part, due to the absence of ground truth *in vivo* passive myocardial stiffness values. Here, using 3D printing, we created soft, homogenous, isotropic, hyperelastic heart phantoms of varying geometry and stiffness and simulate diastolic filling by incorporating the phantoms into an MRI-compatible left ventricular inflation system. We estimate phantom stiffness from MRI and pressure data using inverse finite element analyses based on a neo-Hookean model. We demonstrate that our identified softest and stiffest values of 215.7kPa and 512.3kPa agree well with the ground truth of 226.2kPa and 526.4kPa. Overall, our estimated stiffnesses revealed a good agreement with the ground truth ($< 5.8\%$ error) across all models. Our results suggest that MRI-driven computational constitutive

modeling can accurately estimate synthetic heart material stiffnesses in the range of 200-500kPa.

## 6.2 Introduction

Impaired cardiac filling contributes to the pathophysiology of many cardiac diseases [180] which can eventually degenerate into heart failure. This multifaceted clinical syndrome is characterized by the heart's inability to supply adequate blood to meet the metabolic demands of the body [181]. Heart failure is a global public health issue with significant mortality and morbidity [182] and high socioeconomic burden [183]. The high prevalence of heart failure [184] is also projected to continue increasing over the next decade [185].

Diastolic dysfunction, which could be due to abnormal active relaxation [186] and/or diastolic chamber stiffness [187], is observed in nearly half of heart failure patients, who are said to have heart failure with preserved (*i.e.* relatively normal) ejection fraction (HFpEF) [188]. In these patients, though ejection fraction is maintained, multiscale remodeling leads to impaired LV diastolic filling resulting in inadequate cardiac output. Passive stiffness of myocardium, the cardiac muscle tissue, regulates diastolic filling [187] and increased passive myocardial stiffness is a significant remodeling mechanism implicated in the etiology of HFpEF [32]. To improve understanding and management of HFpEF, a clinical method to reliably identify changes in passive myocardial stiffness is a necessity [189]. Moreover, other diseases where passive myocardial stiffness is altered due to pathophysiological remodeling of the cardiomyocytes and extracellular matrix (e.g. hypertrophic cardiomyopathy) can benefit from a method for clinical estimation of passive myocardial stiffness.

Medical imaging-based computational constitutive modeling enables estimation of patient-specific *in vivo* passive myocardial stiffness [34, 175], a biomechanical metric with significant prognostic value for HFpEF patients [190]. Cardiac MRI sequences can be used to acquire data needed for computational modeling: high resolution cardiac geometry, cardiac motion and local displacements, and boundary conditions. While subject-specific cardiac microstructural organization can be obtained using *in vivo* cDTI [157, 191], the method

is still limited by issues such as low spatial resolution and signal-to-noise ratio. Ventricular pressures, which are also necessary for computational modeling can be acquired invasively through catheterization or estimated non-invasively through doppler echocardiography [192, 193].

Numerous studies combine cardiac MRI, ventricular pressures, and inverse finite element modeling to estimate *in vivo* passive myocardial mechanical behavior and stiffness in computational models [144, 194, 195, 3, 9, 161, 196, 197, 158, 142, 146, 168, 160]. Through these patient-specific simulations, certain metrics of cardiac function that are otherwise difficult to measure directly, such as wall stresses and strains, can be estimated. This personalized understanding of cardiac function can reveal disease mechanisms that can be therapeutic targets [131]. However, before these techniques for estimating *in vivo* passive myocardial stiffness can be adopted clinically they need to be thoroughly validated.

Although the patient-specific data needed to estimate *in vivo* passive myocardial stiffness are unavoidably subject to uncertainty [198, 159, 199, 164], we lack detailed studies evaluating the accuracy and precision of MRI-based *in vivo* passive myocardial stiffness estimation. Moreover, the data acquisition and analysis techniques employed for MRI-driven stiffness estimation can vary significantly due to lack of a consensus. There is, however, a dearth of studies on the accuracy and precision of these MRI-based myocardial stiffness estimation methods. This lack of validation is largely because of the absence of ground truth *in vivo* passive stiffness estimates. Overcoming this validation gridlock necessitates the development of a highly controlled *in vitro* stiffness estimation framework that incorporates heart phantoms of known stiffness properties.

We addressed this unique validation challenge by developing soft, MRI-visible heart phantoms of known stiffness. The phantoms were incorporated within an *in vitro* MRI-compatible flow loop to fill the phantom's LV, simulating cardiac diastolic filling. Using MRI-driven computational modeling, the acquired MRI and ventricular pressure data of the phantom during inflation was used to estimate the material stiffness of the phantom's ventricular wall. To quantify the accuracy of our method, the estimated stiffness was then compared to ground truth measures of the phantom material stiffness obtained through tensile testing. We used heart phantoms of varying stiffness to quantify our ability to identify

stiffness changes. Phantoms of different geometries were also used to assess the identifiability of stiffness independent of geometry. In this work, we validate a material stiffness estimation framework in soft, homogenous, isotropic hyperelastic heart-like phantoms of varying stiffness and geometrical complexity. In doing so, we demonstrate the feasibility and intrinsic challenges of using MRI-driven computational constitutive modeling for accurate identification of material stiffness in heart-like phantoms that resemble *in vivo* passive myocardial stiffness.

## 6.3 Materials and Methods

We describe the heart phantom development (section 6.3.1), the ground truth characterization of the phantom material's mechanical properties (section 6.3.2), the *in vitro* ventricular filling setup, acquisition of the heart phantom's unloaded geometry, filling volumes and pressures (section 6.3.3), the data analysis strategy (section 6.3.4), and lastly, the *in silico* calibration of material parameters using the experimentally derived LV pressure-volume relation (section 6.3.5).

### 6.3.1 Phantom Development

Our study used four different heart phantoms. Three of these had the same geometry ($g_i$), but different material stiffness ($s_j$) (referred to as $\boldsymbol{M}_{g1,s1}$ $\boldsymbol{M}_{g1,s2}$ $\boldsymbol{M}_{g1,s3}$), with model $\boldsymbol{M}_{g1,s1}$ being the softest and model $\boldsymbol{M}_{g1,s3}$ being the stiffest. To understand the possible effects of geometry on the accuracy of the stiffness estimates, the final model ($\boldsymbol{M}_{g2,s1}$) was manufactured with material of a similar stiffness to model $\boldsymbol{M}_{g1,s1}$, but with a different geometry.

We generated 3D geometric heart models by segmenting (MITK v2018.04.2) high-resolution T1-weighted images from healthy *ex vivo* porcine subjects (restored to *in vivo* mid-diastasis geometry) [171] (Figure 6.1a). The geometric models were designed to include basal ventricular ports and an apical anchor to facilitate loading and motion stabilization (Fusion 360, Autodesk) (Figure 6.1b) The heart phantoms were designed to primarily be composed of the LV and RV, but some extent of the atrial myocardium was included to ease the addition of the basal ports needed for volume control and loading.

The heart phantoms were manufactured to match, as closely as feasible, both the mechanical and MRI relaxation properties of human ventricular myocardium. Direct 3D printing of suitably soft materials is currently infeasible; therefore, the heart phantoms were cast using a combination of epicardial and endocardial molds. To develop the phantoms, we employed a lost-wax casting technique that has been used to manufacture anatomically detailed heart phantoms [172].

A negative epicardial mold was created using the epicardial surface of the 3D geometric heart models. The ventricular blood pool segmentations were used to create LV and right ventricle (RV) blood pool casts (endocardial molds). The epicardial and endocardial molds were converted into stereolithography files and 3D printed (Ultimaker 3 Extended) using tough polylactic acid and water-soluble polyvinyl alcohol, respectively. The "myocardial" space between the nested molds was then injected with a blend of silicone elastomers (Sylgard 184 and 527, Dow Corning). The models were then left to cure in the 3D printed mold for 48 hours at room temperature (Figure 6.1c), after which the mold parts were removed and the blood pool casts were dissolved in water. Finally, the heart phantoms were fitted with adapter parts (Figure 6.2b) at the basal ventricular ports for attachment of tubing to connect the phantoms to the flow loop.

Using this technique, we produced subject-specific heart phantoms of high geometric fidelity and both passive stiffness and MRI relaxation properties [200] that approximate ventricular myocardium. The stiffness of Sylgard blends can be tuned by varying the mass composition of its constituents [8], therefore we were also able to produce heart phantoms with a range of stiffness. Three silicone elastomer blends containing a mass ratio of Sylgard 184 to Sylgard 527 of 20:80, 25:75, and 30:70 respectively, were used to produce phantom materials with increasing stiffness.

### 6.3.2 Ground Truth Material Characterization

The ground truth stiffness of the silicone elastomer blends used to manufacture the phantoms were assessed via uniaxial tensile testing. Testing samples were produced in parallel with the phantom development and were subjected to identical curing conditions as the associated heart phantoms. Samples of uniform width (12 mm) were cut out from cured

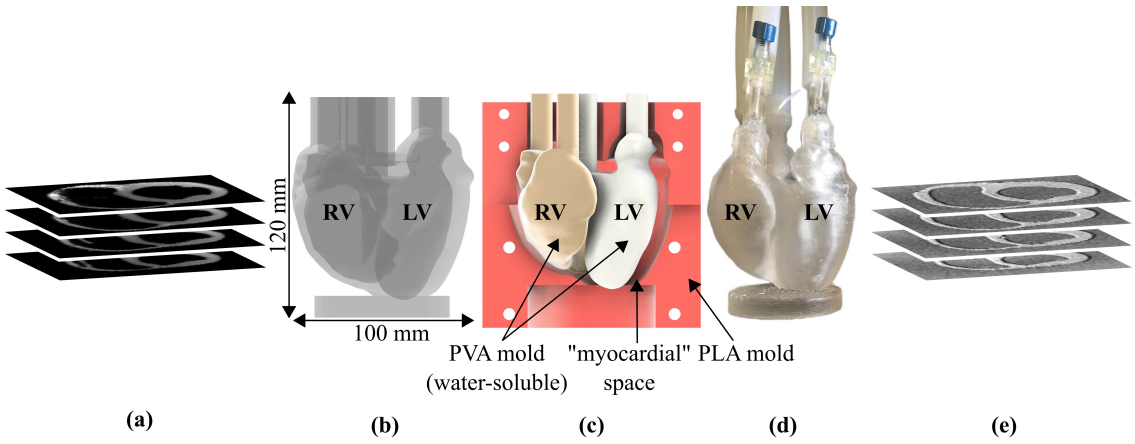


Figure 6.1: Phantom Manufacturing Process: (a) *Ex vivo* porcine T1-weighted images (at mid-diastasis) were segmented to develop the geometric heart model. (b) The model was fitted with basal ports for volume loading and an apical anchor was added to fix the apex during experiments. (c) The phantom was manufactured by curing a silicone elastomer blend in a 3D printed PLA/PVA mold for 48 hours. The LV and RV blood pool casts were printed with water-soluble PVA. (d) The subject-specific phantom was fitted with 3D printed barbed connectors and tubing for loading and pressure measurements. (e) Unloaded 3D SPGR images of the 3D printed phantom shows high geometric agreement with porcine subject images.

sheets of uniform thickness (3mm) and were subsequently mechanically tested according to a modified ASTM D412 standard using a material testing machine (Instron 5848 Microtester, 100N load cell). The samples were mounted by first clamping the specimen to the upper grip, zeroing the load cell, then clamping the specimen to the lower grip while taking care to avoid load application. The grip-to-grip lengths of the mounted samples were used as the gauge lengths since the samples were attached such that the width and thickness in the grip-to-grip region was uniform. We did not directly measure the strains in the samples because clamping an extensometer onto these soft samples was found to appreciably affect the forces measured. Instead, to calculate the principal stretch in each sample, the extension from the tensile tester was used with the gauge length, which was set to 65 mm.

We considered the potential effects of viscoelasticity in our samples by assessing the effects of strain rates on the mechanical behavior and didn't find significant strain rate effects which is consistent which previous work in literature that assessed strain rate effects on the mechanical behavior of Sylgard [179, 174]. Hence, we modeled the phantom materials as effectively hyperelastic. The test was performed at ambient conditions with a strain rate of 10 % per second which was the closest we could get to the maximum strain rate in the *in vitro* phantoms.

For each model, three separate samples were tested, and the averaged stress-stretch data across all three samples was used as the ground truth. All silicone elastomer blends were assumed to be nearly incompressible [174, 201] and were modeled with an incompressible Neo-Hookean material model:

$$\psi = C_1(I_1 - 3) - p(J - 1) \tag{6.1}$$

$C_1$ is the Neo-Hookean material parameter, and $I_1$ is the first invariant of the right Cauchy-Green deformation tensor defined as $\mathbf{C} = \mathbf{F}^T\mathbf{F}$. The Lagrange multiplier ($p$) is introduced to enforce incompressibility. $J = \det(\mathbf{F})$ where $\mathbf{F}$ is the deformation gradient tensor. For incompressible materials, $J = 1$. With the left Cauchy-Green deformation tensor defined as $\mathbf{b} = \mathbf{F}\mathbf{F}^T$, the Cauchy stress tensor reads:

$$\boldsymbol{\sigma} = \frac{2}{J}\boldsymbol{F}\frac{\partial \psi}{\partial \boldsymbol{C}}\boldsymbol{F}^T = 2C_1\boldsymbol{b} - p \tag{6.2}$$

For uniaxial extension of an incompressible material with stretch ratio $\lambda_1 = \lambda$, it follows that the principal stress in the stretch direction is:

$$\sigma_{11} = 2C_1 \left( \lambda^2 - \frac{1}{\lambda} \right) \tag{6.3}$$

The phantom material stiffness can be represented as the tangent modulus, which is the derivative of the principal Cauchy stress with respect to the stretch.

The Neo-Hookean material parameter, $C_1$, was estimated using nonlinear regression of the experimentally measured principal Cauchy stress versus stretch data and used to define the ground truth material stiffness.

We calculated the coefficient of variation (standard deviation/mean $\times$ 100%) in principal Cauchy stress to quantify the variation across the three samples tested for each model. We also calculated the coefficient of determination ($R^2$) between the Neo-Hookean model predicted principal Cauchy stress and the experimentally determined principal Cauchy stress to quantitatively assess the fit of the material model to the experimental data.

### 6.3.3 Imaging Experiments

We developed an MRI-compatible flow loop (Figure 6.2) to simulate diastolic filling of the heart phantoms' LV, to enable acquisition of each phantom's filling kinematics using MRI, and to allow for measurement of ventricular filling pressures using pressure transducers. The phantoms were connected to the flow loop at the basal ports using custom-built, 3D printed barbed connectors designed with inlets for inserting pressure transducers(Figure 6.2b). All models were fixed at the apical anchor and placed in a container filled with blood-mimicking fluid of similar MRI relaxation properties, density, and viscosity to blood (40% glycerol, 60% water, 0.75 mL/L ferumoxytol). The phantom box, which was adapted with flow ports to attach tubing was subsequently connected to an MRI-compatible, programmable flow pump (CardioFlow 5000 MR, Shelley Medical Imaging Technologies). The same blood-mimicking fluid was used to inflate the LV and fill the RV of the phantoms. A similar flow circuit was used in our lab to assess flow dynamics in synthetic aorta models using MRI [202].

The flow pump was programmed to deliver square flow waveforms (peak flow rate =

300 ml/s) to inflate each heart phantom's LV. The phantoms' RV was fluid filled at the start of the experiment and sealed such that its volume remained constant throughout the experiment. Filling pressures were acquired in the MRI scanner room during volumetric loading, immediately prior to scanning because the pressure transducers, though MRI safe, can induce image artifacts if used during imaging and imaging can produce pressure signal artifacts [203]. During the inflation experiment, both intraventricular pressures were simultaneously acquired (PowerLab, LabChart 8, ADInstruments) using MRI-compatible pressure transducers (Micro-Tip SPR 350S, Millar). Pressures for ten loading cycles were acquired, with the averaged pressure cycle used as the loading pressure. The coefficient of variation in the pressure waveform across the ten measured cycles was calculated for each experiment and was used to define the uncertainty in the pressure measurements.

All *in vitro* images were collected on a 3T MRI (Skyra, Siemens) using a 32-channel spine and 18-channel body matrix coil. The reference (unloaded) geometry of the phantom was acquired using a 3D spoiled gradient echo (3D SPGR) spanning the entire volume. Cine gradient echo (GRE) images were acquired during LV inflation to capture phantom motion and to obtain filling volumes. Imaging parameters are detailed in Table 6.1. Ventricular filling pressures were synced with the image acquisition using a trigger.

Table 6.1: Imaging parameters for 3D SPGR and GRE Cine

| | 3D SPGR | GRE Cine |
|---|---|---|
| TE/TR | $2.17/5.5ms$ | $3.06/19.89ms$ |
| FA | $20°$ | $12°$ |
| Spatial Resolution | $1 \times 1 \times 1mm^3$ | $1 \times 1 \times 4mm^3$ |
| Number of Slices/Phases | $144/NA$ | $30/75$ |
| Matrix | $220 \times 320$ | $128 \times 128$ |

### 6.3.4 Data Analysis

The myocardium was semi-automatically segmented from the 3D SPGR images of the unloaded phantom. First, a morphological segmentation [204, 205] was performed in ImageJ (v1.53q), then the final segmentation was produced after a few morphological operations

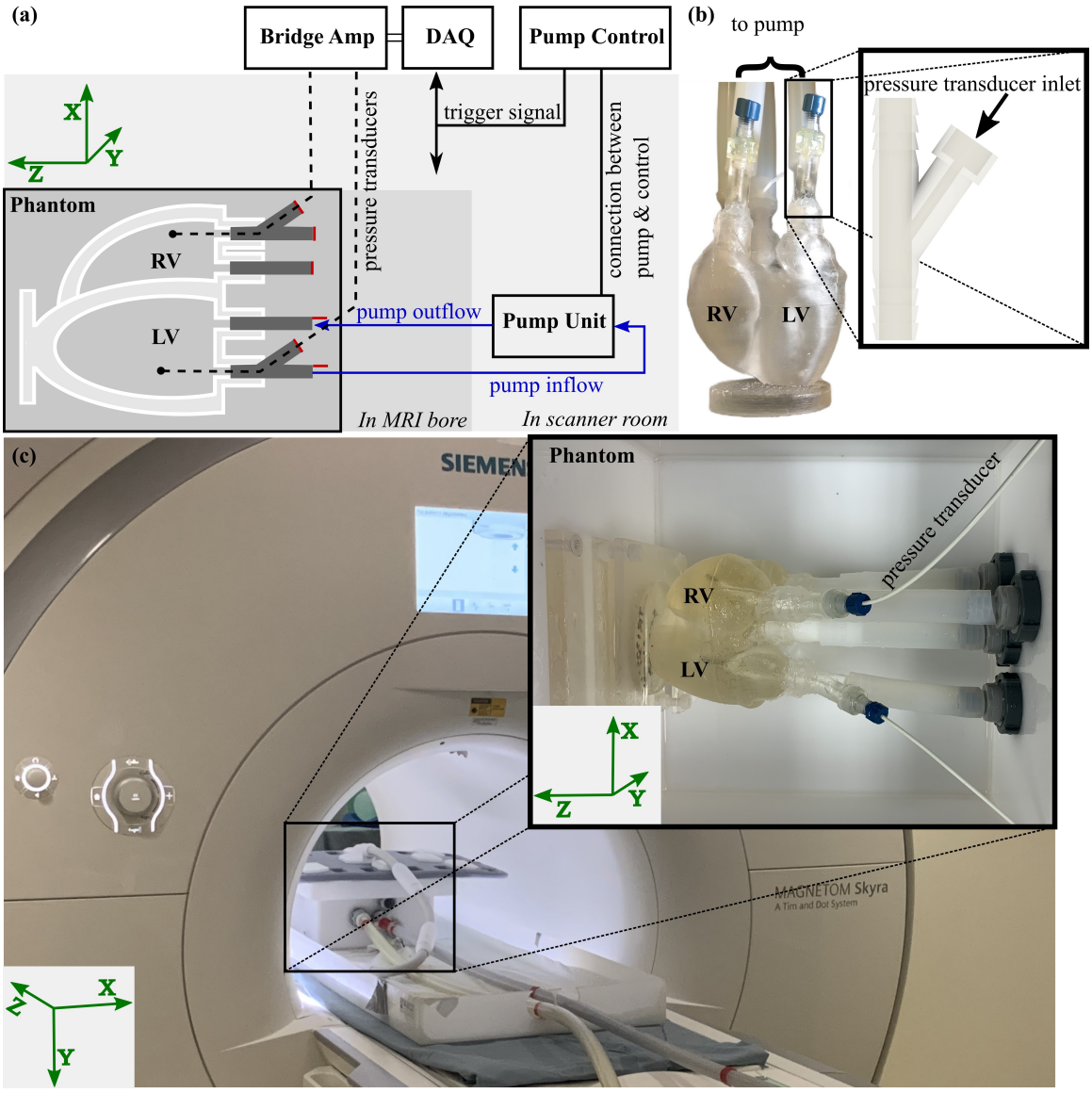


Figure 6.2: Experimental Setup Schematic. (a) The heart phantom is connected to an MRI-compatible flow loop and fixed at the apex and basal ports. The programmable flow pump is used to inflate the phantom's left ventricle (LV), while the right ventricle (RV) is kept at constant volume. Pressure transducers were used to record the intraventricular filling pressures simultaneously. (b) The heart phantom was adapted with tubing for connection to the flow loop. The custom-built barbed adapter with inlets for the pressure transducer is highlighted. (c) Phantom inflation setup inside scanner. Inside of the phantom box is highlighted.

and some manual cleanup. The segmentation masks were then used to generate the reference geometric models that were used for *in silico* modeling (Figure 6.3a-c).

The LV volumes during inflation were determined by semi-automatically segmenting the LV blood pool from the GRE cine images of the phantom during inflation using Otsu's method [206] and manual cleanup (ImageJ). For each model, the segmentation was performed on the volumetric cine short axis image stack at different phases, then the LV volume at each inflation phase was extracted to determine the inflation volume over time (Figure 6.3d-f).

The synced measured pressures and filling volumes were used to obtain an LV pressure-volume relationship that was used for stiffness estimation.

### 6.3.5 *In silico* modeling and stiffness quantification

First, the geometric heart phantom models in the reference configuration (see section 6.3.4) were modified by virtually closing off the top surface of the ventricular basal ports. This was done to create enclosed cavities for efficient computation of the ventricular volume evolution. The 3D geometric models were meshed with standard quadratic tetrahedral elements using GMSH [207] (Figure 6.3c). Following a mesh sensitivity study, final mesh characteristics for each model are detailed in Table 8.2. The finite element model was then built to be solved using FEBio [120].

Based on our tensile testing experiments (see section 6.3.2), it was shown that the silicone elastomer phantom materials can be adequately modeled using the Neo-Hookean strain energy function. Additionally, we assumed our simulated material to be nearly incompressible following prior mechanical characterization studies on Sylgard 527 and 184 [174, 201]. We enforced incompressibility in the simulation by setting the bulk modulus to be 100 times the Neo-Hookean material parameter. Across all the models, incompressibility was well enforced, with most elements having a relative element volume at peak inflation of 1.000 and the worst elements deviating from this peak inflation volume by a maximum of 0.003. The boundary and loading conditions of the computational heart phantoms were set to match the experimental conditions. The apical anchor was kinematically constrained in all directions, while the top surfaces of the ventricular basal ports were kinematically constrained from

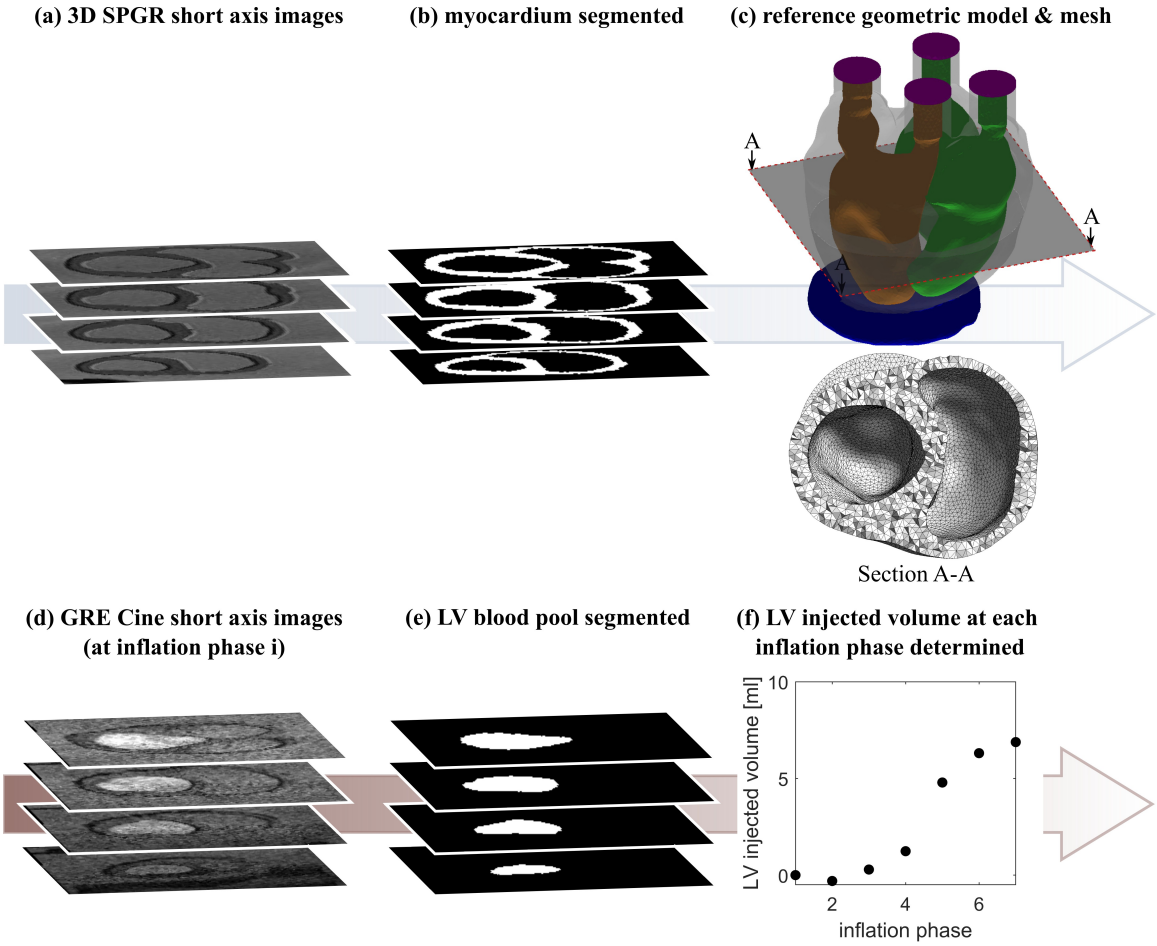


Figure 6.3: (a) 3D SPGR short-axis images. (b) Myocardium segmentation is used to develop (c) the reference geometric models that are meshed with quadratic tetrahedral elements. In the finite element model, the LV endocardial surface was inflated with measured LV pressure (orange), the RV endocardial surface was inflated with measured RV pressure (green), the top surface of the basal ports were fixed in the longitudinal direction (purple), and the apical anchor was fixed in all directions (blue). (d) GRE Cine short axis image stack at an inflation phase. (e) The LV blood pool from each inflation phase of the GRE cine images is segmented. (f) The volume at each inflation phase is extracted from the segmentation and is used to determine the LV injected volume at each cardiac phase.

moving in the longitudinal direction. The recorded LV and RV intraventricular pressures were measured simultaneously during the imaging experiment. During simulation, these pressures were applied to the LV and RV endocardial surfaces and the resulting phantom deformations were computed.

We set up a computational optimization scheme to deduce the phantom material stiffness from the MRI-derived LV volumes and catheter-based pressure measurements. More specifically, with FEBio as the forward solver, we used the constrained Levenberg-Marquardt optimization algorithm [208](Figure 6.4) to find the appropriate constitutive parameter ($C_1$) that minimized the error between the simulated and experimental LV volume evolution. The material parameter optimization algorithm minimizes an objective function ($\varphi$), which is the sum of the squared difference between the experimental and simulated LV volume at each inflation phase. $C_1$ is iteratively updated to minimize $\varphi$ until the reduction in $\varphi$ falls below a specified tolerance ($1e^{-3}$ kPa).

For all models, the initial guess was set as $C_1 = 25$ kPa. We placed constraints on the possible values of $C_1$ to ensure it remained within reasonable limits (1 kPa $< C_1 <$ 100 kPa). Inverse parameter estimates using the upper and lower bounds of the measured pressure uncertainty were also performed to account for the uncertainty in material parameters due to pressure measurement.

The mesh size chosen for the inverse simulation was determined from a mesh convergence study and was found to lead to a variance of $< 1\%$ LV peak inflation volume with further mesh refinement, an accuracy that was deemed appropriate for our study.

Table 6.2: Mesh Properties

| | $M_{g1,s1}$ | $M_{g1,s2}$ | $M_{g1,s3}$ | $M_{g2,s1}$ |
|---|---|---|---|---|
| Number of Nodes | 221714 | 227874 | 228105 | 243540 |
| Number of Elements | 134309 | 138415 | 138446 | 150475 |
| Degrees of Freedom | 402927 | 415245 | 415338 | 451425 |
| Average Element Volume ($mm^3$) | 1.89 | 1.90 | 1.86 | 1.94 |

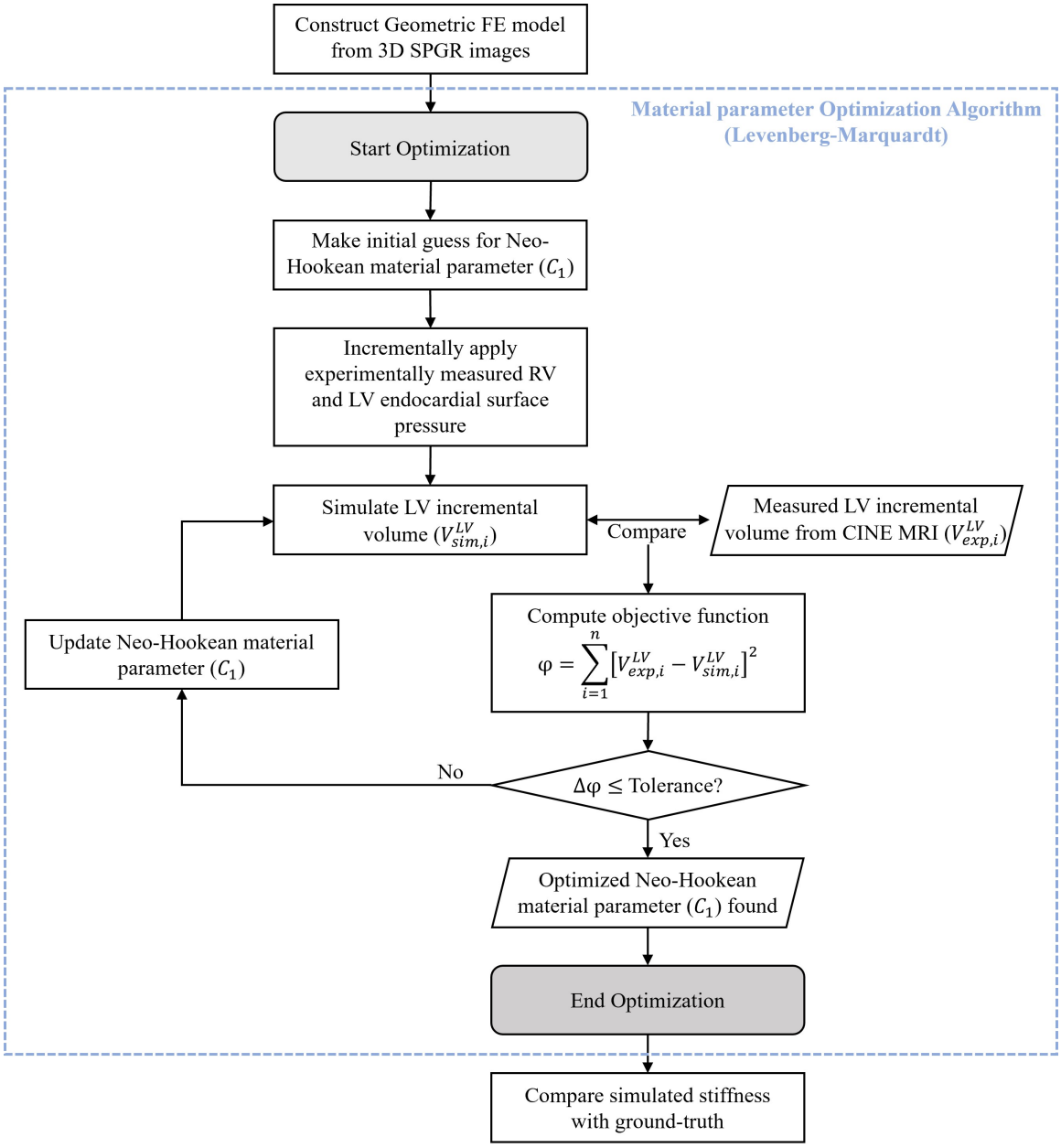


Figure 6.4: Material parameter optimization flowchart. The goal is to minimize the objective function ($\varphi$), which is the sum of the squared difference between the experimental LV volume ($V_{exp}^{LV}$) and simulated LV volume ($V_{sim}^{LV}$) at each inflation phase ($i$). The parameter optimization algorithm iteratively updates the $C_1$ Neo-Hookean material parameter to minimize $\varphi$ until the reduction in $\varphi$ ($\Delta\varphi$) falls below a specified tolerance which was set to $1e^{-3}$ kPa.

## 6.4 Results

### 6.4.1 Ground Truth Material Characterization

Principal Cauchy stress versus stretch for representative samples of the silicone elastomer blend used to cast models $M_{g1,s1}$, $M_{g1,s2}$, $M_{g1,s3}$, and $M_{g2,s1}$ are shown in Figure 6.5. The uncertainty arises from the standard deviation of the three samples subjected to uniaxial tensile testing.

A Neo-Hookean material model was fit to the average (from three tests) stress-stretch data to obtain the ground truth material parameters. Ground truth stiffness at the peak simulated principal stretch in the *in silico* phantoms (stretch, $\lambda = \lambda_{peak}$) is reported for each model (Table 6.3). The coefficient of variation of the experimentally determined Cauchy stress at $\lambda_{peak}$ across the three samples was 10.4%, 4.3%, 2.5% and 2.8% for models $M_{g1,s1}$, $M_{g1,s2}$, $M_{g1,s3}$, and $M_{g2,s1}$, respectively. The $R^2$ between the Neo-Hookean model predicted principal Cauchy stress and the experimentally determined principal Cauchy stress was over 0.99 for all models tested (Figure 6.5).

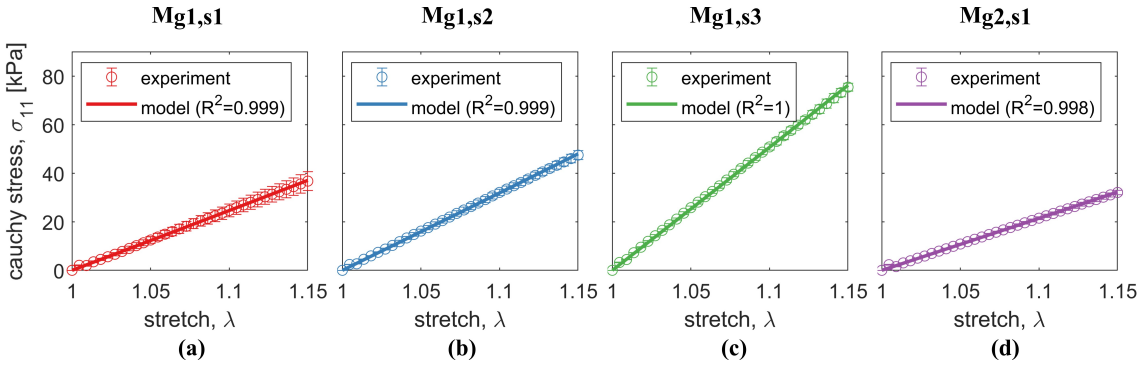


Figure 6.5: Cauchy stress ($\sigma_{11}$) versus stretch ($\lambda$) from the ground truth uniaxial tensile tests for (a) model $M_{g1,s1}$ (red) (b) model $M_{g1,s2}$ (blue) (c) model $M_{g1,s3}$ (green) (d) model $M_{g2,s1}$ (purple). The standard deviation arises from tests across three samples.

### 6.4.2 Inflation pressure-volume relations

The measured LV inflation pressure and the LV inflation volumes obtained from segmenting the cine MRI images are shown in Figure 6.6. The experimentally measured LV pressure

versus LV injected volume (inflation volume – initial volume) for all the models is shown in Figure 6.6.

The peak LV inflation pressures were 31.6 mmHg, 33.2 mmHg, 78.7 mmHg, and 24.1 mmHg and the peak LV injected volumes were 6.3 mL, 5.9 mL, 8.5 mL, and 8.1 mL for models $M_{g1,s1}$, $M_{g1,s2}$, $M_{g1,s3}$, and $M_{g2,s1}$, respectively. The peak RV measured pressures were 2.1 mmHg, 1.3 mmHg, 1.5 mmHg, and 0.9 mmHg for models $M_{g1,s1}$, $M_{g1,s2}$, $M_{g1,s3}$, and $M_{g2,s1}$, respectively.

### 6.4.3 Simulated Mechanical Behavior

The computed phantom deformation and the variation in displacement magnitude at maximum inflation is shown for all four models (Figure 6.7). $\lambda_{peak}$ (peak simulated principal stretch) in the phantoms were 1.11, 1.09, 1.13, and 1.10 for models $M_{g1,s1}$, $M_{g1,s2}$, $M_{g1,s3}$, and $M_{g2,s1}$, respectively.

Figure 6.6 (fourth row) shows that the calibrated LV pressure-volume (dotted line) corresponds well with the experimentally determined LV pressure-volume (points). Specifically, the calibrated LV pressure-volume is from a forward simulation using the optimized material parameters. The uncertain region around this simulated LV pressure-volume curve corresponds to the forward simulations using the optimized material parameters calibrated using the upper and lower uncertainty limits of the experimentally derived LV pressure-volume relationship.

The principal Cauchy stress versus stretch from a simulated tensile test using the optimized (simulated) material parameters is shown for all models (dotted line) (Figure 6.8) and compared with the ground truth mechanical behavior in uniaxial tension (solid line). From Figure 6.8, we see that there is good agreement in the simulated and ground truth mechanical behavior for all models with the percent difference in Cauchy stress at $\lambda_{peak}$ being 5.2%, -1.0%, 3.3%, and 4.7% for models $M_{g1,s1}$, $M_{g1,s2}$, $M_{g1,s31}$, and $M_{g2,s1}$, respectively. Table 6.3 reports the ground truth and simulated stiffness (tangent modulus) at $\lambda_{peak}$ for all models. The percentage difference between the simulated and ground truth stiffness is $<$ 6% for all the models.

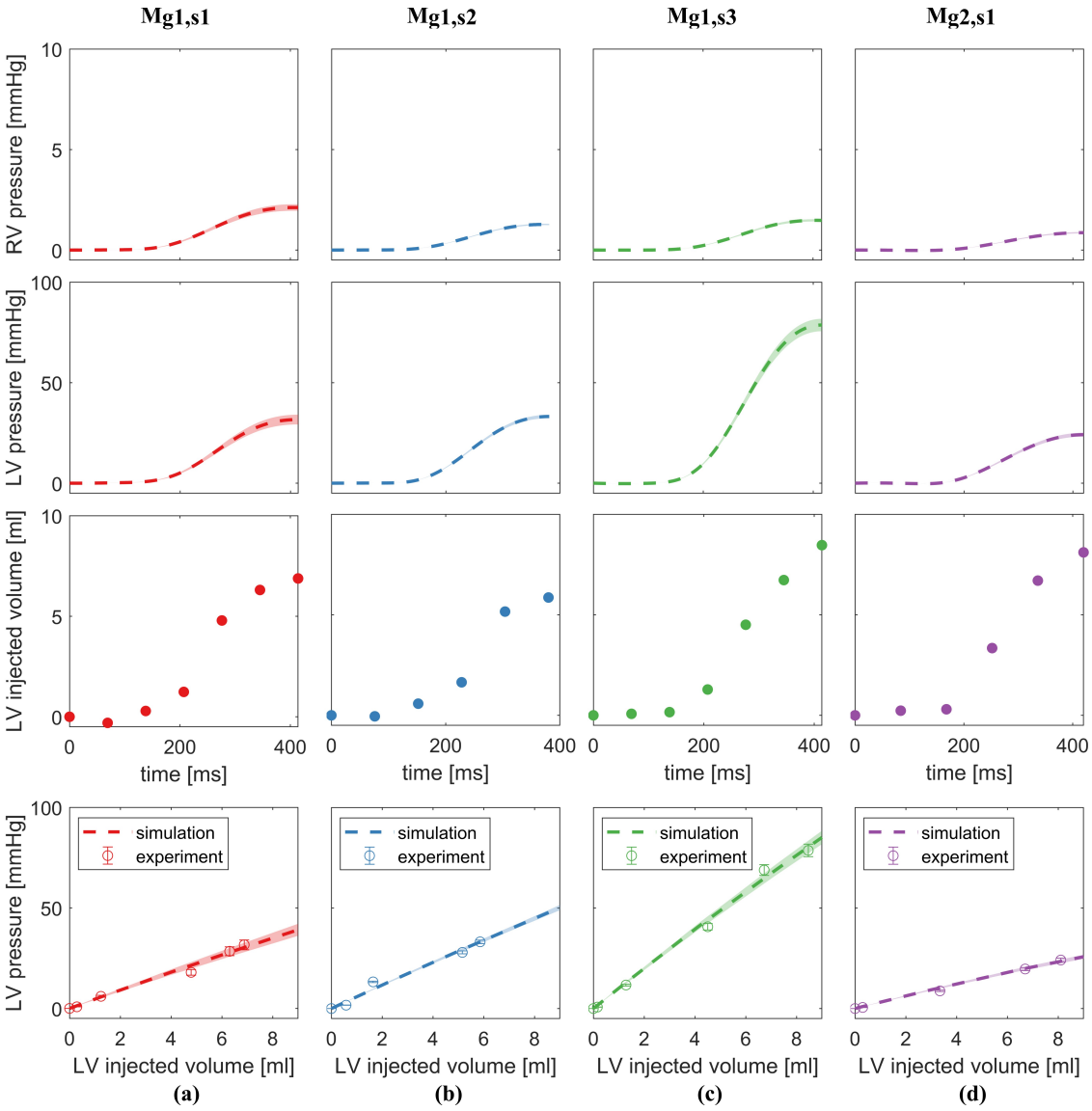


Figure 6.6: Right ventricular (RV) pressure versus time (first row), left ventricular (LV) pressure vs time (second row), LV volume over time (third row), and experimentally measured LV pressure-volume overlayed with calibrated (optimized) LV pressure-volume (fourth row) for (a) model $\boldsymbol{M}_{g1,s1}$ (red) (b) model $\boldsymbol{M}_{g1,s2}$ (blue) (c) model $\boldsymbol{M}_{g1,s3}$ (green) (d) model $\boldsymbol{M}_{g2,s1}$ (purple). Uncertainty arises from the maximum coefficient of variation in measured pressures across all experiments.

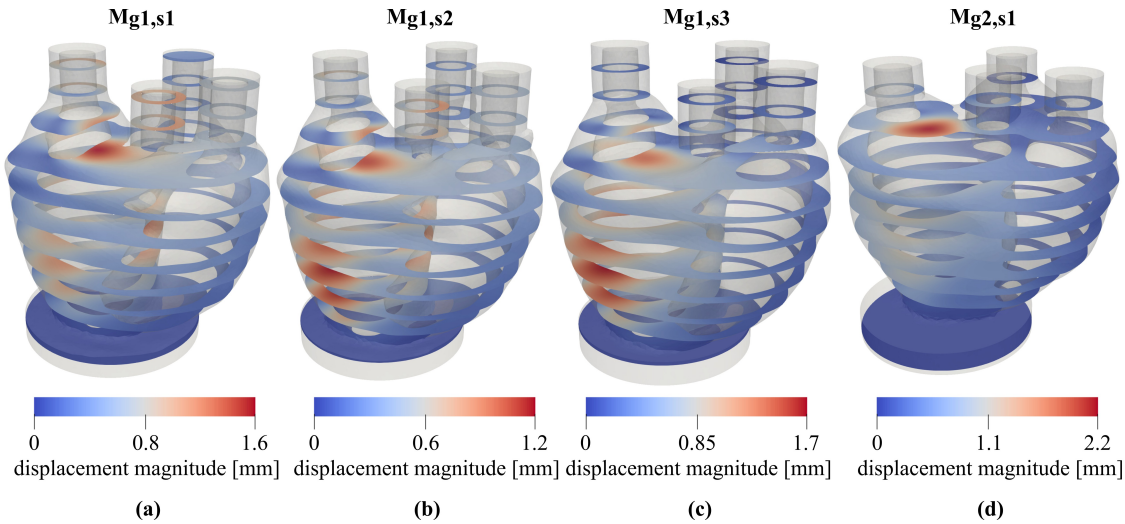


Figure 6.7: Computed phantom deformation at maximum inflation. Short-axis slices show the spatial variation in displacement magnitude for (a) model $M_{g1,s1}$, (b) model $M_{g1,s2}$, (c) model $M_{g1,s31}$, and (d) model $M_{g2,s1}$.

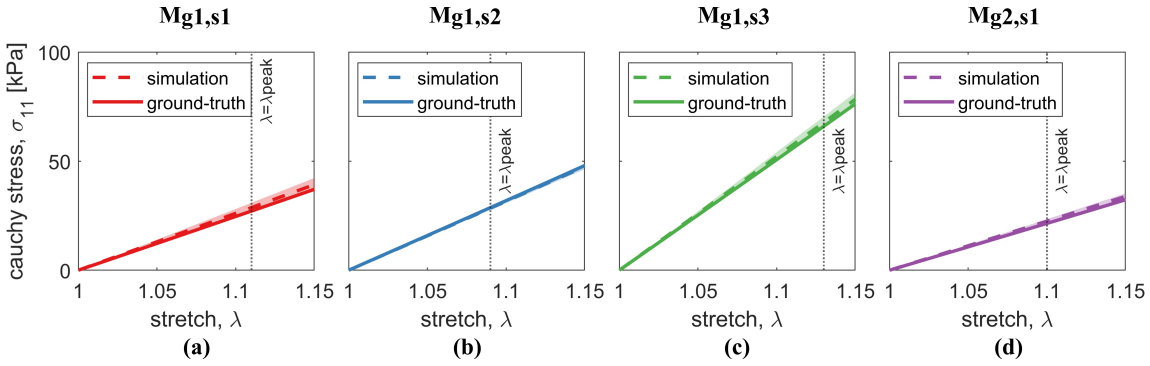


Figure 6.8: Cauchy stress ($\sigma_{11}$) versus stretch ($\lambda$) from uniaxial tensile tests using simulated (optimized) material parameter (dotted lines) and ground truth material parameter obtained from tensile testing (solid lines) for (a) model $M_{g1,s1}$ (red), (b) model $M_{g1,s2}$ (blue), (c) model $M_{g1,s3}$ (green), and (d) model $M_{g2,s1}$ (purple). Uncertainty arises from the maximum coefficient of variation in the measured pressures across all experiments. The peak simulated principal stretch ($\lambda_{peak}$) in the models is indicated.

Table 6.3: Simulated versus Ground Truth Mechanical Behavior

| | | $C_1$ [kPa] | Stiffness [kPa] ($\lambda = \lambda_{peak}$) | $\sigma_{11}$ [kPa] ($\lambda = \lambda_{peak}$) |
|---|---|---|---|---|
| | Ground truth | $40.9 \pm 4.3$ | $248.2 \pm 26.2$ | $27.3 \pm 2.8$ |
| $M_{g1,s1}$ | Simulation | $43.3 \pm 3.3$ | $262.7 \pm 20.1$ | $28.7 \pm 2.2$ |
| | % difference | 5.9% | 5.8% | 5.2% |
| | Ground truth | $53.0 \pm 2.1$ | $320.2 \pm 12.6$ | $28.6 \pm 1.2$ |
| $M_{g1,s2}$ | Simulation | $52.4 \pm 4.0$ | $316.9 \pm 24.2$ | $28.4 \pm 2.2$ |
| | % difference | -1.0% | -1.0% | -1.0% |
| | Ground truth | $84.2 \pm 2.1$ | $512.3 \pm 12.7$ | $65.6 \pm 1.7$ |
| $M_{g1,s3}$ | Simulation | $86.5 \pm 6.6$ | $526.4 \pm 40.2$ | $67.8 \pm 5.2$ |
| | % difference | 2.7% | 2.7% | 3.3% |
| | Ground truth | $35.6 \pm 1.0$ | $215.7 \pm 6.0$ | $21.5 \pm 0.6$ |
| $M_{g2,s1}$ | Simulation | $37.4 \pm 2.9$ | $226.2 \pm 17.3$ | $22.5 \pm 1.7$ |
| | % difference | 4.9% | 4.9% | 4.7% |

## 6.5 Discussion

### 6.5.1 Contributions

Increased passive myocardial stiffness is a cardiac remodeling mechanism associated with a significant structural change underlying the progression of HFpEF [187, 32]. There exist several studies leveraging cardiac MRI data and ventricular pressure measurements for the estimation of patient-specific passive myocardial stiffness. However, most of these studies have a common limitation - the estimated passive myocardial stiffness estimates were not comprehensively validated. This lack of validation is due, in part, to the lack of ground truth *in vivo* passive myocardial stiffness estimates available for comparison. Nevertheless, before computational models can be adopted as clinical prognostic tools, there is a need for thorough validation.

We quantitatively assessed the accuracy of MRI-driven computational modeling for myocardial stiffness estimation in four soft silicone phantoms with realistic cardiac morphometry and known ground truth mechanical behavior and stiffness. In models $M_{g1,s1}$, $M_{g1,s2}$, $M_{g1,s31}$, and $M_{g2,s1}$, the stiffness estimation errors were 5.8%, -1.0%, 2.7%, and 4.9%. These

low errors demonstrate the feasibility of highly accurate passive myocardial material stiffness estimation using an MRI-driven computational constitutive modeling framework. Additionally, we developed a phantom manufacturing procedure that can reproduce phantoms of high geometric fidelity, myocardium mimicking stiffness, and heart-like MRI relaxation properties [200]. For all models, the estimated ground truth material stiffnesses were consistent with previous work evaluating the stiffness of Sylgard blends[8] (see supplementary Figure 6.9). Additionally, across all samples subjected to uniaxial tensile testing, we were able to reproduce nearly identical principal stress vs stretch relationships and reaction force vs displacement curves in simulated tensile tests (see supplementary Figures 6.10 and 6.11). We can also tune the material stiffness of the phantoms by manufacturing them with different silicone elastomer mass compositions (see section 6.3.1). We incorporated the heart phantoms within an MRI-compatible LV diastolic filling setup and acquired the MRI and pressure data necessary for estimating the phantoms' LV myocardial stiffness using a finite element model. Through finite element modeling and material parameter optimization, we obtained the mechanical (Neo-Hookean) behavior and stiffness of the phantoms and compared these simulated mechanical metrics with ground truth values obtained through tensile testing.

Augenstein *et al*. [154] developed an experimental method for validating passive material properties of a cylindrical silicone gel phantom obtained through MRI-based computational constitutive modeling. Their simulated constitutive parameters were compared to parameters obtained independently from rotational shear testing and showed very good agreement. We extend their study by using geometrically accurate biventricular heart phantoms for validation. Additionally, we perform the validation in heart phantoms of varying stiffnesses and geometries to understand the identifiability across a range of stiffnesses independent of geometry. Our work also differs, in that Augenstein *et al*. used MRI tagging obtained displacements in their objective function, whereas we used the LV pressure-volume relationship for calibrating our computational model for material parameter optimization. Material parameter optimization using local displacements enables the construction of a richer objective function that incorporates more of the deformation kinematics, which better constrains the solution. However, MRI data for estimating local displacements (e.g. from MRI tagging or cine DENSE) is not always available in subjects or in experiments. While

we used only the inflation pressure-volume relationship to calibrate the computational models, we compared the simulated end-inflation shapes with the image-obtained end-inflation shapes and found good agreement across all models (Supplementary Figure 6.12)

Zhu *et al*. [209] used an anatomically accurate biventricular heart phantom to validate the displacements obtained from a MRI-based computational phantom. They compared displacements from a finite element forward simulation with a ground truth obtained from marker tracking using cameras. Our study extends theirs by using more physiologically relevant loading conditions in that we applied pressures on the endocardial surface, whereas the phantom used in their study applies displacement boundary conditions on the apex to control the phantom motion. In addition, their work was focused on validating simulated displacements as opposed to material parameters or material stiffness.

### 6.5.2 Deviations of *in vitro* system from *in vivo* heart

Our phantom materials are a practical approximation of myocardium, but lack some key biomechanical features [4]. Notably, unlike myocardium, which is anisotropic due to the underlying microstructure, our material is isotropic. Additionally, the phantom material's nonlinearity in the relevant stretch range differs from that which is exhibited by myocardium. Specifically, unlike myocardium which exhibits an exponential-like stress-strain relationship in uniaxial tension, our material more closely resembles a linearly elastic material in the stretch range considered.

Our phantom materials are also stiffer than healthy human ventricular myocardium. We sought to manufacture the softest heart phantom with a material of similar stiffness to healthy human myocardium. Emig *et al*. [177] compiled myocardial stiffness values from a range of studies and showed that the passive (diastolic) tangent stiffness of human adult hearts ranged from 8 to 70 kPa. However, caution should be taken when interpreting the stiffness values due to the wide range of estimation techniques used, myocardium constitutive laws employed, and the lack of standardization of the stretch at which the stiffnesses were estimated and reported across all studies. Nevertheless, the softest phantom we could produce demonstrated a stiffness of 215.7 kPa, which is significantly stiffer than

healthy human ventricular myocardium. Our phantom manufacturing procedure can reproduce phantoms with stiffnesses closer to that of human myocardium ($\lessapprox$ 100 kPa), but the far inferior workability of the softer phantoms meant our choice of material was limited to phantoms with stiffness $\gtrapprox$ 200 kPa. Wang *et al*. [9] and Zile *et al*. [33] showed that in HFpEF patients, there can be myocardial stiffness increases of two to three times that of a control group. Given the range of passive myocardial stiffness of human adults (8 to 70 kPa [177]), model $M_{g2,s1}$, the softest phantom (stiffness 215.7 kPa) has comparable stiffness to the passive myocardial stiffness expected in HFpEF. The stiffest heart phantom (model $M_{g1,s3}$) had a stiffness of 512.3 kPa, which is closer to the stiffness of myocardial infarcts [210].

The boundary conditions of the *in vitro* phantoms were modeled appropriately in the *in silico* phantoms, but some of the applied constraints differ from those which are seen in the *in vivo* heart. For example, the basal ports were fixed longitudinally whereas *in vivo* we expect some valve plane motion. Secondly, in all the models, the apex was kinematically constrained in all directions unlike in the *in vivo* heart, where the apex freely rotates. Additionally, as the RV volume was kept constant due to the lack of a second MRI-compatible pump to simultaneously inflate the phantom's RV, the septal mechanics are different from that which is expected *in vivo*.

Some additional complexities that exist in the *in vivo* beating heart could not be modeled by our *in vitro* system. For example, our system doesn't include pericardial interactions, which have been shown to influence myocardial mechanics [211, 160]. Moreover, through our manufacturing procedure, we are unable to replicate the prestresses which *in vivo* myocardium is known to exhibit [137]. Inclusion of the influence of these myocardial prestresses which are thought to facilitate diastolic filling [138] and homogenize the ventricular wall stresses [137] can significantly impact passive myocardial stiffness estimation [138, 139, 140]. Due to these prestresses, the geometry obtained from medical images don't represent an actual unstressed reference configuration for finite element analysis even though the images may represent an externally unloaded state. Though there exist inverse methods for identifying the truly unstressed configuration [141], full validation of these computational inverse prestress estimation procedures is still lacking.

Our heart phantoms do not replicate exactly, the *in vivo* heart. Nevertheless, we demonstrate our ability to use an MRI-driven computational constitutive modeling framework to

identify the material parameters of soft heart-like phantoms of varying stiffness and different complex geometries. This is an important demonstration of feasibility before future *in vivo* passive myocardial stiffness estimation.

### 6.5.3 Computational Modeling Framework

Although there are many permutations of methods to acquire and analyze the input data needed for myocardial stiffness estimation, for our study, we considered only one MRI-based framework for myocardial stiffness estimation. Other methods may introduce uncertainties not considered in this study. Additionally, we acquired relatively high-resolution imaging data which may not be feasible in clinical settings. Future work could consider the effects of downsampling the data to more clinically relevant resolutions and assessing the effects of data resolution on the stiffness estimation.

### 6.5.4 Inflation Volumes

Our maximum inflation volumes in the phantoms were 6.3 mL, 5.9 mL, 8.5 mL, and 8.1 mL for models $M_{g1,s1}$, $M_{g1,s2}$, $M_{g1,s3}$, and $M_{g2,s1}$, respectively. We are interested in modeling the ventricular passive filling which results from the atrial kick, responsible for 20% [212] of total LV filling during diastole. The normal stroke volume of healthy humans is 50 to 100 mL [213], therefore to accurately model passive filling, our injection volumes should be between 10 and 20 mL. We are only modeling a part of the diastolic filling volume differences, due to the mechanical limitations of our inflation system. However, despite this fact, we show that even at these lower kinematic differences, we can already accurately estimate the stiffness of the heart phantoms at physiological stretch conditions.

### 6.5.5 Conclusion

We quantitatively evaluated the accuracy of MRI-driven computational modeling to estimate myocardial stiffness using personalized soft heart phantoms of varying stiffness and geometry. Our study obtained excellent agreement with an independently obtained ground truth stiffness and demonstrates the utility of MRI-driven constitutive modeling to estimate

myocardial stiffness in the beating heart, non-invasively and *in vivo*.

## 6.6 Acknowledgements

F.K. receives research support from the Stanford Bio-X Stanford Interdisciplinary Graduate Fellowship. This project was funded, in part, by NIH R01 HL131823 to D.B.E.

## 6.7 Declarations

### 6.7.1 Conflicts of interest

No benefits in any form have been or will be received from a commercial party related directly or indirectly to the subject of this manuscript.

## 6.8 Supplementary Information

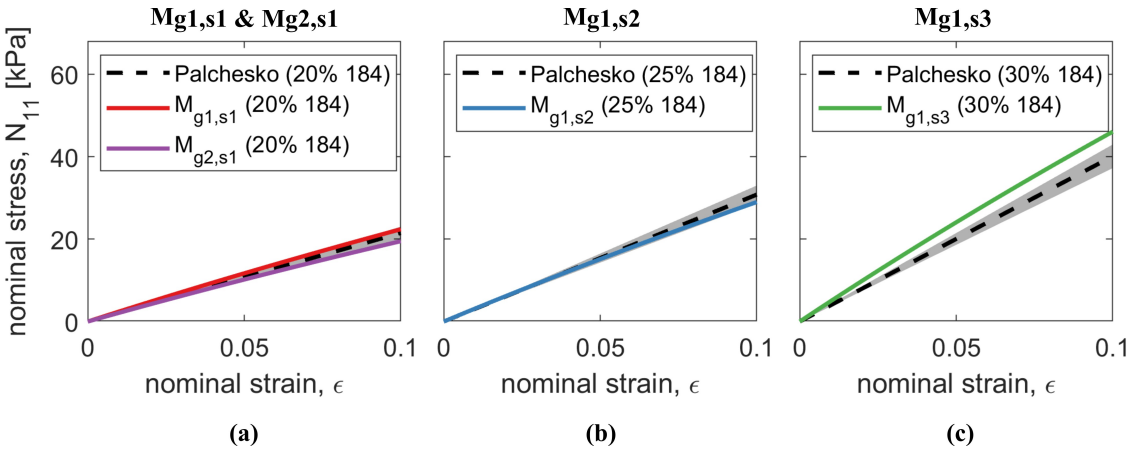


Figure 6.9: Comparison between measured ground truth mechanical behavior and mechanical behavior of sylgard blends predicted by Palchesko *et al.* [8]

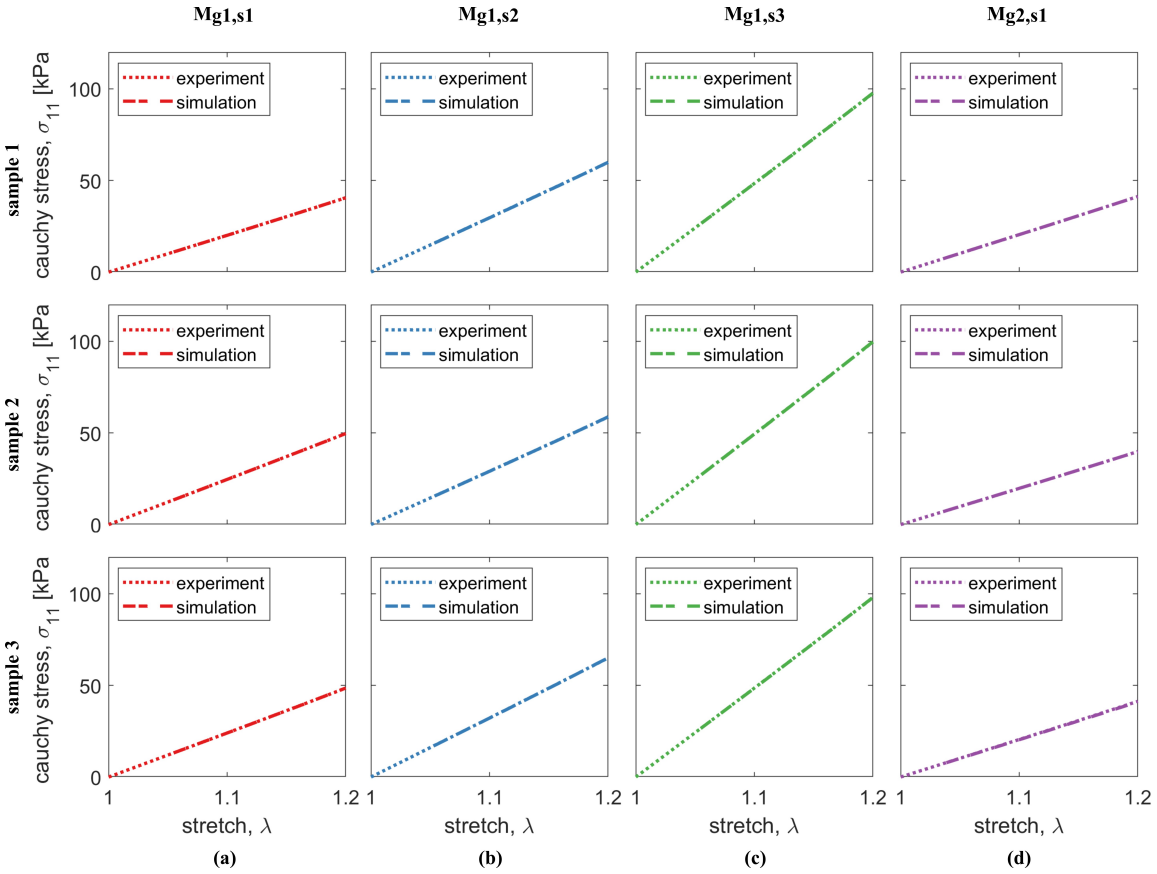


Figure 6.10: FEM-based error analysis on uniaxial tensile testing specimen used to evaluate ground truth mechanical behavior of the models. The tensile test was simulated using the experimentally determined loading conditions and material parameters and a comparison between the simulated and ground truth principal Cauchy stress vs stretch is shown for all models and samples tested

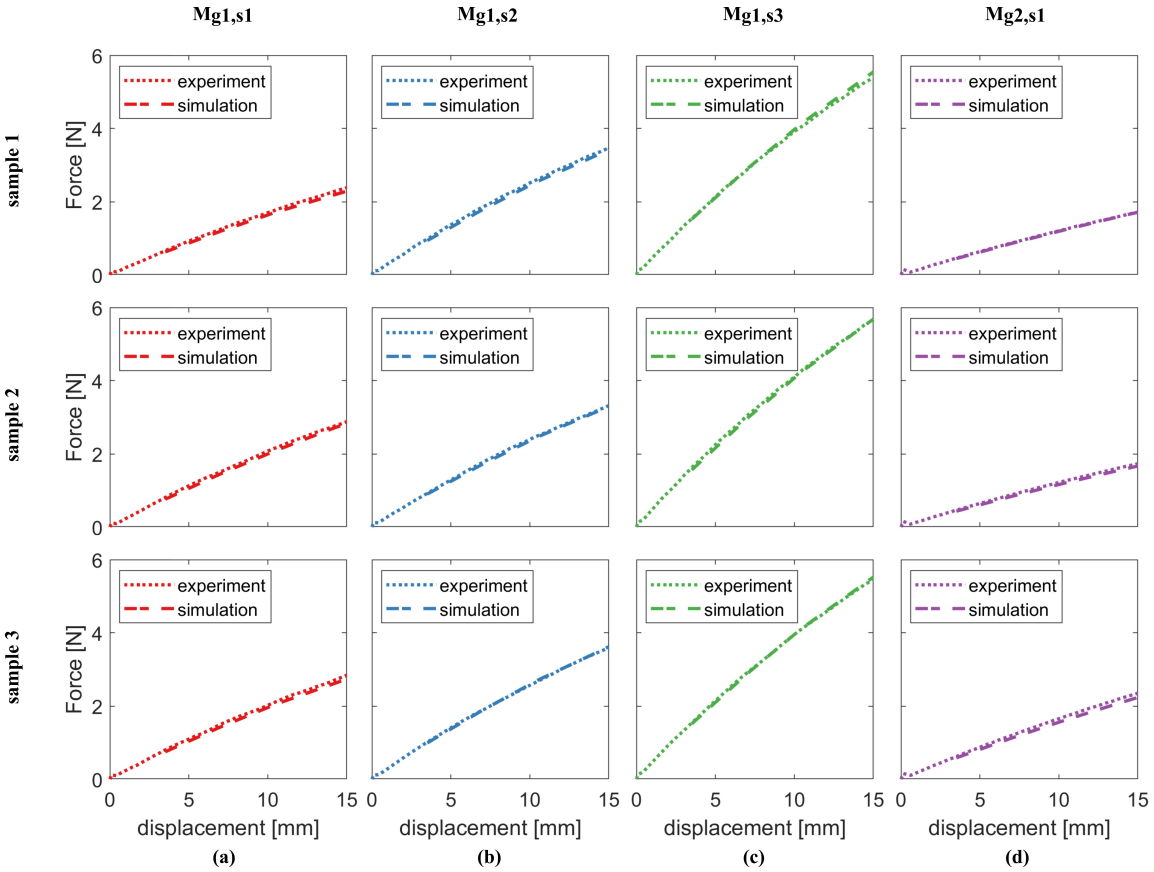


Figure 6.11: FEM-based error analysis on uniaxial tensile testing specimen used to evaluate ground truth mechanical behavior of the models. The tensile test was simulated using the experimentally determined loading conditions and material parameters and a comparison between the simulated and ground truth reaction force vs displacement in the principal direction is shown for all models and samples tested

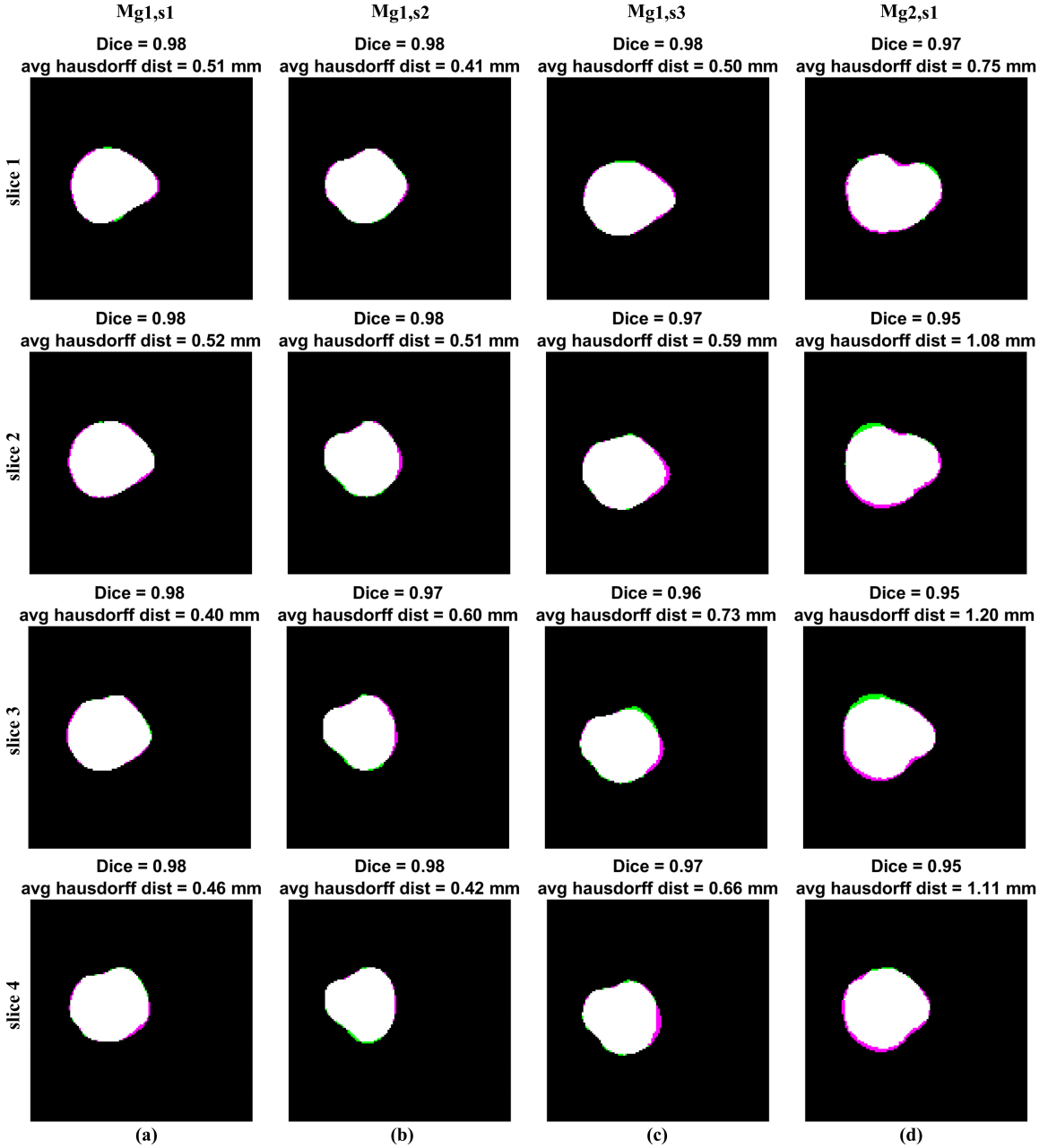


Figure 6.12: Comparison between simulated and image-derived end-inflation LV shape. Representative slices show the overlap between the binary masks (white region), the deviation of the simulated shape from the overlap region (purple), and the deviation of the image-obtained shape from the overlap region (green). Overall, there is very good spatial overlap (high dice scores) and surface shape agreement (low average Hausdorff distances between the LV endocardial surfaces)

# Chapter 7

# Evaluating Passive Myocardial Stiffness using *in vivo* cine, cDTI, and tagged MRI

## 7.1 Abstract

Increased passive myocardial stiffness is implicated in the pathophysiology of many cardiac diseases, and its *in vivo* estimation can improve management of heart disease. MRI-driven computational constitutive modeling has been used extensively to evaluate passive myocardial stiffness. This approach requires subject-specific data that is best acquired with different MRI sequences: conventional cine (e.g. bSSFP), tagged MRI (or DENSE), and cardiac diffusion tensor imaging. However, due to the lack of comprehensive datasets and the challenge of incorporating multi-phase and single-phase disparate MRI data, no studies have combined *in vivo* cine bSSFP, tagged MRI, and cardiac diffusion tensor imaging to estimate passive myocardial stiffness. The objective of this work was to develop a personalized *in silico* left ventricular model to evaluate passive myocardial stiffness by integrating subject-specific geometric data derived from cine bSSFP, regional kinematics extracted from tagged MRI, and myocardial microstructure measured using *in vivo* cardiac diffusion tensor imaging. To demonstrate the feasibility of using a complete subject-specific imaging dataset for passive myocardial stiffness estimation, we calibrated a bulk stiffness parameter of a transversely isotropic exponential constitutive relation to match the local kinematic field extracted from tagged MRI. This work establishes a pipeline for developing subject-specific

biomechanical ventricular models to probe passive myocardial mechanical behavior, using comprehensive cardiac imaging data from multiple *in vivo* MRI sequences.

## 7.2 Introduction

Altered passive myocardial stiffness is evident in several cardiovascular diseases, including heart failure with preserved ejection fraction [187], hypertrophic cardiomyopathy [214], aortic stenosis [215], and myocardial infarction [216]. As passive myocardial stiffness is an important biomarker of global heart function, a robust method for its reliable identification can improve heart disease management.

Recent research on biomechanical modeling of passive ventricle mechanics has focused on uncertainty quantification [166], validation [217], impact of boundary conditions [159] and residual strains [138], improving the optimization strategy [218], and improving the identifiability of passive myocardial material parameters [219]. Owing to research progress, medical image-based computational constitutive modeling has been applied to clinical data, demonstrating its utility as a clinical prognostic tool [34].

The subject-specific data needed to estimate passive myocardial stiffness using inverse finite element (FE) modeling are best acquired using multiple imaging techniques. For example, the gross geometry of the heart is best determined from cine bSSFP images, regional diastolic kinematics can be measured using cine DENSE imaging or tagging, and cardiac microstructural organization can now be measured non-invasively using *in vivo* cardiac diffusion tensor imaging (cDTI). However, no studies, to our knowledge, have combined *in vivo* cine bSSFP, tagged MRI, and cDTI for estimation of subject-specific passive diastolic mechanical behavior partly due to the lack of comprehensive, high-fidelity datasets. Moreover, there is no established methodology for integrating multi-phase *in vivo* geometric, functional, and kinematic data, and single-phase *in vivo* cDTI into a single subject-specific *in silico* model for mechanics modeling.

As demonstrated by a recent study by Stimm *et al.* [157], advancements in *in vivo* cDTI acquisition and post-processing have made incorporating subject-specific *in vivo* microstructure into FE models feasible. While the authors estimated passive myocardial stiffness by matching experimentally measured volumes, progress in MRI-driven kinematics

modeling [155] means we can optimize for material parameters using richer objective functions that incorporate the local myocardial deformation kinematics. In addition, absence of a kinematic field meant the authors could not account for the potentially different microstructural configurations between the cDTI acquisition phase (i.e. end-systole) and the phase at which the mechanics model was generated (i.e. diastasis).

Herein, we acquired high fidelity *in vivo* MRI data, including cine bSSFP, tagged MRI, and cDTI, in a healthy adult subject (male, 24 years, BMI 26). We present a framework for integrating multiple sets of subject-specific MRI data to develop an *in silico* FE mechanics model of the subject's left ventricle (LV). Then we demonstrate the feasibility of using local kinematics to calibrate the *in silico* model to estimate passive myocardial stiffness.

## 7.3 Methods

### 7.3.1 Image Acquisition

The entire MRI dataset, consisting of cine bSSFP, cine diffusion tensor imaging (cDTI) and tagged MRI, was acquired in a 90-minute scanning session using a 3T Siemens MRI scanner at the Veterans Affairs Palo Alto Health Care System (Skyra, Siemens).

*In vivo* cine bSSFP short axis images were acquired with retrospective gating (TE/TR = 1.49/35.0 ms; FA = 35°; resolution = 1×1×6 $mm^3$; 6 mm slice thickness; 17 slices; FoV = 250×320; 25 cardiac phases). Short axis grid tagged images were acquired using a retrospectively gated sequence (TE/TR = 3.01/31.50 ms; FA = 12°; 7 mm tag spacing; same resolution, slice thickness, slices, FoV, and cardiac phases as cine bSSFP). In long axis, two line tagged images were acquired with tag lines oriented longitudinally and radially (TE/TR = 2.8/46.08 ms; FA = 10°; resolution = 1×1×10 $mm^3$, 10 mm slice thickness, 3 slices).

*In vivo* cDTI were acquired after the tagged MRI using a free-breathing, slice-following, spin-echo Echo Planar Imaging (EPI) navigator-based sequence with symmetric first and second-order motion-compensated diffusion gradients [220] (TE/TR = 86/3 R-R intervals; resolution = 2×2×8 $mm^3$; 8 mm slice thickness; 13 slices; b-values = [0, 350] s/$mm^2$; number of diffusion directions = 10; Averages = 6; Partial Fourier = 6/8; GRAPPA = 2). Blip-up (right-left) and blip-down (left-right) datasets were acquired at end-systole.

### 7.3.2 Geometric Analysis (cine bSSFP)

The cine bSSFP imaging data were processed using Cardiac Image Modeller (CIM, version 8.2, University of Auckland, New Zealand), which provided semi-automated segmentation of the LV surfaces over the entire imaging cycle in the form of FE models. Specifically, landmark points at the LV basal centroid, right ventricular basal centroid and LV apex were manually identified and subsequently used to construct a prior FE model in an orthogonal cardiac coordinate system. The origin of the cardiac coordinate system was located at one-third of the longitudinal distance from the LV base to apex landmark points. The x, y, and z axes were directed towards the LV apex, right ventricular centroid, and posterior LV wall respectively. LV endocardial and epicardial surfaces were interactively generated for all cardiac phases in Cardiac Image Modeler by fitting the prior FE model to guide-points placed by the analyst. The reference ("load free") FE mechanics model geometry was chosen as the diastasis phase corresponding to the start of atrial systole [221], and the timing was determined by the mitral valve motion from the cine bSSFP long-axis images (Fig. 7.1: middle).

### 7.3.3 Kinematics Analysis (tagged MRI)

The grid-tagged images were rigidly registered to the cine bSSFP images. Then, a set of myocardial points were defined in the diastasis phase of the grid-tagged images by masking out the LV myocardium using the diastasis LV surface model mentioned above. This was achievable because the cine bSSFP and tagged images were temporally aligned. Subsequently, we used a validated neural-net-based algorithm [222] to track the local myocardial displacements across the cardiac cycle relative to the diastasis phase (Fig. 7.1: right).

### 7.3.4 Microstructural Analysis (cDTI)

The raw *in vivo* cDTI data were first processed to correct geometric distortion induced by off-resonance phase accrual while traversing k-space over long echo times. Specifically, following 2D rigid registration of the blip-up and blip-down EPI datasets, the complete dataset was distortion corrected using TOPUP and the FSL package [223]. Diffusion tensors

at each image voxel were then reconstructed from the distortion corrected diffusion weighted images using least squares linear regression.

To embed the *in vivo* cDTI diffusion tensors measured at end-systole into the LV geometric model at diastasis, a two-step approach was proposed. Step-1 was designed to translate and rotate the raw diffusion tensor data from end-systole to diastasis using the kinematic field derived from tagged MRI analysis. Step-2 was designed to interpret the reconfigured diffusion tensor data within the LV geometric FE model at diastasis using a field-based fitting approach [224]. Prior to Step-1, the cDTI data at end-systole were masked by, first kinematically deforming the cine bSSFP LV geometric model from diastasis to end-systole using the tagged MRI-obtained kinematic field, and then intersecting the deformed model with the cDTI images. Subsequently, at each voxel within the myocardium, a deformation gradient tensor between the end-systole and diastasis frame was interpolated from the kinematics field and decomposed into its rotational and stretch components. The diffusion tensors at end-systole were then subsequently reoriented to their diastasis configurations using the corresponding rotation tensors. The material point positions of the voxels were also translated based on the kinematics field.

To verify this approach of using the kinematics field to reorient myofiber direction data, we simulated the inflation of an LV FE model with a rule-based fiber field description. We then used the FE-simulated kinematics field to reorient the fiber vectors to their deformed states and compared them with the end-inflation directions predicted by the FE solver. The comparison revealed good agreement with an average angle difference of $0.8\pm0.8°$and fiber location difference of $0.007\pm0.006$ mm (mean$\pm$std). While in this work the microstructure was reconfigured from the end-systolic to diastasis state, during which the LV undergoes isovolumic relaxation followed by rapid filling, the LV FE model used to validate the microstructure reorientation approach was under passive inflation. However, validation of the method for reconfiguring LV microstructure using deformation kinematics, does not require exact replication of the underlying deformation.

After transforming the *in vivo* DTI data into the appropriate configuration for integration with the geometric model, a microstructural orientation field, defined by the fiber, imbrication, and sheet (Euler) angles, was initialized at each node (a total of 34 nodes) of the FE model and interpolated over the FE model using tricubic Hermite basis functions. The fiber

angles, defined as the elevation angle between a fiber vector that was projected to the surface of the FE model and the local circumferential direction, were initially set to be $+60°$ for the endocardial nodes, and $-70°$ for the epicardial nodes, and the imbrication angles, defined as the angle between the fiber vector and its projection to the surface of the FE model, were set to $0°$. A local representation of this microstructural orientation field, expressed as the fiber vector $\mathbf{f}_{(v)}$, at each voxel $v$ was computed. The nodal angular parameters describing the fiber vector field were optimized to maximize the alignment of $\mathbf{f}_{(v)}$ with the direction of maximal diffusion. The objective function was expressed as the dot product of $\mathbf{f}_{(v)}$ with the projection of the diffusion tensor ($\mathbf{D}_{(v)}$) in the direction of $\mathbf{f}_{(v)}$. Only fiber and imbrication angles were fitted. Optimization of the fiber orientation field was performed using the software package CMGUI [225] following procedures outlined by Freytag *et al.* [224]. The 3D description of the fiber orientation at diastasis was subsequently incorporated into the *in silico* LV model (Fig. 7.1).

### 7.3.5 Constitutive relation

The myocardium was modeled using a transversely isotropic exponential constitutive relation [98] described by the following strain energy density function:

$$\begin{aligned} \psi &= \frac{1}{2}c(e^{Q}-1) \\ Q &= b_1E_{ff}^2 + b_2(E_{ss}^2 + E_{nn}^2 + 2E_{sn}^2) + b_3(2E_{fs}^2 + 2E_{fn}^2) \end{aligned} \tag{7.1}$$

where $c$, $b_1$, $b_2$, and $b_3$ are the constitutive parameters and $E_{ij}$ denotes the components of the Green-Lagrange strain tensor ($\boldsymbol{E}$). Subscripts $f$, $s$, and $n$ correspond to the fiber, sheet, and sheet-normal directions respectively.

Here, we fit $c$, the parameter that scales the overall passive myocardial stiffness, and kept the anisotropy parameters constant at physiologically realistic values of $b_1 = 8.61$, $b_2 = 3.67$, $b_3 = 25.77$ [9].

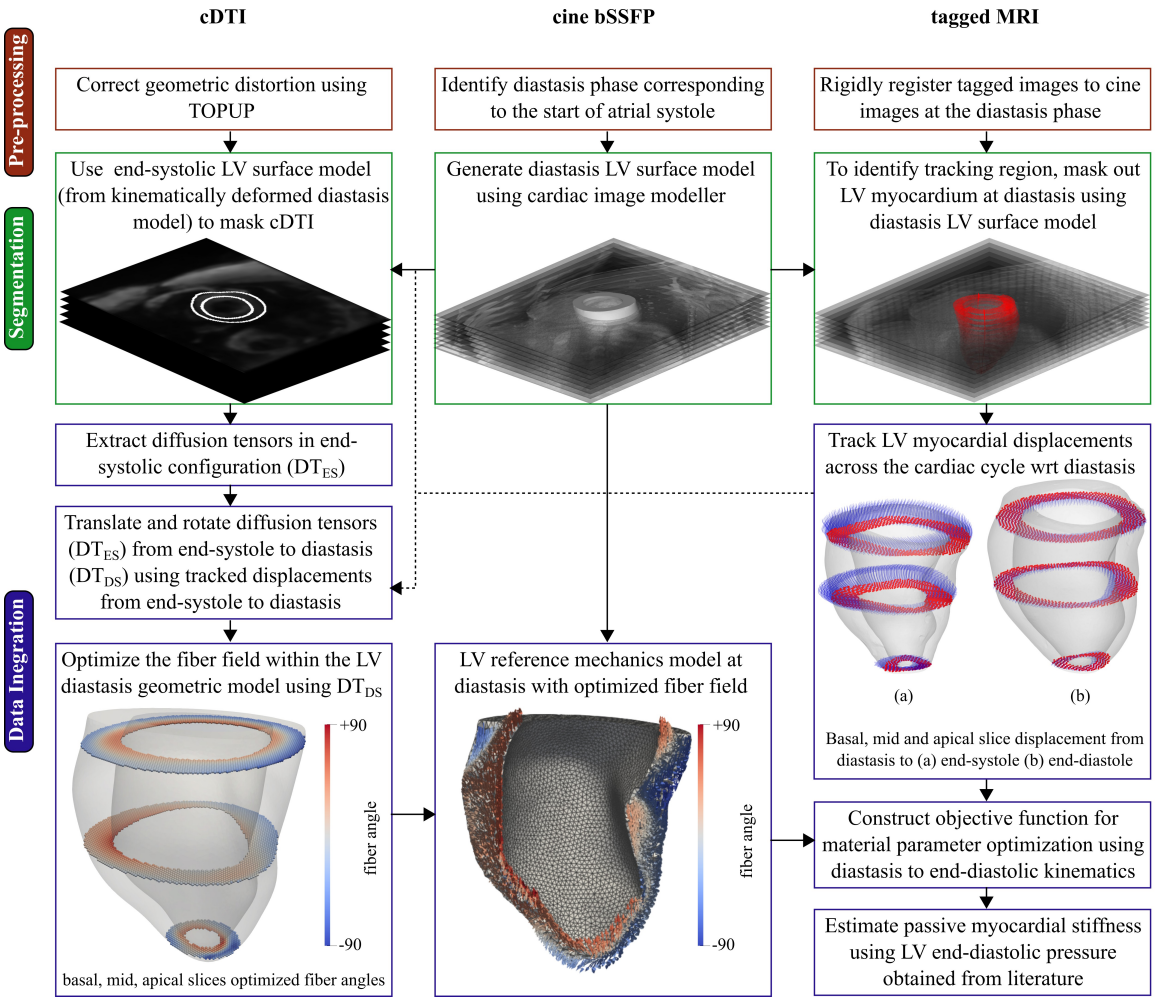


Figure 7.1: Pipeline for integrating cine bSSFP, cDTI and tagged MRI data into a mechanics model for estimating passive myocardial stiffness.

### 7.3.6 Mechanics Analysis

In this work, the FE mechanics analysis was limited to the passive filling phase between diastasis and end-diastole. The LV diastasis geometric model was used as the reference mechanics model, (section 7.3.2) and was meshed with standard quadratic tetrahedral elements using GMSH [207] (1.9 $mm^3$ average element volume, 129,763 nodes, 81,295 elements, 389,289 degrees of freedom). The FE analysis was done using FEBio [120]. In the absence of LV cavity pressure recordings, the *in silico* LV was passively inflated by applying an endocardial pressure of 1 kPa, which is consistent with the expected LV end-diastolic

pressure in healthy humans [9, 226]. Uncertainty in the simulated material parameter due to a 0.2 kPa uncertainty in LV end-diastolic pressure was also investigated.

Dirichlet boundary conditions were applied at the base of the *in silico* model by constraining the simulated basal surface displacement from diastasis to end-diastole to match that which was obtained from the tagged MRI-obtained kinematics. Then, the experimentally determined LV local displacements from diastasis to end-diastole were used to inversely estimate the optimal transversely isotropic exponential constitutive relation [98] (see section 7.3.5). An initial estimate of the material parameters was obtained from Wang *et al.* [9]. Then, we iteratively updated the $c$ constitutive parameter to minimize the differences between the experimentally measured nodal displacements (tagged MRI) and the simulated pressure-induced nodal displacements.

## 7.4 Results

### 7.4.1 Microstructure fitting

The fitted LV myocardial fiber fields at diastasis and end-systole are shown in Fig. 7.2. Each fiber vector is color-coded by the fiber angle (i.e. the elevation angle with respect to the short axis plane). Here, we see a variation of fiber orientation with mostly positive fiber angles on the endocardial surface, and fibers with mostly negative fiber angles on the epicardial surface, in accordance with the expected transmural variation in fiber orientation [34].

### 7.4.2 *In silico* stiffness calibration

The optimal bulk stiffness parameter ($c$) of the transversely isotropic constitutive relation (see section 7.3.5) was 5.8$\pm$1.2 kPa based on a LV filling pressure range of 1$\pm$0.2 kPa. The mechanical behavior of the fitted constitutive relation in biaxial extension is shown (Fig. 7.3a) and compared with results from the Wang *et al.* study on passive stiffness of human myocardium [9]. In Fig. 7.3b, we show the computed deformed LV and the spatial variation in principal stretch after deformation (mean$\pm$95%CI = 1.03$\pm$0.03 n.d.).

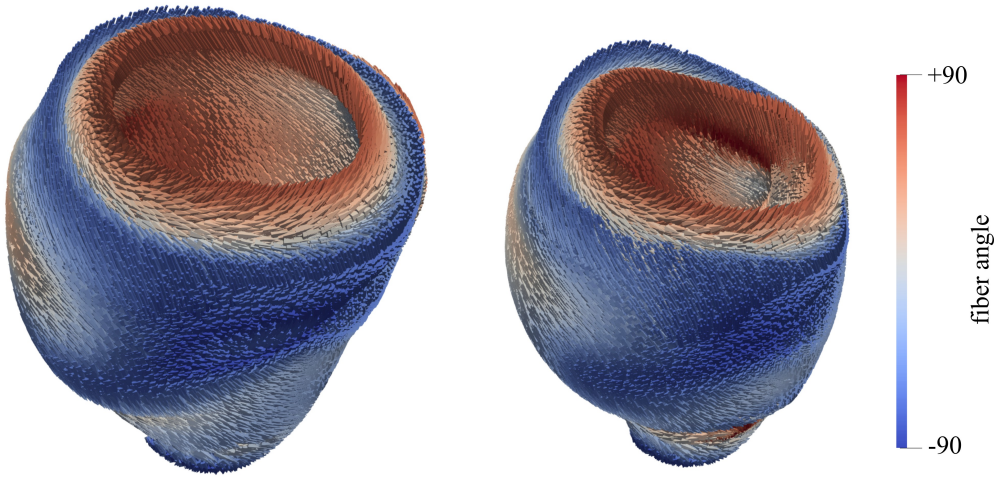


Figure 7.2: *In vivo* cardiac DTI fiber vectors after field-fitting showing the LV myocardium mesh elements colored by the fiber angle at (a) diastasis (b) end-systole.

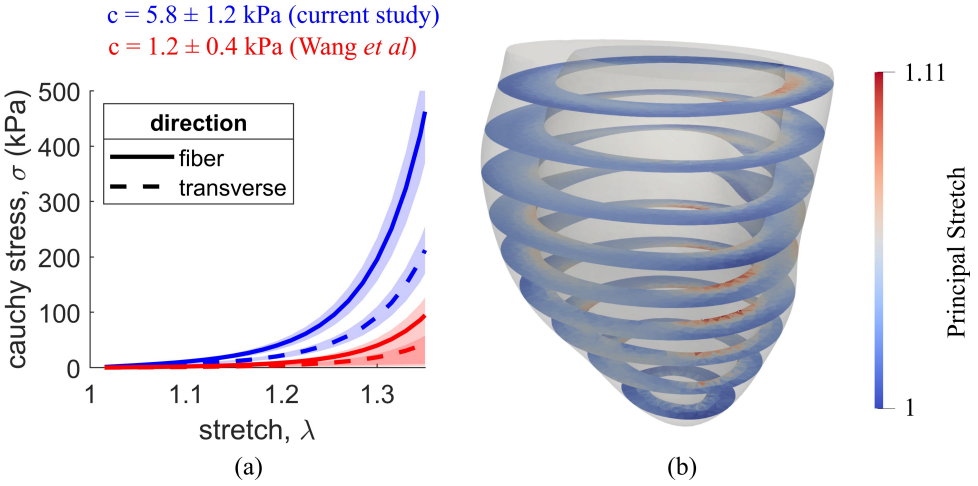


Figure 7.3: (a) Myocardial mechanical behavior in simulated equibiaxial extension based on *in vivo* passive myocardial stiffness estimations from this study (blue) and from Wang *et al.* (red) [9]. Uncertainty arises from the range of LV end-diastolic pressure used for the material parameter optimization (1±0.2 kPa). (b) LV computed deformation at maximal inflation. Short-axis slices depict the spatially varying principal stretch.

## 7.5 Discussion

Evaluation of *in vivo* passive myocardial stiffness can improve understanding and management of heart diseases with associated myocardial stiffness alterations. Though several previous studies have leveraged medical imaging and FE modeling to evaluate passive myocardial mechanical behavior and stiffness, none to our knowledge have combined *in vivo* bSSFP, tagged MRI, and cDTI to build the FE models as we have done here.

In most regions of the LV model, the fitted microstructural field (Fig. 7.2) follows the expected transmurality, with positive fiber angles on the endocardial surface and negative fiber angles on the epicardial surface. However, in some regions this expected variation did not hold, likely due to poor cDTI image quality (and subsequently bad diffusion tensors) in these regions. Given the sensitivity of mechanical behavior to microstructure [34], the evaluated material parameter was affected by these regions with anomalous fiber orientations.

Our calibrated passive myocardial stiffness is about 5 times stiffer than the Wang *et al.* study's results [9] (Fig. 7.3), which could be partly due to fundamental differences between our study and theirs. For example, while Wang *et al.* used catheter pressure measurements, here we used an LV end-diastolic pressure for healthy humans from literature. Additionally, in their study, passive myocardial stiffness was estimated by matching experimentally measured surfaces, whereas, we matched end-diastolic local kinematics. Moreover, Wang *et al.* did not prescribe the microstructure using *in vivo* cDTI.

### 7.5.1 Limitations

We only fitted the bulk passive stiffness parameter ($c$) of the transversely isotropic constitutive relation while keeping the anisotropy parameters constant. Of note, fitting multiple parameters introduces additional considerations related to the identifiability and uniqueness of the parameter set. Secondly, we lacked LV diastolic pressure measurement, which also precluded use of kinematic data from multiple cardiac phases for parameter estimation. Nevertheless, since calibrating constitutive parameters using a generic end-diastolic pressure from literature was feasible, we expect the framework to generalize to the use of personalized diastolic pressures. Future work would explore model calibration using non-invasive estimates of diastolic pressures, such as those derived from Doppler ultrasound

imaging [192]. Lastly, we did not consider myocardial *in vivo* pre-stresses [138]. Methods for unloading the pre-stressed geometry exist [141], but they rely on assumptions about the material parameters and have not been extensively validated.

### 7.5.2 Conclusion

We have established a method to integrate multi-phase *in vivo* cardiac cine bSSFP and tagged MRI with single-phase *in vivo* cDTI for the development of a subject-specific LV mechanics model. We then demonstrated the feasibility of calibrating the leading material parameter of a transversely isotropic exponential constitutive relation using local deformation kinematics. This framework can also be readily extended to model subject-specific contractile mechanics. Future work will focus on extensive validation and model sensitivity analysis to understand uncertainty in the results. We will also evaluate the passive myocardial stiffness estimation approach in additional human subjects.

### 7.5.3 Acknowledgements

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# Chapter 8

# Characterizing Passive Myocardial Stiffness in Healthy Humans with MRI

## 8.1 Abstract

Altered passive myocardial stiffness is evident in the pathophysiology of several cardiovascular diseases, making it an important biomarker of heart health. Biophysical modeling of the heart enables evaluation of the material parameters governing passive myocardial stiffness. Using an MRI and inverse finite element modeling approach, the subject-specific data required include: gross cardiac geometry from conventional cine (e.g. bSSFP), regional diastolic kinematics from tagged MRI (or DENSE), microstructural configuration from cardiac diffusion tensor imaging, and diastolic pressure(s) invasively measured or non-invasively estimated. Though the necessary MRI data is ideally acquired with separate MRI sequences, due to the lack of comprehensive datasets and the challenge of combining multi-phase and single-phase imaging data, there is a lack of studies combining *in vivo* cine bSSFP, tagged MRI, and cardiac diffusion tensor imaging to estimate passive myocardial stiffness. Moreover, the sensitivity of passive myocardial stiffness estimation to the input data hasn't been adequately assessed. Thus, the objectives of this work were to: (1) develop a framework for integrating multiple sets of subject-specific *in vivo* MRI data into *in silico* finite element left ventricular models for estimating passive myocardial stiffness, (2) use the framework to characterize the passive material parameters of multiple healthy subjects, and

(3) evaluate the sensitivity of passive myocardial material parameter estimation to loading conditions and myofiber orientation. This work contributes towards the establishment of a range of reference values for passive myocardial material parameters in healthy human subjects.

**Key terms:** cardiac mechanics, *in vivo* cardiac MRI, heart failure, passive myocardial stiffness

## 8.2 Introduction

Heart failure, identified 25 years ago as an emergent epidemic in the USA [11], currently remains a major public health issue [227] with high and increasing prevalence [184], considerable socioeconomic burden [183], and significant mortality and morbidity [182]. This complex cardiovascular syndrome is characterized by the heart's inability to supply enough blood to meet the body's metabolic demands [181]. Patients are clinically categorized into two major groups: heart failure with preserved (HFpEF) and reduced (HFrEF) ejection fraction. In HFrEF, pharmacologic and device-based interventions that interrupt adverse cardiac remodeling have proven effective [228]. On the other hand, evidence-based therapeutic options for HFpEF are limited, and management is mostly targeted at related symptoms (e.g. hypertension) [29, 30]. The success of HFrEF treatment demonstrates that mechanism-focused interventions have the potential to improve outcomes in HFpEF, which necessitates better understanding of the underlying cardiovascular remodeling mechanisms unique to HFpEF.

LV chamber stiffening is a consistent finding in HFpEF and can be assessed clinically via pressure-volume analyses. However, with this technique, contributions from myocardial hypertrophy and passive myocardial stiffening cannot be separated. These two remodeling mechanisms are associated with different alterations to the cardiomyocyte and extracellular matrix and so are important to differentiate. Moreover, abnormal passive myocardial stiffness is evident in several other cardiovascular diseases including aortic stenosis [215], hypertrophic cardiomyopathy [214], and locally in myocardial infarction [216], meaning measures of passive myocardial stiffness could be an important biomarker of heart health.

Passive myocardial stiffening primarily affects ventricular relaxation during atrial contraction as the ventricles passively fill under the action of atrial systole. A significant focus of patient-specific cardiac biomechanical modeling has been the development of methods for reliable identification of subject-specific passive myocardial stiffness independent of anatomic and loading effects.

Several groups have demonstrated the utility of MRI-based inverse finite element modeling (FEM) for the identification of subject-specific passive myocardial stiffness [175, 34]. Some recent contributions to MRI-based modeling of heart mechanics have been related to validation [217], uncertainty quantification [229, 166], assessing the impact of boundary conditions [159, 160] and residual strains [138, 140], improvement of the optimization strategy [218, 230], development of material models [116, 231], and improving the identifiability of passive myocardial stiffness [1, 219]. Provided cardiac geometry, microstructural organization, boundary conditions, kinematics, and end-diastolic (filling) pressure(s), continuum balance laws, and optimization techniques can be leveraged in FEM to inversely obtain the parameters of the constitutive model governing the material's mechanical behavior. End-diastolic pressure(s) can be measured through catheterization or non-invasively estimated (e.g. from mitral flow velocity [232]). Cardiac surface geometry is best determined from conventional cine balanced steady-state free-precession (bSSFP) MRI as it offers exceptional signal to noise ratio and high contrast between myocardium and blood, enabling excellent dileneation of endocardial and epicardial borders [68]. Local deformation kinematics can be acquired from tagged MRI [233] or cine DENSE MRI [234], and cardiac microstructure can be non-invasively measured using *in vivo* cDTI [235]. However, to our knowledge, no study has integrated subject-specific *in vivo* cDTI, cine bSSFP, and tagged MRI into personalized *in silico* passive mechanics models. This is partly due to the challenge of acquiring comprehensive, high-fidelity imaging data using multiple cardiac MRI sequences, and the lack of an established framework for combining the disparate data into a single personalized *in silico* mechanics model. To determine changes in cardiac dysfunction, there is also a need to adequately characterize the range of healthy *in vivo* myocardial behavior. However, most available clinical imaging datasets are acquired from subjects with cardiovascular conditions.

Herein, we acquired high fidelity *in vivo* MRI data, including cine bSSFP, tagged MRI,

and cDTI, in healthy adult subjects (N=7) with no cardiovascular disease (male & female, 22-32 years, BMI 22-28). We developed a framework for integrating the multiple sets of subject-specific MRI data into *in silico* FEM models of the subjects' left ventricle (LV). We then use the pipeline to characterize the passive myocardial stiffness of each subject by calibrating the personalized *in silico* model using local kinematics. Furthermore, we performed an analysis to understand the sensitivity of passive mechanics to both "myofiber" orientations and loading conditions. In so doing, we contribute towards the establishment of a range of reference values for passive myocardial stiffness in healthy human subjects.

## 8.3 Methods

In this section, first we summarize the cardiac MRI acquisition used for all seven subjects (Section 8.3.1). Then we describe development of the FE forward mechanics models, including determination of the LV reference geometric models from the cine bSSFP images (Section 8.3.2), extraction of local deformation kinematics from tagged MRI, personalization of the LV myocardial microstructure using cDTI (Section 8.3.4), as well as rule based descriptions of the microstructure (Section 8.3.5). We also discuss the constitutive relation used to model passive myocardial stiffness in this study (Section 8.3.6) and the material parameter estimation using inverse FEM (Section 8.3.7).

### 8.3.1 Image Acquisition

This study was approved by the local Institutional Review Board and all subjects signed statements of informed consent. For each subject (Table 8.1), a complete MRI dataset, consisting of cine bSSFP, cine diffusion tensor imaging (cDTI) and tagged MRI (Figure 8.1), was acquired during a 90-minute exam using a 3T MRI scanner (Skyra, Siemens). All images were acquired to span from the LV base to apex.

*In vivo* cine bSSFP short axis images were acquired with retrospective gating (TE/TR = 1.49/35.0 ms; FA = 35°; resolution = 1×1×6 $mm^3$; 4 mm slice gap; 25 cardiac phases). Short and long axis tagged images were acquired with the same spatial and temporal resolution as the cine bSSFP, also using a retrospectively gated sequence. In the short axis, grid

tagged images were obtained (TE/TR = 3.01/31.50 ms; FA = 12°; resolution = 1×1×6 $mm^3$; 4 mm slice gap; 25 cardiac phases; 7 mm tag spacing) and in the long axis, images with radially oriented line tags were acquired (TE/TR = 2.8/46.08 ms; FA = 10°; resolution = 1×1×10 $mm^3$, 10 mm slice thickness, 3 slices). The first long axis image bisected a central region of the interventricular septum and the LV free wall, another intersected the posterior RV insertion point and the anterior papillary muscle, and the third crossed the anterior RV insertion point and posterior papillary muscle.

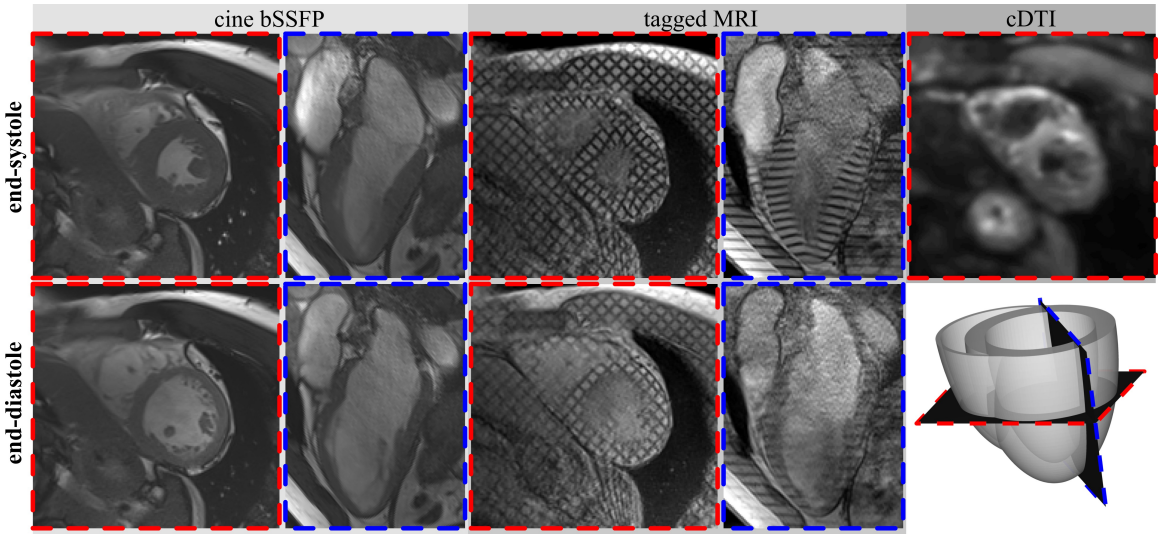



Figure 8.1: Representative cine bSSFP, tagged MRI, and cDTI (b=0) images from one subject showing both short and long axis views and end systole and end diastole.

Table 8.1: Subjects' Clinical Indices

| Subject # | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
|---|---|---|---|---|---|---|---|
| Sex | Female | Male | Male | Male | Female | Male | Female |
| Age | 25 | 29 | 32 | 24 | 22 | 25 | 26 |
| BMI [$kg/m^2$] | 22.7 | 27.8 | 23.1 | 26.9 | 21.9 | 27.5 | 24.9 |
| LV mass [g] | 83.8 | 128.1 | 131.6 | 182.3 | 69.7 | 170.9 | 81.3 |
| EDV | 131.5 | 151.2 | 165.3 | 226.1 | 89.9 | 196.7 | 118.4 |
| ESV | 48.1 | 68.9 | 70.7 | 84.6 | 40.7 | 96.3 | 43.3 |
| EF | 63.4 | 54.4 | 57.2 | 62.6 | 54.7 | 51.1 | 63.4 |
| WT [mm] | 5.7±0.6 | 7.7±0.8 | 7.5±0.9 | 8.3±0.6 | 6.2±0.4 | 8.5±1.0 | 6.5±0.8 |

*In vivo* cDTI were acquired after the tagged MRI acquisition using a navigator-based free-breathing, slice-following, spin-echo Echo Planar Imaging (EPI) sequence with symmetric first and second-order motion-compensated diffusion gradients [220] (TE/TR = 86/3 R-R intervals; resolution = 2×2×8 $mm^3$; 2 mm slice gap; b-values = [0, 350] $s/mm^2$; number of diffusion directions = 15; Averages = 5; Partial Fourier = 6/8; GRAPPA = 2). Blip-up (right-left) and blip-down (left-right) datasets were acquired at end-systole with a trigger delay.

### 8.3.2 Geometric Analysis from Cine bSSFP

The first pre-processing step necessary for some of the cine bSSFP data was slice-to-slice alignment. In cine MRI, each slice is acquired in a single breath-hold during which images of the entire cardiac cycle are obtained. Slice misalignment can arise due to breath hold differences and subject motion. We developed a contour-based slice alignment approach that minimizes the distance between the long axis and short axis endocardial contours. First, the long axis slices were aligned by ensuring that the endocardial contours intersect. Each short axis slice was then aligned by applying the best-fit (least squares) translation to fit the short axis endocardial contour to the points where the aligned long axis endocardial contours intersect the short axis planes. The alignment was repeated on all the short axis slices until all the long and short axis images were co-registered (Figure 8.2). Slice alignment correction was only necessary in subjects one, four, and seven.

The cine bSSFP images were then processed using Cardiac Image Modeller (CIM, version 8.2, University of Auckland, New Zealand), which provided semi-automated segmentation of the LV surfaces over the entire imaging cycle in the form of FE models (Figure 8.3). Specifically, landmark points at the LV basal centroid, right ventricular basal centroid and LV apex were manually identified and subsequently used to construct a prior FE model in an orthogonal cardiac coordinate system. The origin of the cardiac coordinate system was located at one-third of the longitudinal distance from the LV base to apex landmark points. The x, y, and z axes were directed towards the LV apex, right ventricular centroid, and posterior LV wall respectively. LV endocardial and epicardial surfaces were interactively generated for all cardiac phases in CIM by fitting the prior FE model to guide-points placed

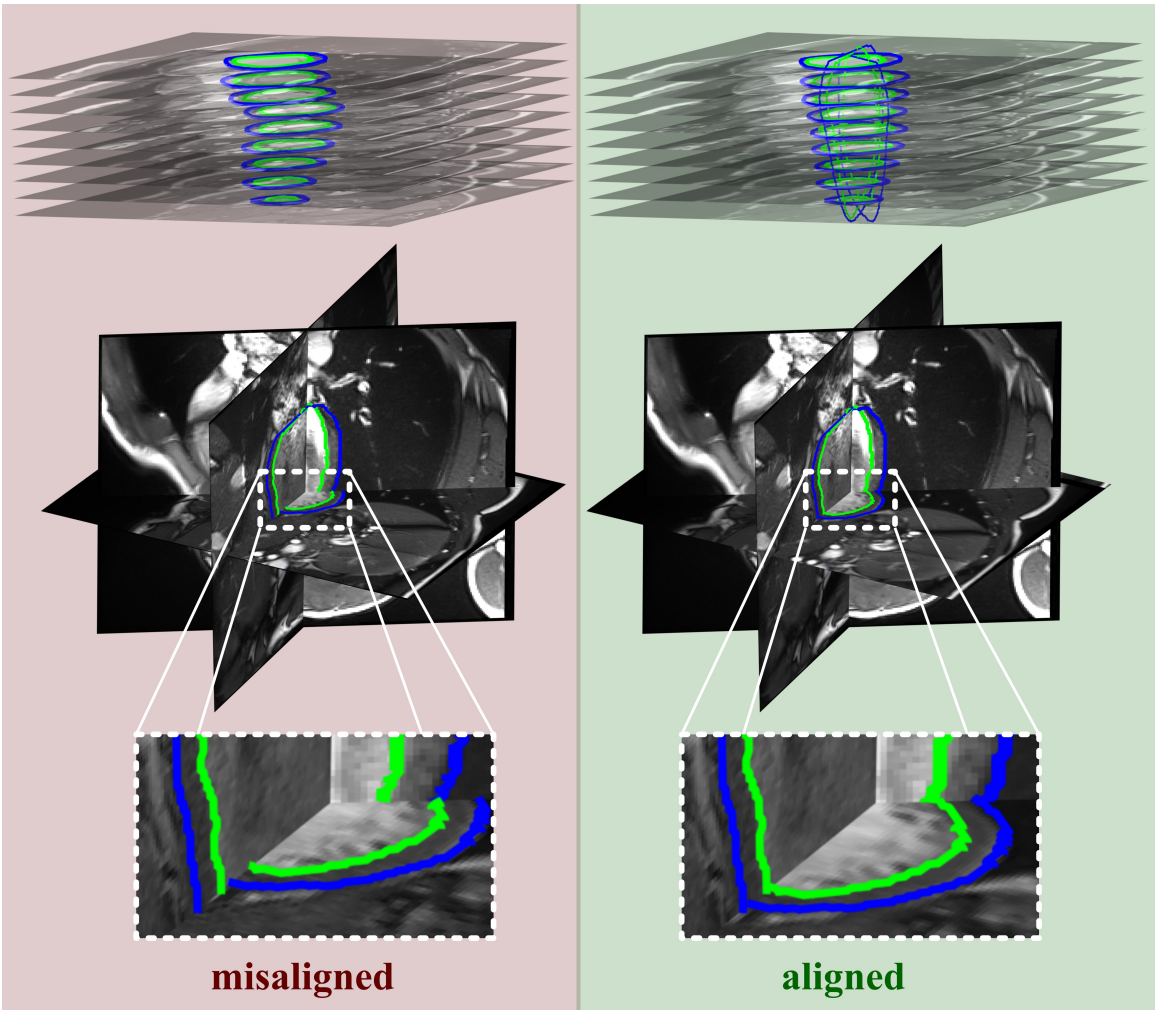



Figure 8.2: Cine bSSFP slice alignment. Slice misalignment due to breath hold differences and subject motion was corrected using a contour-based approach that ensured the short and long axis slices were aligned.

by the analyst. The reference ("load free") FE mechanics model geometry was chosen as the end-diastasis phase corresponding to the start of atrial systole [221], and the timing was determined by the mitral valve motion from the cine bSSFP long-axis images.

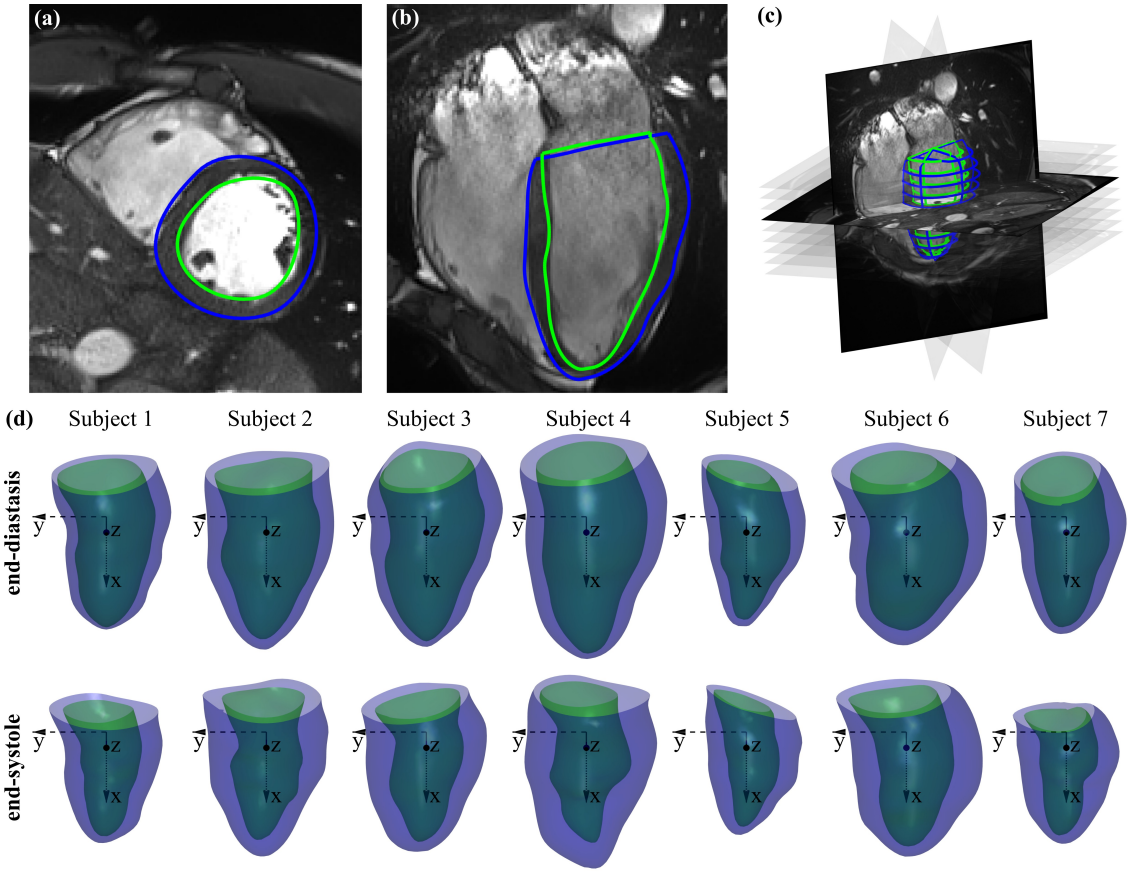



Figure 8.3: Cine bSSFP model generation. A prior FE model was fit to guide points placed on the (a) short and (b) long axis images. (c) Intersection of the model with all the images. (d) Final end-diastasis and end-systolic surface models for all subjects. The endocardial surface is colored green and the epicardial surface is colored blue.

### 8.3.3 Kinematics Analysis from Tagged MRI

First the kinematic and geometric data were registered. This was especially important in the subject cases with slice misalignment. The short-axis grid-tagged image slices were registered to the cine bSSFP images by aligning the epicardial contours. This 2D registration

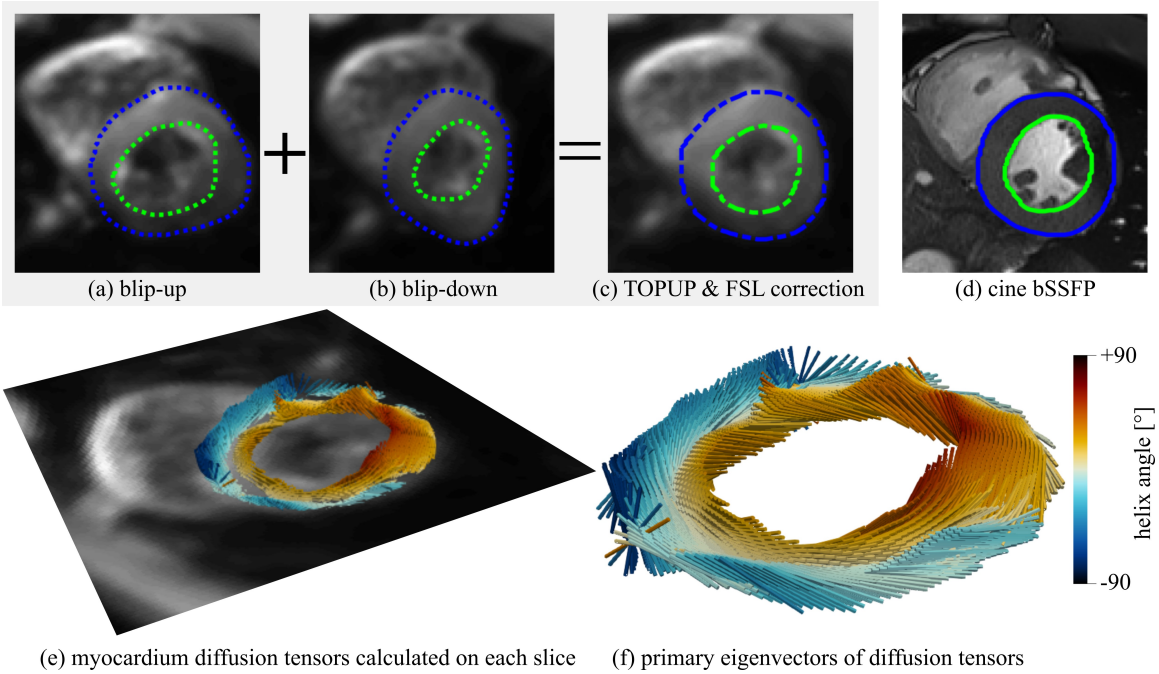
was possible because the tagged and cine bSSFP image datasets were spatially and temporally aligned. The long-axis tagged data was registered by aligning the epicardial outline with the bSSFP-obtained epicardial end-diastasis surface model. The long and short-axis registrations of the tagged data only consisted of in-plane translation. Given that the reference geometric model for the FEM was developed during end-diastasis, the registration was performed on the end-diastasis images and the resultant transformation was applied to images acquired at all other cardiac phases.

Following registration of the tagged data to the cine bSSFP images, a set of myocardial points were defined in the end-diastasis phase of the grid-tagged images by masking out the LV myocardium using the end-diastasis LV surface model. Subsequently, we used a validated neural-net-based algorithm [222] to track the local myocardial displacements across the cardiac cycle relative to the end-diastasis phase.

### 8.3.4 Personalized Microstructure Analysis from cDTI

For each subject, the blip-up and blip-down cDTI datasets (see section 8.3.1) underwent 2D rigid registration owing to shifts of the heart during the breath-held acquisitions. Subsequently, the five averages of the blip-up and blip-down datasets were retrospectively combined to determine a complete blip-up/blip-down dataset, which was distortion corrected using TOPUP and the FSL package (Figure 8.4) [236, 223]. All subsequent cDTI analyses were done using the distortion corrected images. After registering the distortion corrected diffusion weighted images to the end-systolic bSSFP images by aligning the epicardial contours, the bssfp-obtained end-systolic LV surface model was intersected with the diffusion weighted images to identify regions belonging to the LV myocardium. Then diffusion tensors at each image voxel within the LV myocardium were reconstructed using least squares linear regression.

A two-step approach was implemented to embed the *in vivo* cDTI diffusion tensors measured at end-systole into the end-diastasis LV geometric model. The first step was to reconfigure (reposition and reorient) the raw diffusion tensor data from end-systole to end-diastasis using the kinematic field derived from tagged MRI analysis. The second step was to interpret the reconfigured diffusion tensor data within the LV geometric FE model




(a) blip-up (b) blip-down (c) TOPUP & FSL correction (d) cine bSSFP

(e) myocardium diffusion tensors calculated on each slice (f) primary eigenvectors of diffusion tensors

Figure 8.4: cDTI geometric distortion is corrected by combining the (a) blip-up and (b) blip-down images using TOPUP and the FSL package. (c) The geometrically corrected images correspond better to the (d) cine bSSFP images. Manually contoured epicardial (blue) and endocardial (green) outlines are overlayed on the images. (e) the myocardial diffusion tensors are calculated on each slice of the geometrically corrected images. (f) The primary eigenvectors of the diffusion tensors for a representative slice colored by the fiber angle (helix angle) are shown.

at end-diastasis using a field-based fitting approach [224].

The diffusion tensor reconfiguration for the first step was made feasible by the kinematic field derived from tagged MRI analysis. The end-diastasis LV geometric model was deformed to end-systole using the kinematic field to determine the end-diastasis positions of the diffusion tensors. Then, the end-systolic diffusion tensor locations were registered to this kinematically deformed model and then translated to their end-diastasis positions using the kinematics field. To determine the orientation of the diffusion tensors during end-diastasis, at each voxel within the myocardium, a deformation gradient tensor between the end-systole and end-diastasis cardiac phase was interpolated from the kinematics field. The rotational component of the deformation gradient tensor was obtained from polar decomposition and then used to transform each diffusion tensor to the end-diastasis orientation. To verify this approach of using the kinematics field to reorient myofiber direction data, we simulated the inflation of an LV FE model with a rule-based fiber field description. We then used the FE-simulated kinematics field to reorient the fiber vectors to their deformed states and compared them with the end-inflation directions predicted by the FE solver. The comparison revealed good agreement with an average angle difference of $0.8\pm0.8°$and fiber location difference of $0.007\pm0.006$ mm (mean$\pm$std). Similar kinematics-based diffusion tensor reconfiguration approaches have been implemented previously [237, 238].

After transforming the *in vivo* DTI data into the appropriate configuration for integration with the geometric model, a microstructural orientation field defined by the fiber, imbrication, and sheetlet(Euler) angles, was initialized at each node of a FE model of the LV at end-diastasis (a total of 34 nodes) and interpolated over the FE model using tricubic Hermite basis functions. The fiber angles, defined as the elevation angle between a fiber vector that was projected to the surface of the FE model and the local circumferential direction, were initially set to be $+60°$ for the endocardial nodes, and $-70°$ for the epicardial nodes, and the imbrication angles, defined as the angle between the fiber vector and its projection to the surface of the FE model, were set to $0°$. A local representation of this microstructural orientation field, expressed as the fiber vector $\mathbf{f}_{(v)}$, at each voxel $v$ was computed. The nodal angular parameters describing the fiber vector field were optimized to maximize the alignment of $\mathbf{f}_{(v)}$ with the direction of maximal diffusion. The objective function was expressed as the dot product of $\mathbf{f}_{(v)}$ with the projection of the diffusion tensor ($\mathbf{D}_{(v)}$) in the direction

of $\mathbf{f}_{(v)}$. Only fiber and imbrication angles were fitted. Optimization of the fiber orientation field was performed using the software package CMGUI [225] following procedures outlined by Freytag *et al.* [224]. The 3D description of the fiber orientation at end-diastasis was subsequently incorporated into the *in silico* LV model.

### 8.3.5 Rule-based Microstructure Description

Subject-specific microstructure, such as that which can be obtained through *in vivo* cDTI is only recently available in patients or research subjects [157, 239, 234]. Moreover, processing and integrating cDTI-obtained microstructural information into inverse FE mechanics models can be prohibitively cumbersome. This has motivated the development of several mathematical models of myofiber orientation [10, 150, 151, 152, 153] which are used in the absence of personalized *in vivo* estimates of cardiac microstructure.

The widely used Laplace-Dirichlet rule-based (ldrb) algorithm introduced by Bayer *et al.* [10] was used in this work to define alternate rule-based descriptions of the fiber orientation in the LV models. To generate a continuous fiber orientation description throughout the entire LV myocardium, the ldrb algorithm requires a definition of the basal, epicardial and endocardial surfaces, as well as prescription of helical and transverse angles on the epicardial and endocardial boundaries. The helical angle is the degree to which cardiomyoctes are angled out of the short-axis plane, while the transverse is the angle between the cardiomyocytes and the circumferential direction. The helical ($\theta$) and transverse ($\beta$) angles are defined at a transmural depth $d$ normalized from 0 to 1 as:

$$\theta(d) = \theta_{endo}(1-d) + \theta_{epi} * d \tag{8.1a}$$

$$\beta(d) = \beta_{endo}(1-d) + \beta_{epi} * d. \tag{8.1b}$$

For each subject-specific model, we tested 9 separate rule-based fiber descriptions which spanned a range of $\theta_{endo}$ from 30° to 90° and $\theta_{epi}$ from −30° to −90° (Figure 8.5). These cases are referred to as $R_{i,j}$ with $i$ and $j$ referring to $\theta_{endo}$ and $\theta_{epi}$ respectively. For example, $R_{60,-90}$ represents a rule-based description with $\theta_{endo} = 60°$ and $\theta_{epi} = -90°$. Across all models, $\beta$ was set to 0 on the epicardial and endocardial surfaces, since there is evidence that

the transverse angle stays relatively constant at zero throughout the myocardial wall [240].

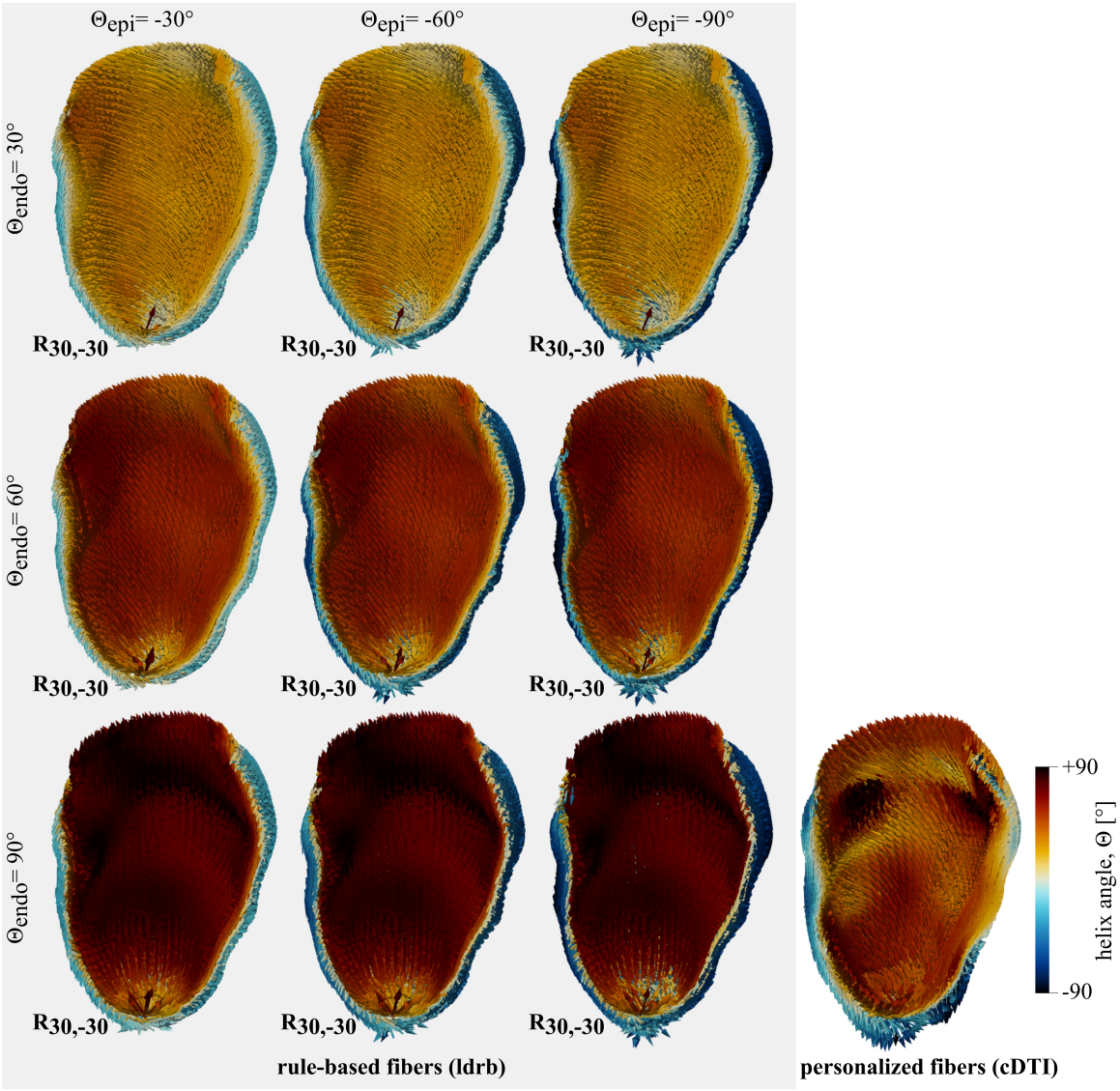



Figure 8.5: Different fiber orientations tested for each subject. Myofiber orientations were personalized using *in vivo* cDTI and alternatively described using the rule-based ldrb algorithm [10]. The fiber orientations for the personalized and the range of rule-based fiber orientations evaluated are shown. $\theta$ refers to the helix angle.

### 8.3.6 Constitutive Relation

The myocardium was modeled using as a hyperelastic incompressible material using a transversely isotropic exponential constitutive relation [98] described by the following strain energy density function:

$$\begin{aligned} \psi &= \frac{1}{2}c(e^{Q}-1) \\ Q &= b_f E_{ff}^2 + b_t(E_{ss}^2 + E_{nn}^2 + 2E_{sn}^2) + b_{ft}(2E_{fs}^2 + 2E_{fn}^2) \end{aligned} \tag{8.2}$$

where the $c$ parameter scales the overall passive myocardial stiffness and the $b_f$, $b_t$, and $b_{ft}$ parameters assign different exponential responses in the fiber ($f$) direction, across transverse ($t$) planes orthogonal to the fiber, and along the shear plane between the fiber and transverse direction ($ft$). $E_{ij}$ denotes the components of the Green-Lagrange strain tensor ($\boldsymbol{E}$) in a local myocardial coordinate system consisting of the fiber ($f$), sheetlet($s$) and sheetletnormal ($n$) directions.

We used the reformulation proposed by Xi *et al.* which expresses the strain energy law in terms of $c$, $\alpha$, $r_f$, and $r_t$ [3]. Following this reformulation, the material behavior is readily interpreted as the homogenous stiffness scaling parameters ($c$ and $\alpha$), and the anisotropy ratios in the fiber and transverse directions ($r_f$ and $r_t$). The reformulation also encapsulates the non-uniqueness and correlation of parameters of the transversely isotropic law in the space of $c$ and $\alpha$. The material parameters of the reformulated constitutive relation are described in terms of parameters of the original formulation as:

$$\alpha = b_f + b_t + b_{ft} \tag{8.3}$$

$$r_f = \frac{b_f}{\alpha} = 1 - r_t - r_{ft} \tag{8.4}$$

$$r_t = \frac{b_t}{\alpha} \tag{8.5}$$

$$r_{ft} = \frac{b_{ft}}{\alpha}. \tag{8.6}$$

### 8.3.7 Passive Material Parameter Estimation

The FE mechanics analysis was limited to the passive filling phase from end-diastasis to end-diastole. The LV end-diastasis geometric model was used as the reference mechanics model, (section 8.3.2) and was meshed with standard quadratic tetrahedral elements using GMSH [207]. The final mesh characteristics were determined following a mesh sensitivity study and are detailed in Table 8.2. The FE model was then built to be solved in FEBio [120].

Table 8.2: Mesh Properties

| Subject # | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
|---|---|---|---|---|---|---|---|
| Number of Nodes | 37268 | 55631 | 55494 | 76484 | 32226 | 74933 | 37868 |
| Number of Elements | 22420 | 35370 | 35018 | 49039 | 19549 | 48305 | 23069 |
| Element Volume ($mm^3$) | 3.54 | 3.40 | 3.43 | 3.48 | 3.42 | 3.33 | 3.42 |

Clinically, a normal LV end-diastolic pressure is typically characterized as ranging from 4-12 mmHg with an average of approximately 8 mmHg [9, 241, 242]. Therefore, in the absence of measured subject-specific LV cavity pressure recordings, the *in silico* LV was passively inflated by applying an endocardial pressure of 8 mmHg. Motivated by a study by Hasaballa *et al.* which quantified the beat-to-beat variation in LV end-diastolic chamber stiffness in controls [243], the uncertainty in the simulated material parameters due to a $\pm$20% uncertainty in LV end-diastolic pressure was also investigated. Dirichlet boundary conditions were applied to constrain the longitudinal displacements of the *in silico* model's basal nodes to match the tagged MRI-obtained kinematics. In the short axis, only nodes belonging to the epicardial ring were fixed, while the remaining nodes were unconstrained. Then, the experimentally determined LV local displacements from end-diastasis to end-diastole were used to inversely estimate the optimal parameters of the reformulated transversely isotropic exponential constitutive relation (see section 8.3.6). Note that our *in silico* model is two-state, only considering the reference state at end-diastasis and the maximum inflation state at end-diastole. This was done to avoid assuming the form of the diastolic pressure waveform given the notable variability among subjects [162].

Fung-type exponential constitutive relations, are known to have non-unique solutions. In the case of the the reformulated transversely isotropic relation used here, there exists a

set of nearly kinematically equivalent solutions in the space of $c$ and $\alpha$. Here, we present the results of the passive mechanical behavior as optimal $c$-$\alpha$ coupling curves which encapsulate the set of non-unique solutions. The anisotropy ratios were kept constant at physiologically realistic values of $r_f = 0.55$, $r_t = 0.25$, and $r_{ft} = 0.2$ given that the stiffest behavior is expected to be along the fiber direction. As demonstrated by Xi *et al.* the $c$-$\alpha$ coupling curve can be described by the relationship $\alpha c^a = b$. Given the linear nature of the coupling curve in log-space, with the form of $a \log c + \log \alpha = \log b$, to find the optimal $c$-$\alpha$ coupling curve that characterizes the non-unique passive material properties, at least two parameter optimizations are required at different $\alpha$ values.

For each subject, to determine the optimal $c$-$\alpha$ coupling curve, simulations were performed to find the corresponding minimal-residual $c$ values of three logarithmically spaced $\alpha$ from 10-200. Specifically, for each alpha value, after initializing the $c$ material parameter using results from Nasopoulou *et al.* [1], we iteratively updated $c$ to minimize differences between the experimentally measured nodal displacements (tagged MRI) and the simulated pressure-induced nodal displacements. Subsequently, after identifying three pairs of optimal $c$ and $\alpha$, a straight line was fit to the three pairs of optimal $c$ and $\alpha$ points in log-space to identify the set of nearly kinematically equivalent solutions to the passive mechanics identification problem in the space of $c$-$\alpha$. The parameter identification methodology, as well as the procedure for developing the forward FE models, is summarized in Figure 8.6.

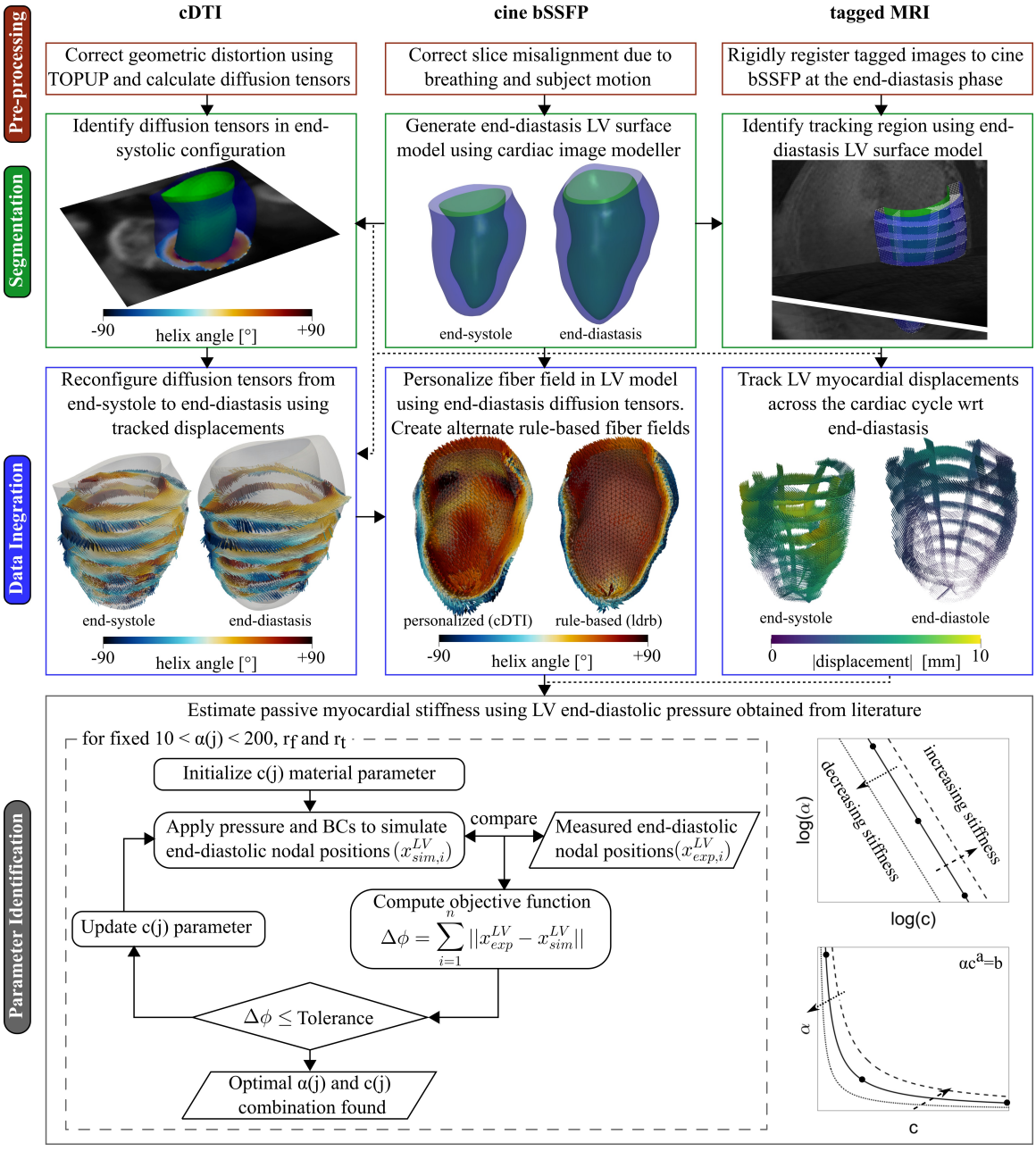



Figure 8.6: Pipeline for integrating geometry (cine bSSFP), microstructure (cDTI), and kinematics (tagged MRI) data into a finite element mechanics model for estimating passive myocardial stiffness. In the case of rule-based fiber orientations, the cDTI column is extraneous. The material parameter estimation procedure using the forward FE mechanics model and the kinematic data is also summarized.

## 8.4 Results

Across all subjects, the identified optimal $c$ and $\alpha$ value pairs provided a good match between the end-diastolic nodal positions predicted by the FE simulation and that which was measured from the tagged MRI (Figure 8.7). An optimal $c$-$\alpha$ coupling curve which provided a good fit to the pairs of optimal $c$ and $\alpha$ points ($R^2 > 0.99$) was identified in all subjects (Figure 8.8). This provides further evidence for the reported coupling between $c$ and $\alpha$ reported by Xi *et al.* [3].

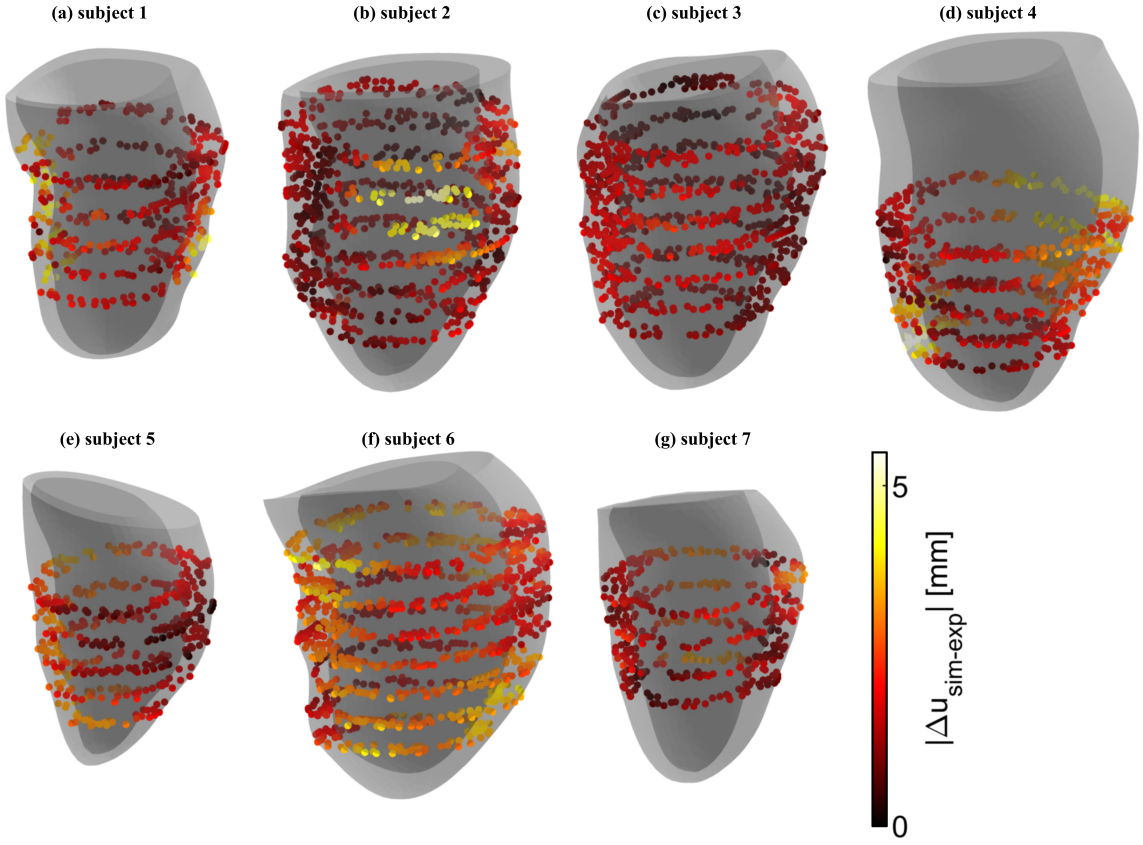



Figure 8.7: Displacement error between experimentally determined nodal positions and simulated nodal positions using one optimal $c$ and $\alpha$ combination. All models shown here had personalized fibers except for Subject 4 that did not have adequate cDTI data. $R_{60,-60}$ model shown for Subject 4 instead.

In figure 8.8 we overlay all subjects' optimal $c$-$\alpha$ coupling curves for the range of rule-based fiber models tested, and for the personalized fiber model with uncertainty due to a

±20% end-diastolic pressure variability. We also show the mechanical behavior in uniaxial tension using the optimal material parameters at $\alpha = 45$ for all the subjects (Figure 8.9). Note that the poor cDTI acquisition in subject 4 prevented the development of a personalized fiber model of this subject. Hence, we used the $R_{60,-60}$ model instead when comparing with the other subjects' models with personalized fibers. We more directly compare the observed variation in the estimated material parameters due to the loading pressure and rule-based fiber variability, in figure 8.10. Specifically, we show the average and standard deviation of the estimated $c$ at $\alpha = 45$ across the range of end-diastolic pressures tested in the personalized model and we show the same for the range of rule-based fiber orientations tested. In all subjects, the parameter variation due to the end-diastolic pressure uncertainty exceeded the variation due to rule-based fiber orientations variability.

Using the estimated parameters (at $\alpha = 45$), we calculated the fiber cauchy stress and stretch. The distribution of fiber stress and stretch are shown in figure 8.11 and the spatial map of fiber stress in different regions of the the LV is shown for all subjects (Figure 8.12). Of note, there wasn't significant variation in the fiber stress and stretch distribution for different values of $c$ and alpha that fall along the optimal coupling curve (<0.6% difference on average).

Table 8.3 shows the estimated $c$ parameter at $\alpha = 45$ for all subjects. The $c$ ranged from 1.2 kPa in subject 4 to 6.1 kPa in subject 1. The average fiber cauchy stress, fiber stretch and displacement error between the FE-predicted nodal positions and the tagged MRI-obtained target nodal positions are also listed. For all the metrics in table 8.3, we also show the minimum and maximum range from alternate simulations using the different rule-based descriptions of microstructure evaluated in this study (Figure 8.5).

To put the subjects' estimated parameters in context, results from three previously published studies that performed passive myocardial parameter estimation in healthy humans are compared. In figure 8.8h, we overlay the parameters obtained in Nasopoulou *et al.* [1], Wang *et al.* [2], and Xi *et al.* [3] over the optimal $c$-$\alpha$ coupling curves for all the subjects. We also do this comparison with the literature in terms of principal Cauchy stress and stretch in a simulated idealized unixial tensile test using the material parameters. Specifically, using the optimal $c$-$\alpha$ coupling curve for a subject, we found the $c$ parameter that corresponds to

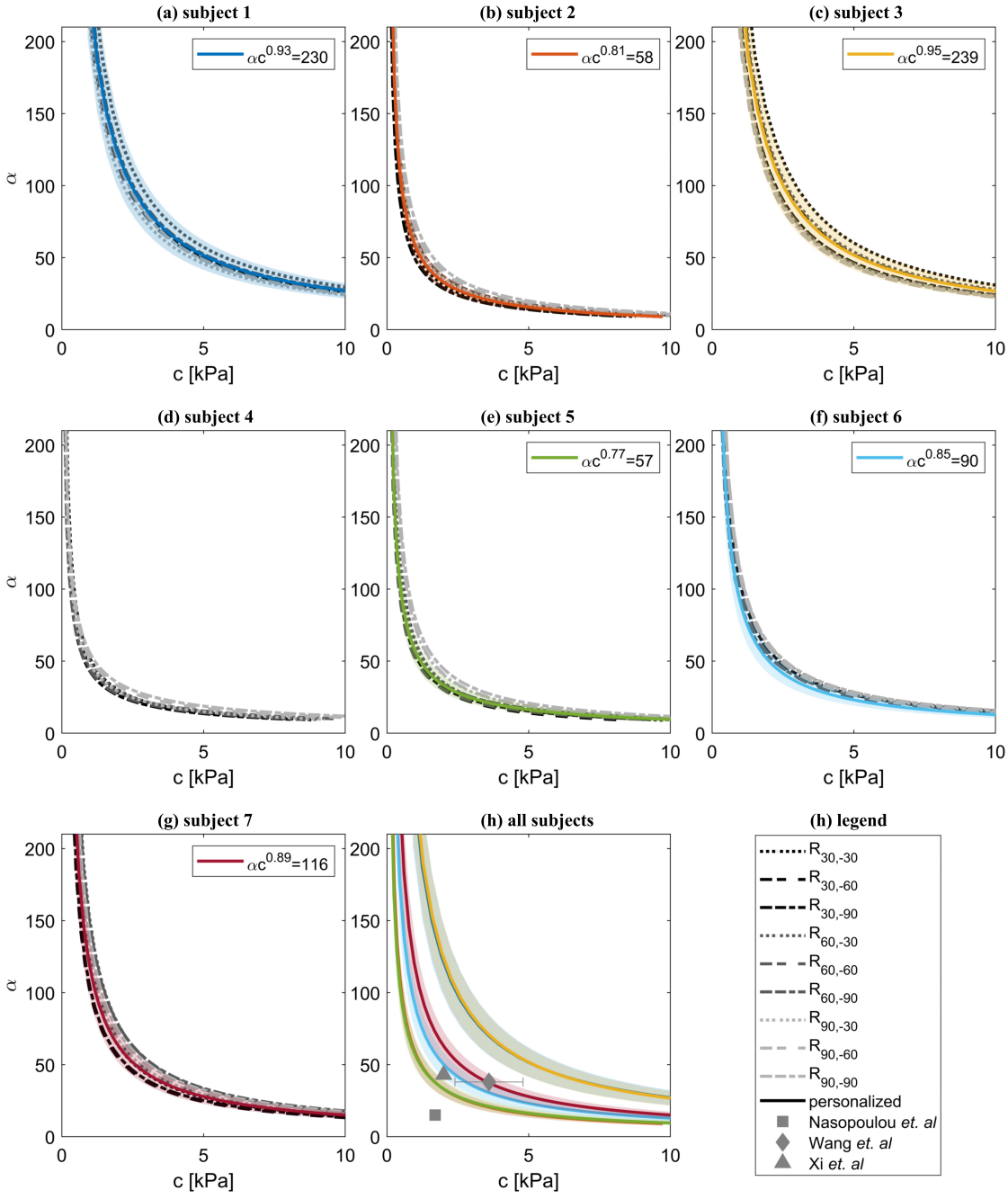



Figure 8.8: Optimal $c$-$\alpha$ coupling curves for all subjects. Solid colored lines are for models with personalized (cDTI) fibers and the uncertain region around this line (colored and transparent) represents the upper and lower bounds of the behavior due to a $\pm 20\%$ variation in LV end-diastolic pressure. Literature results of passive myocardium parameter estimation in healthy humans are shown relative to results from this study (h). Subject 4 did not have adequate cDTI data, so personalized mechanics results aren't shown.

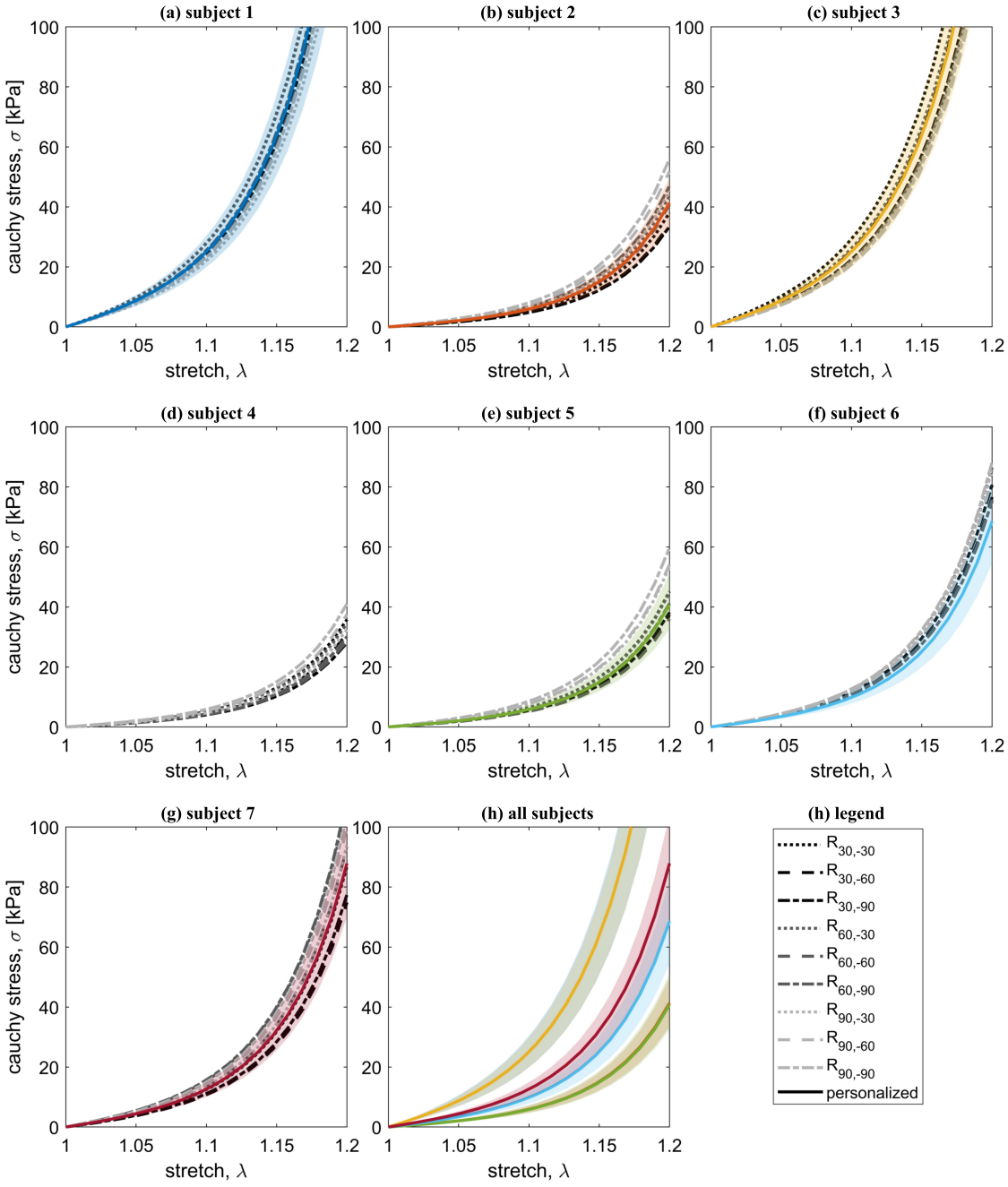



Figure 8.9: Cauchy stress vs stretch from idealized uniaxial tension along the fiber direction for $\alpha = 45$. Color and line coding are same as in figure 8.8

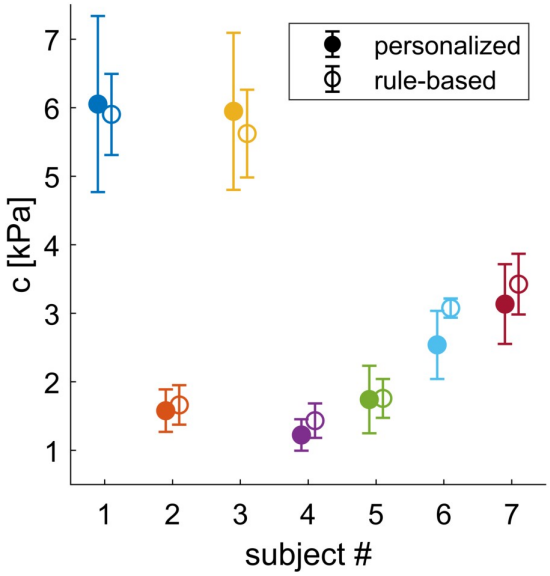



Figure 8.10: Comparison of the effects of rule-based fibers and ±20% end-diastolic pressure uncertainty (within beat-to-beat range). The solid marker represents the average and standard deviation of estimated $c$ at $\alpha = 45$ using the personalized fiber models across the range of pressures tested. The clear marker represents the average and standard deviation of estimated $c$ at $\alpha = 45$ across the range of rule-based descriptions of the fiber orientation evaluated

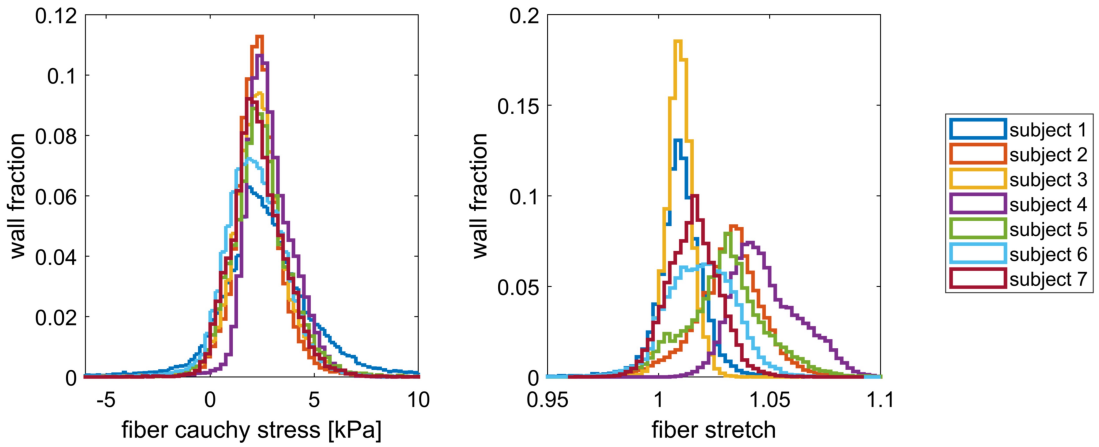



Figure 8.11: Fiber cauchy stress and stretch distribution in the LV myocardium at end-diastole. Shown for models with personalized (cDTI) fiber orientations except for subject 4 ($R_{60,-60}$ shown instead)

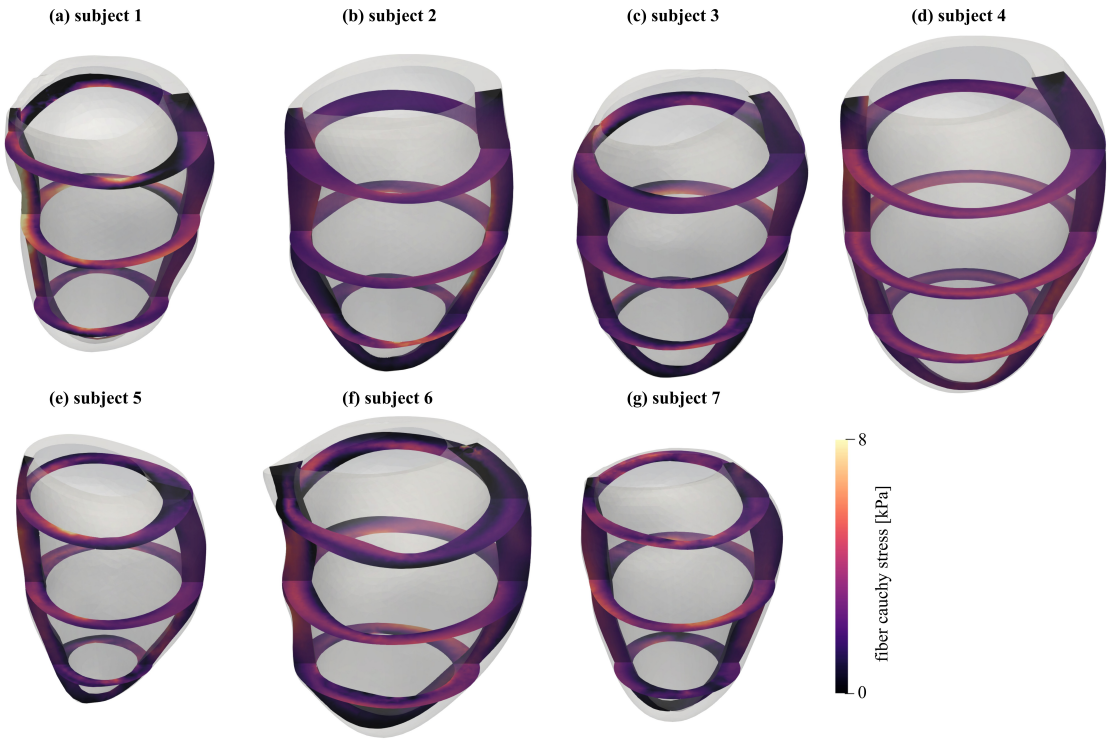



Figure 8.12: Fiber cauchy stress map at apical, equatorial, basal, and long axis slices of the LV. Shown for models with personalized (cDTI) fiber orientations except for subject 4 ($R_{60,-60}$ shown instead)

Table 8.3: Passive material parameter estimation results: $c$ material parameter at $\alpha = 45$, average displacement error between simulation and experimental nodal positions ($|\Delta u_{sim-exp}|$), average fiber cauchy stress ($|\sigma_{fiber}|$), and average fiber stretch ($|\lambda_{fiber}|$) for all subjects. Outside the square brackets are results from models with the personalized fibers orientations and inside the square brackets are the maximum and minimum values across models with the rule-based fibers. Results of passive material parameter estimation in healthy humans from literature included [1, 2, 3]. [a]Scaling constant defined as half $c$ parameter, so values reported here are doubled for consistency

| Case | $\alpha$ | $c$ kPa | $\lvert\Delta u_{sim-exp}\rvert$ mm | $\lvert\sigma_{fiber}\rvert$ kPa | $\lvert\lambda_{fiber}\rvert$ |
|---|---|---|---|---|---|
| subject 1 | 45 | 6.1 [6.8 5.2] | 1.93 [1.96 1.87] | 2.64 [2.91 2.38] | 1.010 [1.013 1.009] |
| subject 2 | 45 | 1.6 [2.1 1.3] | 1.60 [1.95 1.52] | 2.32 [2.64 1.86] | 1.032 [1.046 1.018] |
| subject 3 | 45 | 6.0 [7.0 5.0] | 1.31 [1.37 1.20] | 2.36 [2.66 1.99] | 1.009 [1.013 1.008] |
| subject 4 | 45 | [2.0 1.2] | 2.41 [2.90 1.92] | 2.86 [3.01 2.19] | 1.049 [1.054 1.023] |
| subject 5 | 45 | 1.6 [2.3 1.5] | 2.00 [2.37 1.95] | 2.44 [2.75 2.12] | 1.030 [1.043 1.020] |
| subject 6 | 45 | 2.5 [3.3 2.9] | 2.53 [2.78 2.64] | 2.19 [2.38 1.92] | 1.019 [1.021 1.012] |
| subject 7 | 45 | 3.2 [4.0 2.7] | 1.74 [1.86 1.71] | 2.34 [2.86 2.16] | 1.016 [1.026 1.012] |
| Nasopoulou *et al.* | 15 | 1.7 | 3.35 | - | - |
| Wang *et al.* | 38 | $3.6 \pm 1.2$[a] | - | - | - |
| Xi *et al.* | 43 | 2.0[a] | 1.78 | - | - |

a given $\alpha$ from the literature study. The obtained optimal $c$ and $\alpha$ combination for the subject was then used to simulate the mechanical behavior (cauchy stress vs stretch) in uniaxial tension along the fiber direction. This mechanical behavior across all subjects is compared with the literature results (Figure 8.13)

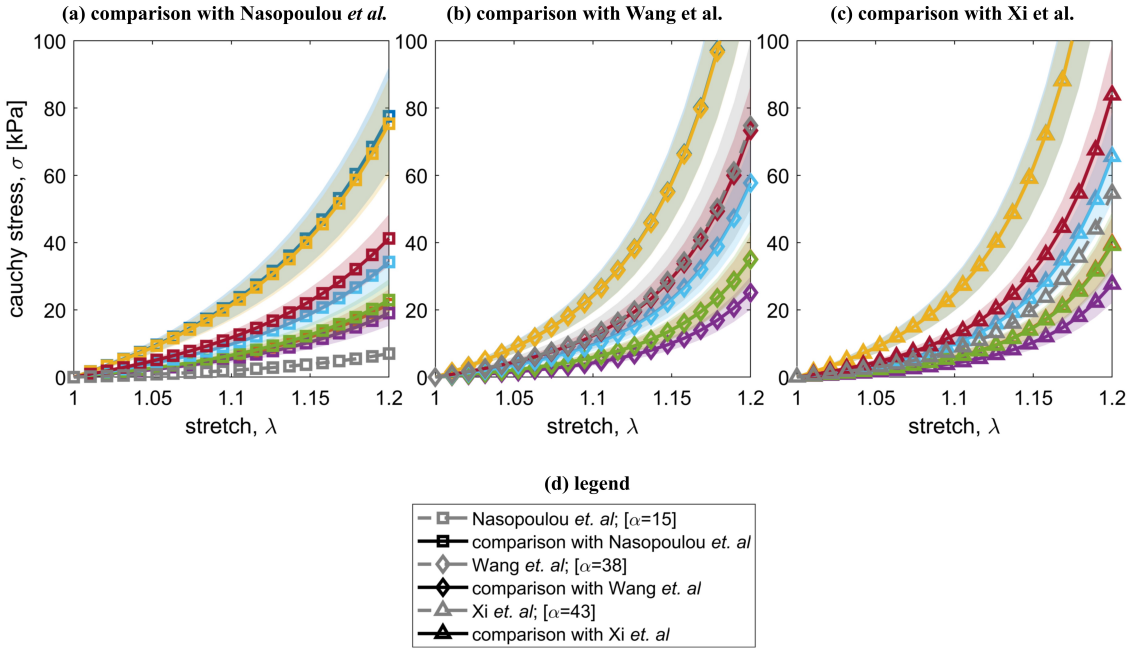



Figure 8.13: Comparison of estimated material parameters for all subjects with results from (a) Nasopoulou *et al.* [1], (b) Wang *et al.* [2] and (c) Xi *et al.* [3] who performed passive material parameter estimation in healthy humans. Cauchy stress vs stretch from idealized uniaxial tension along the fiber direction is shown using the corresponding $c$ for $\alpha$ from the previous studies. Color and line coding are same as in figure 8.8. Personalized fiber model results shown for all subjects except for subject 4, ($R_{60,-60}$ shown instead)

## 8.5 Discussion

Altered passive myocardial stiffness is a component of maladaptive cardiac remodeling implicated in the etiology of several heart diseases including heart failure. However, the specific underlying mechanisms and the extent to which the changes contribute to the progression of heart failure are poorly understood. This challenge has motivated the development

of cardiac biophysical models to estimate the material parameters governing the mechanical behavior and stiffness of *in vivo* passive myocardium. Several previous studies have leveraged medical imaging and inverse FE modeling to evaluate passive myocardial material parameters. However, none to our knowledge, have combined *in vivo* bSSFP, tagged MRI, and cDTI to build the FE models as we have done here, despite the fact that these cardiac MRI methods independently provide key information necessary for material parameter estimation: cardiac surface geometry (bSSFP), deformation kinematics (tagged MRI or cine DENSE), and microstructure (cDTI).

In the absence of cine bSSFP data, some studies have extracted the reference geometry for FE modeling from tagged MRI [144, 244], eliminating the need to register kinematic and geometric data. However, surface geometries extracted from tagged MRI may overestimate wall thickness due to the difficulty associated with dileneating endocardial borders from papillary muscles. In the absence of local kinematic data, some authors have matched end-daistolic surfaces or volumes [157, 145], but this method may not adequately constrain the material parameter solution. Subject-specific microstructure, such as that which can be obtained through *in vivo* cDTI is rarely ever available in patients or research subjects especially with full LV apex to base coverage and in conjunction with cine bSSFP and tagged MRI data. Thus, mathematical models of the myofiber orientation are commonly used in the FE models instead [10, 150, 151, 153, 152]. While these mathematical fiber descriptions are convenient, there is a lack of studies on the effects of rule-based vs personalized fiber orientations on the final passive myocardial material parameters.

### 8.5.1 Passive myocardial material parameter estimation

In this study, the parameter estimation is based on a single LV state at end-diastole, which compounds the non-unique solution issue associated with Fung-type exponential constitutive relations such as the Guccione transversely isotropic relation. Despite the availability of multiple tagged MRI imaging frames between end-diastasis and end-diastole, absence of LV diastolic pressure measurement precluded the use of kinematic data from multiple cardiac phases for parameter estimation without assuming the form of the diastolic pressure waveform. To better encapsulate the non-uniqueness and correlation of material parameters,

for each subject, we obtained an optimal $c$-$\alpha$ coupling curve for physiological anisotropy ratios. This presentation of the material parameters differs from the more common approach in the literature where only one parameter is calibrated (typically $c$) while the others are fixed [9, 2]. These optimal $c$-$\alpha$ coupling curves can be extended to uniquely identify $c$ and $\alpha$ provided corresponding pressure data from at least one additional cardiac phase between end-diastasis and end-diastole, using the approach introduced by Nasopoulou *et al.* [1].

Results from the Wang and Xi study of healthy *in vivo* passive myocardial material parameters fall within the range of material parameters we obtained, but Nasopoulou's estimates are appreciably softer than the range which we have obtained here (figures 8.8&8.13). The discrepancy in material parameters could be reflective of biological variability in the passive myocardial stiffness of the healthy subjects tested in our respective studies; it may also be due to different modeling assumptions. Ultimately, extending this myocardial stiffness estimation framework to patients with cardiac conditions characterized by altered passive myocardial stiffness will enable better understanding of the accuracy and precision required to adequately measure important clinical changes to myocardial properties. For now, the results presented in this study provide an important reference for healthy passive myocardial material parameters that is consistent with prior work.

### 8.5.2 Effects of fiber fields and loading pressure

Previous work analysing the sensitivity of passive cardiac mechanics forward simulations have revealed key insights on the impact of uncertain inputs on certain quantities of interest like the LV cavity volume, myocardial stresses and strains, wall thickness: (1) the overall stiffness scaling material parameter has the greatest influence on quantities of interest [245, 229, 246], (2) myocardial stress is sensitive to wall thickness [165], (3) while myocardial stresses and strains are sensitive to fiber orientation [247], shear strains are more sensitive to fiber orientation than normal strains [248], and (4) fiber dispersion can influence passive mechanics [249].

There are comparatively fewer inverse uncertainty quantification analyses assessing the identifiability and variation of inversely estimated material parameters given the uncertainty in the experimental data (eg pressures, fiber orientations, kinematics). This task differs

from forward uncertainty quantification where the goal is to establish how uncertainties in FE mechanics model inputs affect the model outputs (quantities of interest). Augenstein *et al.* found that parameter identifiability is affected by the signal to noise ratio of the tagged MRI acquisition. Specifically, more image frames are needed for confidence in estimated parameters with increasing noise in the tagged MRI data [154]. Lazarus *et al.* found that some material parameters cannot be reliably identified without inclusion of high pressure data outside of the physiological range [166]. The work of Palit *et al.* is somewhat similar to ours, in that they investigated the effects of cardiac geometry, LV end-diastolic pressure, and fiber orientations of material parameter estimation [146]. The authors however had no kinematic MRI data, and calibrated material parameters using LV volumes, despite the fact that optimizing for material parameters using local deformation kinematics better constrains the parameter estimation. Additionally, absence of microstructural MRI data meant they couldn't compare between rule-based and personalized fiber orientations. Nevertheless, they found that that the estimated passive material parameters had different degrees of sensitivity to variations in the geometry, pressure and fiber orientations.

A key finding from our study is that while the use of rule-based descriptions in place of personalized fiber orientations does influence the estimated passive material parameters, the deviation is generally less than that which can be attributed to beat-to-beat variation in the LV end-diastolic pressure. As our results suggest, for the task of identifying passive myocardial material parameters, accurate estimation of the diastolic pressure(s) may be of greater priority than highly accurate microstructure evaluation. However, it is important to note that identification of accurate fiber orientation information is still essential, especially for systolic mechanics [248, 250, 167]. Personalized fiber orientation information is also necessary in passive mechanics for quantifying quantities of interest such as myocardial and myofiber stresses and strains [247].

### 8.5.3 Limitations

Cardiac microstructure can be more effectively evaluated using higher performance MRI scanners than that which we used here, enabling better quality *in vivo* personalized fiber orientations and perhaps slightly affecting the results. Though the residual stresses in the *in*

*vivo* heart are well documented [137], we didn't consider their effects, and assumed the diastasis cardiac phase corresponding to the start of atrial systole as the unstressed configuration. Algorithms for unloading the prestressed geometry exist, but they require assumptions about the material parameters and are yet to be fully validated. Moreover, the lack of an end-diastolic pressure trace meant we didn't have the minimum LV pressure, an important piece of information necessary for unloading. Given that usually the end-diastasis pressure is nonzero and positive, we expect that unloading the models to an unstressed state prior to material parameter calibration would lead to a reduction in the estimated passive myocardial stiffness. Additionally, despite evidence of orthotropy in myocardium, we used a transversely isotropic material model. This is because our cDTI acquisition is mostly sensitive to diffusion in the fiber direction, without providing distinct information about sheetlet orientation. When this data becomes available, the passive stiffness estimation framework could easily be extended to incorporate orthotropic material models. It would then be interesting to evaluate differences in mechanical response between the two modeling approaches to understand if transverse isotropy is an adequate approximation of myocardium.

## 8.6 Conclusion

We have established a method to develop subject-specific LV mechanics models for evaluating passive myocardial material parameters by integrating multi-phase *in vivo* cardiac cine bSSFP and tagged MRI with single-phase *in vivo* cDTI (Figure 8.6). We then calibrated the LV FE mechanics models using the local deformation kinematics to estimate the optimal parameters of a reformulated transversely isotropic constitutive model of the myocardium (see section 8.3.6). Furthermore, we assessed independently the sensitivity of passive myocardial parameter estimation to fiber orientations and LV end-diastolic pressures within the beat-to-beat variation range.

# Chapter 9

# Conclusion

Medical image-driven cardiac FE modeling is an emerging clinically viable approach to measuring *in vivo* passive myocardial stiffness, a potent biomarker of heart health implicated in heart failure. The two contributions of this work to the growing field of biophysical modeling of the heart are related to validating MRI-driven myocardial stiffness estimation and using MRI to characterize healthy passive cardiac mechanics. Regarding validation, the main objectives were to: develop a tunable experimental system for validating MRI-based passive myocardial stiffness estimation approaches by using synthetic heart phantoms with realistic cardiac morphometry, and heart-like stiffness and MRI-relaxation properties (chapter 5), and evaluate the accuracy of an MRI and FEM-based framework for estimating passive myocardial stiffness using synthetic heart phantoms of varying stiffness and geometry (chapter 6). Concerning characterizing healthy passive cardiac mechanics using *in vivo* MRI, the objectives were to: establish a tractable pipeline for developing personalized cardiac computational mechanics models for evaluating passive myocardial stiffness from *in vivo* cDTI, cine bSSFP, and tagged MRI (chapter 7), test the *in vivo* passive myocardial stiffness estimation method in healthy human subjects in order to establish normal reference passive myocardial stiffness values, and evaluate the sensitivity of passive mechanics to microstructural configuration. (chapter 8). In the following sections, the outcomes from the two main contributions in the thesis are summarised. Since the limitations have been discussed in detail in the discussion section of each body chapter (chapters 5-8), they are not repeated here. Instead, important issues to address in future work are presented. The

chapter is concluded with a general outlook on the field.

## 9.1 Validating MRI-driven Myocardial Stiffness Estimation

This portion of the thesis is covered in chapters 5 and 6 and the main outcomes were the following:

- Found that materials of tunable heart-like stiffness and MRI-relaxation properties can be developed by varying the mass composition of constituents of blends containing commercially available silicone elastomers, Sylgard 184 and 527. Specifically a linear relationship between the percentage of Sylgard 184 and the stiffness of the blends was determined within the range of 10% to 40% Sylgard 184.

- Established a lost-wax casting method which leverages 3D-printing to manufacture subject-specific heart phantoms using the myocardium-mimicking materials previously discussed. This procedure was used to produce heart phantoms of varying myocardium-mimicking stiffnesses and subject-specific geometries of high fidelity. The ground-truth mechanical behavior of the heart phantom materials were characterized through tensile testing.

- Developed an MRI compatible flow loop controlled by a programmable flow pump to simulate diastolic LV filling in the phantoms. Incorporated the heart phantoms within the flow loop and acquired MRI data of the phantoms including reference geometry, inflation volumes, and ventricular pressures while undergoing LV inflation. Simulated the phantoms' material stiffness using the MRI data and compared the estimates to the independently obtained ground truth.

- Quantitatively assessed the accuracy of MRI-driven computational modeling for myocardial stiffness estimation in four soft silicone heart phantoms of varying realistic cardiac morphometry and known ground truth mechanical behavior and stiffness. Across all models the absolute stiffness estimation errors ranged from 1% to 6%,

demonstrating the feasibility of highly accurate passive myocardial material stiffness estimation using an MRI-driven computational constitutive modeling framework.

An important area of future work would be the development of phantoms that are better representations of the *in vivo* heart. While the heart phantom materials used in this study were a practical approximation of myocardium, they lacked some key biomechanical features including prestresses, exponential nonlinearity in the strain range considered, and anisotropy. Moreover, the silicone elastomers used are insensitive to diffusion imaging. The materials were also stiffer than healthy human myocardium, with the softest phantom more closely representing the stiffness of infarcted myocardium. Additionally, some of the boundary and loading conditions of the *in vivo* heart could not be replicated in our *in vitro* setup including the valve plane motion, apical rotation, synchronized RV filling, and pericardial interactions. Our work is an important demonstration of feasibility and should be extended to more complex *in vitro* experiments incorporating heart phantoms that more accurately replicate *in vivo* conditions.

Future work could also focus on the MRI acquisition used for the phantom studies. For example, while we acquired relatively high-resolution data, we could either acquire lower resolution imaging datasets or downsample the imaging data to more clinically feasible resolutions to assess effects of resolution on stiffness estimation. We could also acquire kinematics data (e.g. MRI tagging) and use reconstructed local deformation kinematics for stiffness calibration instead of using just volumes.

Lastly, the experimental setup could be used to assess the precision and repeatability of passive myocardial stiffness estimation, in addition to accuracy which was assessed here.

## 9.2 Using MRI to Characterize Healthy Passive Cardiac Mechanics

This portion of the thesis is covered in chapters 7 and 8 and the main outcomes were the following:

- Determined the appropriate MRI acquisition parameters to obtain high fidelity *in vivo*

cardiac MRI datasets which include cine bSSFP (surface geometry), tagged MRI (deformation kinematics), and cDTI (myofiber orientations).

- Developed a pipeline for combining the subject-specific geometric, functional, kinematic and microstructural information extracted from the imaging data into a single personalized *in silico* FE mechanics model for evaluating passive myocardial mechanical behavior.

- Applied the passive myocardial material parameter evaluation technique to multiple subjects in order to establish a range of reference values for healthy human *in vivo* passive myocardial mechanical behavior.

- Performed an analysis to understand the sensitivity of passive mechanics to fiber orientations by comparing results from incorporating personalized fibers with those obtained from using different rule based descriptions of the 3D fiber orientation. Also assessed sensitivity of passive mechanics to loading conditions. Across all subjects, the passive mechanical behavior was found to be more sensitive to the loading conditions than the microstructure.

Future work should be focused on quantifying the uncertainty associated with material parameter estimation due to measured inputs including geometry, microstructure, kinematics and hemodynamics. We could also conduct sensitivity analyses on the FE models to better understand the inputs (e.g material parameters, end-diastolic pressure, fiber orientations) with the greatest effect on biomechanical metrics such as strains and stresses. Moreover, in this work, we didn't fit all the parameters of the constitutive relation but focused on the global stiffness parameters and assumed physiologically feasible values for the anisotropy ratios. The influence of variations to the anisotropy ratios can be studied, as this may be an important consideration in disease processes with significant microstructural derangement.

We lacked ventricular pressure measurement which necessitated the use of healthy LV end-diastolic pressure values from literature and precluded the use of multiple cardiac phases for parameter identification. LV filling pressures are ideally obtained through direct measurement by left heart catheterization. However, such measurements are difficult to obtain, especially in healthy subjects and in conjunction with comprehensive cardiac imaging data.

LV filling pressures could however be estimated non-invasively using mitral flow velocities which can be assessed non-invasively by 4D flow MRI or pulsed Doppler ultrasound. Including this acquisition in the imaging dataset would enable the development of fully personalized inverse finite element models for estimating passive myocardial material parameters.

We could also more accurately model the heart. In this work, our FE models only consisted of the LV without accounting for other cardiac structures such as the RV, atria and pericardium. We could include the influence of these other cardiac structures either by fully modeling them or by appropriately incorporating their effects as boundary conditions on the LV models. Though the residual stresses in the *in vivo* heart are well documented, we didn't consider their effects, and assumed the diastasis cardiac phase corresponding to the start of atrial systole as the unstressed configuration. Algorithms for unloading the prestressed geometry exist, but they require assumptions about the material parameters and are yet to be fully validated. Moreover, the lack of an end-diastolic pressure trace meant we didn't have the minimum LV pressure, an important piece of information necessary for unloading.

Despite evidence of orthotropy in myocardium, we used a transversely isotropic material model. This is because our cDTI acquisition is only sensitive to diffusion in the fiber direction, without providing information about sheet orientation. When this data becomes available, the passive stiffness estimation framework could easily be extended to incorporate orthotropic material models. It would then be interesting to evaluate differences in mechanical response between the two modeling approaches to understand if transverse isotropy is an adequate approximation of myocardium.

Another important extension of this work, would be to use the passive myocardial material parameter estimation framework to characterize changes to passive mechanics in heart diseases. This research is motivated by heart failure and the overarching goal is to develop a thoroughly validated clinical tool for the estimation of passive myocardial stiffness to inform management of cardiac diseases with associated alterations to myocardial material properties.

## 9.3 Outlook and Closing Remarks

Medical image-based biophysical modeling of the heart enables rigorous assessment of biomarkers of heart health that cannot otherwise be estimated. This tool holds great promise for cardiology as it provides opportunities to improve understanding of cardiac (patho-)physiology and to inform management of heart disease. However, before the computational tools developed in research labs can fully transition into clinical settings, a number of pertinent issues must be addressed.

Currently, material parameter evaluation with clinical data is a prohibitively long and involved process. Interventions to speed up different portions of the pipeline such as imaging, data extraction, data integration, and FE simulation, would make it a more viable option for clinical use. While current imaging methods can be adequately used to evaluate geometry and kinematics, *in vivo* microstructure evaluation using cDTI is still lagging due to the currently limited imaging quality, resolution, diffusion sensitivity (e.g. to assess sheet orientations), and multi-phase acquisition capabilities. Also of note, there is a lack of consensus on the best approach for material parameter estimation which makes it difficult to compare results across studies. This lack of standardization is manifested in the plethora of material models being used in literature. To inform standardization, it would be important to assess the advantages and disadvantages of existing material models in terms of their "goodness of fit" to experimental data, parameter identifiability given the uncertain and noisy inputs, and physiological meaning of parameters. There is also need for further experimental work including validation and *ex vivo* testing of myocardium to inform the development of constitutive models.

Nevertheless, driven by the contributions of various researchers, the field has grown in leaps and bounds over the last few decades. The current barriers to clinical adoption of biophysical modeling are being tackled by various groups which is promising for the central goal of using these modeling tools to improve patient care.

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